

Optimal Policy Generator: Evidence-Based Policy Recommendations for Jurisdictions

Systematic Generation of Enact/Replace/Repeal/Maintain Recommendations Using Quasi-Experimental Methods and Bradford Hill Criteria

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Abstract

Fifty US states with different laws. Two hundred countries with different policies. Centuries of variation in tax rates, regulations, programs. The data to know what works exists but isn't systematically harvested.

The Optimal Policy Generator measures policy impact on two fundamental welfare dimensions: **real after-tax median income growth** (economic welfare) and **median healthy life years** (health welfare). These two metrics are hypothesized to capture the primary welfare effects of most policies while remaining directly interpretable. No complex conversion factors required.

For any jurisdiction, four outputs: ENACT (evidence-supported policies you lack). REPLACE (policies at the wrong level). REPEAL (policies with net harm). MAINTAIN (policies aligned with evidence). Each recommendation includes expected effects on both metrics, confidence grades, and blocking factors (including freedom/autonomy constraints).

Bradford Hill criteria operationalized into nine quantitative scoring functions. Synthetic control and difference-in-differences isolate policy effects from confounders. Agnostic to which party enacted each policy, only whether it worked. Projected welfare gains under framework assumptions: 5-15% of GDP for typical US states (90% CI: 2-25%), pending retrospective validation.

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Abstract

This specification describes the **Optimal Policy Generator (OPG)**, a proposed framework for producing jurisdiction-specific policy recommendations based on quasi-experimental evidence. OPG measures policy impact on two fundamental welfare dimensions: **real after-tax median income**

growth (economic welfare) and **median healthy life years** (health welfare). These metrics are hypothesized to capture the primary welfare effects of most policies while remaining directly interpretable.

Epistemic status: OPG is an unvalidated methodological proposal. The framework represents a theoretically-motivated approach to evidence aggregation, not a validated predictive tool. Quantitative claims (e.g., “5-15% of GDP welfare gains”) are projections under framework assumptions that require empirical calibration. Terminology like “evidence-supported” indicates consistency with quasi-experimental evidence, not causal proof.

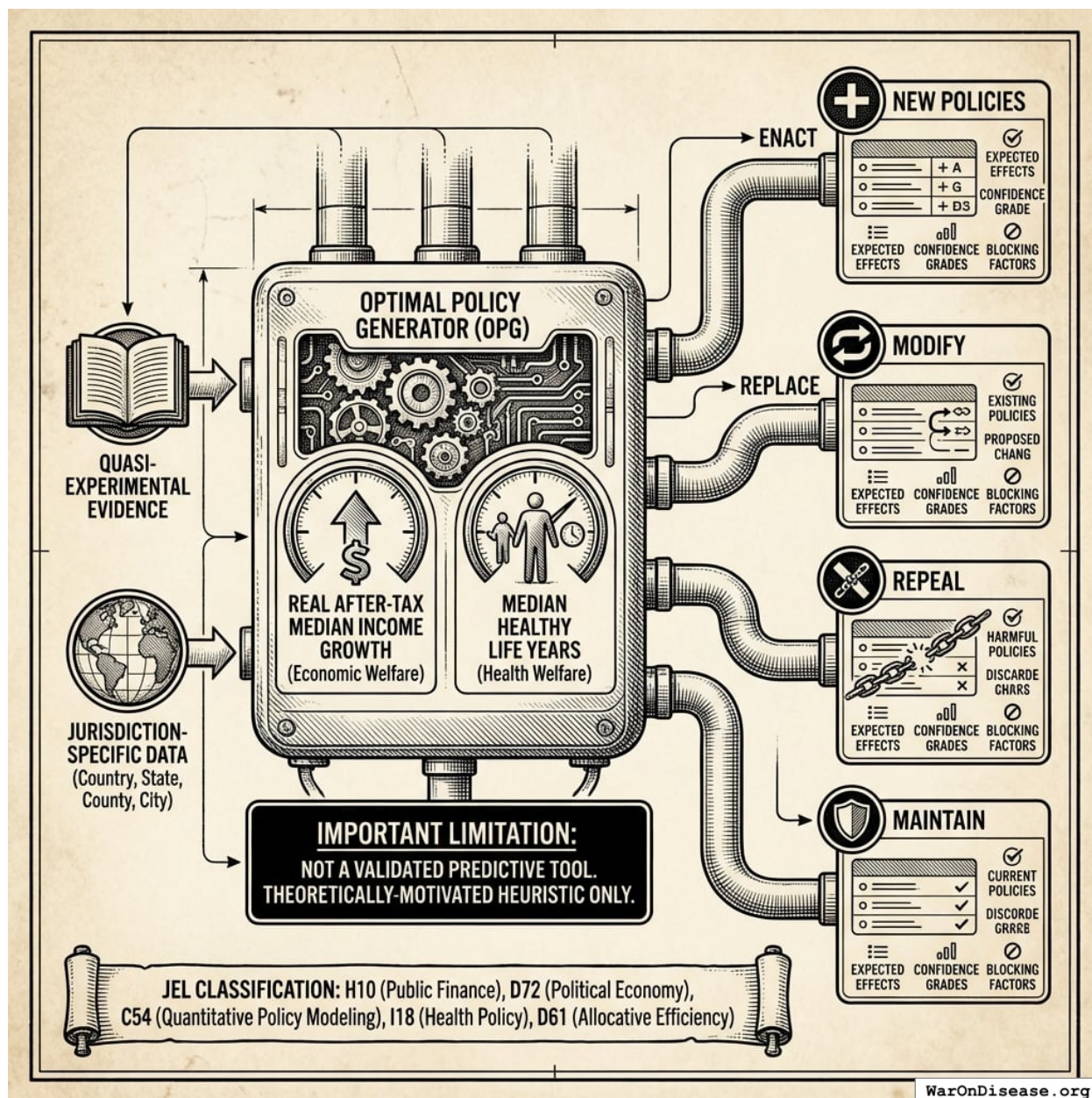


Figure 1: A conceptual diagram of the Optimal Policy Generator (OPG) framework illustrating the flow from evidence and welfare metrics to the four primary policy outputs: Enact, Replace, Repeal, and Maintain.

Important limitation: Until retrospective validation is performed (see Section 20), OPG should be treated as a theoretically-motivated heuristic for policy prioritization, not a validated predictive tool. The quasi-experimental methods provide evidence *consistent with* causation under assumptions that are often untestable.

OPG answers four questions: “What should we add? Change? Remove? Keep?” The framework operates at any jurisdiction level (country, state, county, city) and produces four outputs:

1. **Enact:** New policies the jurisdiction should adopt
2. **Replace:** Existing policies to modify
3. **Repeal:** Harmful policies to remove
4. **Maintain:** Current policies aligned with evidence

Each recommendation includes expected effects on both metrics, confidence grades, and blocking factors (see Section 10.2).

JEL Classification: H10, D72, C54, I18, D61

H10 (Public Finance, Structure and Scope), D72 (Political Economy), C54 (Quantitative Policy Modeling), I18 (Health Policy), D61 (Allocative Efficiency; Cost-Benefit Analysis)

1 The Two Welfare Metrics

OPG measures policy impact using the [two-metric welfare function](#) defined in the Optimocracy Framework:

1. **Real after-tax median income growth** (pp/year) - economic welfare
2. **Median healthy life years** (years) - health welfare

See the Optimocracy paper for full justification of these metric choices, data sources, and the welfare function formula.

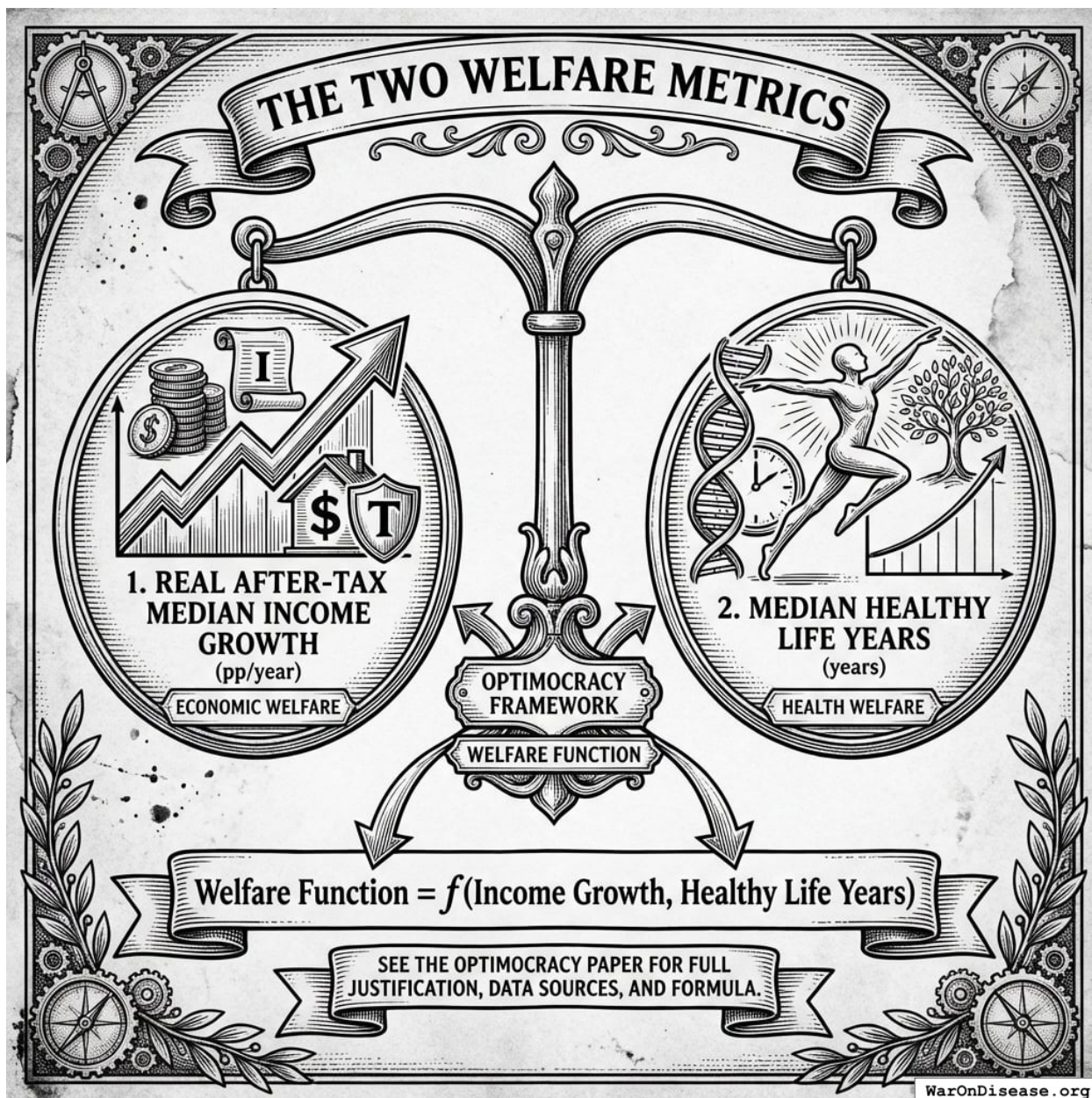


Figure 2: A conceptual diagram illustrating the two pillars of the welfare function: Economic Welfare (income growth) and Health Welfare (healthy life years).

1.1 Why Only Two Metrics?

Simplicity: These two metrics capture the primary welfare dimensions affected by most policies while remaining directly interpretable. No complex conversion factors (VSL, QALY→\$) are needed.

Coverage gap: Freedom and autonomy concerns are handled as [blocking factors](#) rather than adding metric complexity. A policy that improves income and health but restricts freedom is flagged, not silently scored. Environmental impacts and distributional effects are tracked as supplementary indicators where data permits.

1.2 Income Metric Definition

Real after-tax median income growth is defined narrowly as: wages, salaries, and self-employment income, minus taxes paid. This metric captures what appears in household budgets.

What counts as income effects:

- Wage increases from productivity gains (e.g., fewer sick days → measurably higher wages)
- Tax changes that directly affect take-home pay
- Employment effects that translate to wage income

What does NOT count as income effects:

- Healthcare cost savings (these are health system efficiency gains, not personal income)
- Reduced insurance premiums (unless they translate to higher take-home pay via employer pass-through)
- Quality-of-life improvements that don't appear in wages

Implication for policy analysis: This creates genuine tradeoffs that the two-metric framework makes explicit. A tobacco tax, for example, may show:

- **Income effect:** Negative for smokers (direct tax burden), partially offset by productivity gains for those who quit
- **Health effect:** Positive (reduced smoking → longer healthy life)

The framework does not hide this tradeoff by claiming healthcare cost savings are “income gains.” If a policy improves health but costs money, both effects are reported honestly. This is a feature, not a bug: it prevents corner solutions (an infinite tobacco tax would maximize health but devastate income for smokers) and surfaces the welfare tradeoff for democratic deliberation.

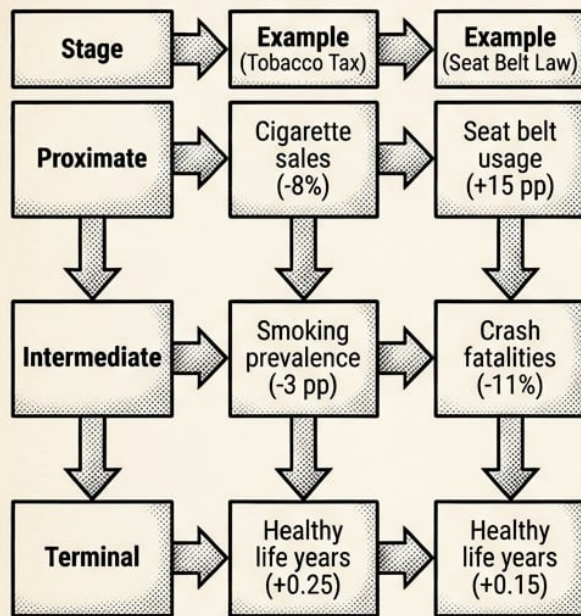
1.3 Outcome Translation Methodology

While OPG uses only two terminal metrics (income growth and healthy life years), evidence often measures **surrogate outcomes** (smoking rates, traffic deaths, crime rates). This section specifies how surrogate outcomes are translated to the terminal metrics.

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While **OPG** uses only two terminal metrics (income growth and healthy life years), evidence often measures **surrogate outcomes** (smoking rates, traffic deaths, crime rates). This section specifies how surrogate outcomes are translated to the terminal metrics.

The translation chain:



Conversion factors must be explicit:

Each translation step requires a documented conversion factor with source:

Conversion	Factor	Source	Uncertainty
1 pp smoking reduction → healthy life years	+0.083 years	CDC life tables,	±30%
1% fatality reduction → healthy life years	+0.015 years	NHTSA data,	±25%
1 pp employment increase → income growth	+0.12 pp/year	BLS wage data	±40%

Uncertainty propagation:

When multiple translation steps are chained, uncertainties compound:

$$\sigma_{\text{terminal}} = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_i^2}$$

For linear translations, this simplifies to:

$$CV_{\text{terminal}} = \sqrt{\sum_i CV_i^2}$$

Where CV is the coefficient of variation at each translation step. A three-step translation with 30% uncertainty at each step yields ~52% uncertainty in the terminal metric.

Important clarification:

The claim that “no complex conversion factors are needed” in the abstract refers to the *terminal metrics themselves* (income and health are directly interpretable, unlike utility or welfare indices). Translation from surrogate outcomes to terminal metrics *does* require conversion factors, which must be documented and include uncertainty bounds.

WarOnDisease.org

Figure 3: A flowchart illustrating the transition from policy interventions to terminal metrics through proximate and intermediate outcomes, highlighting specific conversion factors and the compounding of uncertainty across the chain.

The translation chain:

Policy → Proximate Outcome → Intermediate Outcome → Terminal Metric

Stage	Example (Tobacco Tax)	Example (Seat Belt Law)
Proximate	Cigarette sales (-8%)	Seat belt usage (+15 pp)
Intermediate	Smoking prevalence (-3 pp)	Crash fatalities (-11%)
Terminal	Healthy life years (+0.25)	Healthy life years (+0.15)

Stage	Example (Tobacco Tax)	Example (Seat Belt Law)
Terminal	Healthy life years (+0.25)	Healthy life years (+0.15)

Conversion factors must be explicit:

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Conversion	Factor	Source	Uncertainty
1 pp smoking reduction → healthy life years	+0.083 years	CDC life tables, ¹³⁸	±30%
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2 The Evidence Base: Centuries of Natural Policy Experiments

Core insight: Every jurisdiction that enacted a policy created a natural experiment. The evidence to know what works already exists, scattered across thousands of jurisdictions and hundreds of years. OPG systematically harvests this evidence.

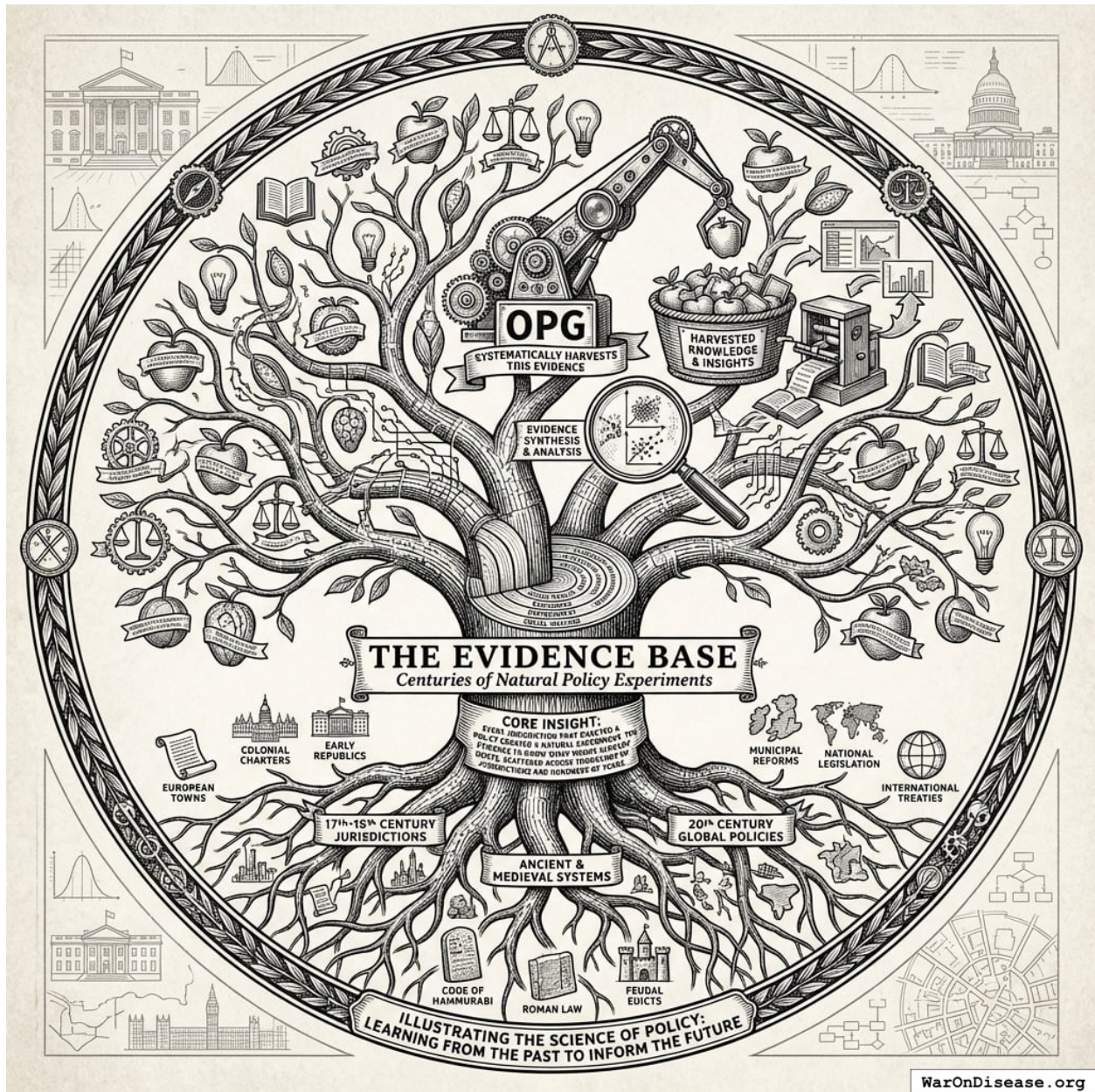


Figure 4: A conceptual visualization showing how diverse policy data from thousands of jurisdictions over centuries flows into a central system for systematic harvesting and analysis.

2.1 Scale of Available Natural Experiments

Level	Jurisdictions	Years	Policy-Years
US States	50	70+	3,500+
Countries	200+	230+	46,000+
EU Regions	300+	50+	15,000+
US Counties	3,000+	50+	150,000+
Cities worldwide	10,000+	varies	millions

Each policy change creates a before/after comparison. Each jurisdiction that didn't adopt creates a control group. This represents a vast, largely untapped evidence base.

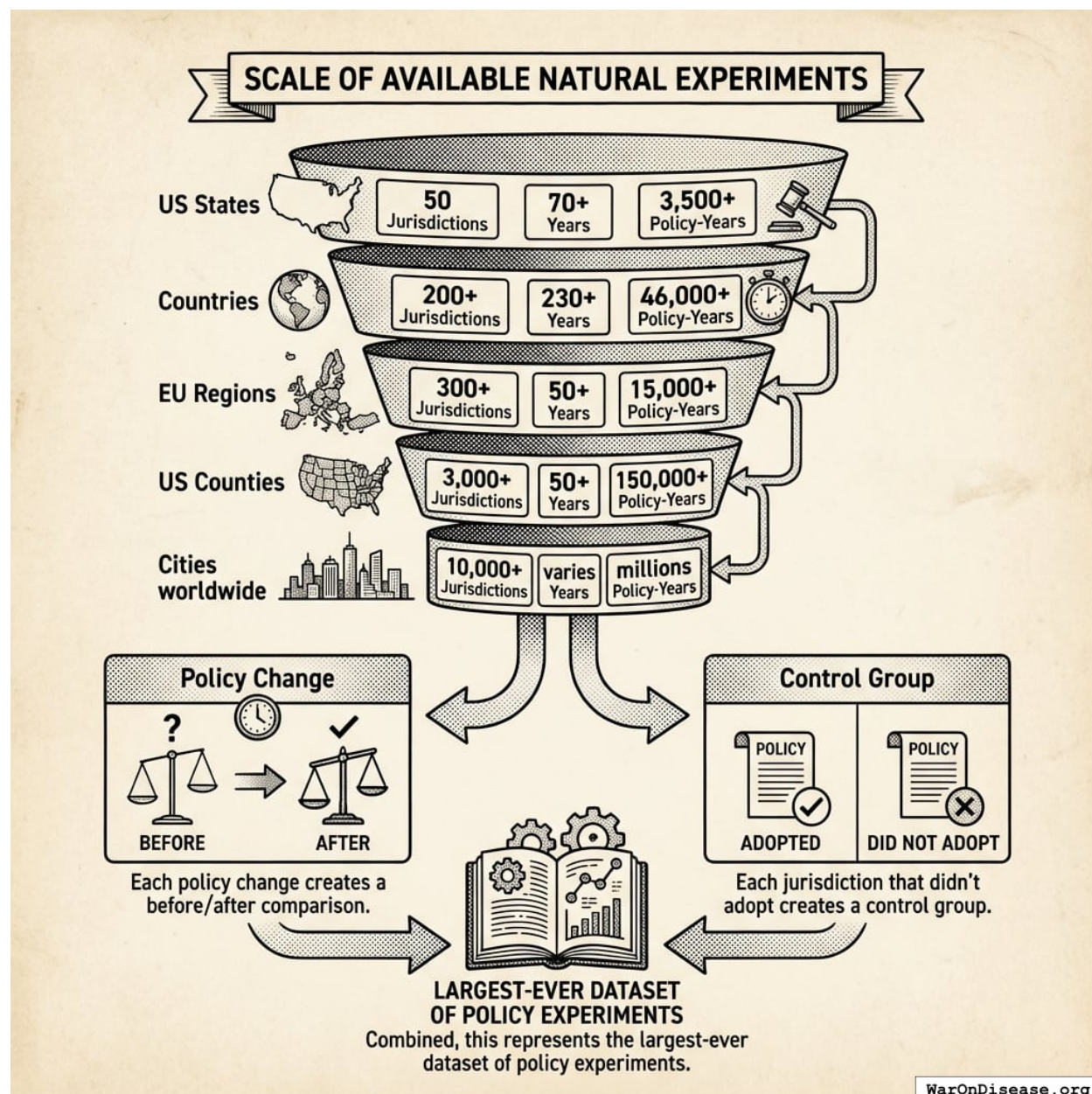


Figure 5: A comparison chart showing the exponential growth in available policy-years across different geographic levels, ranging from 3,500 for US states to millions for cities worldwide.

2.2 The OPG Pipeline

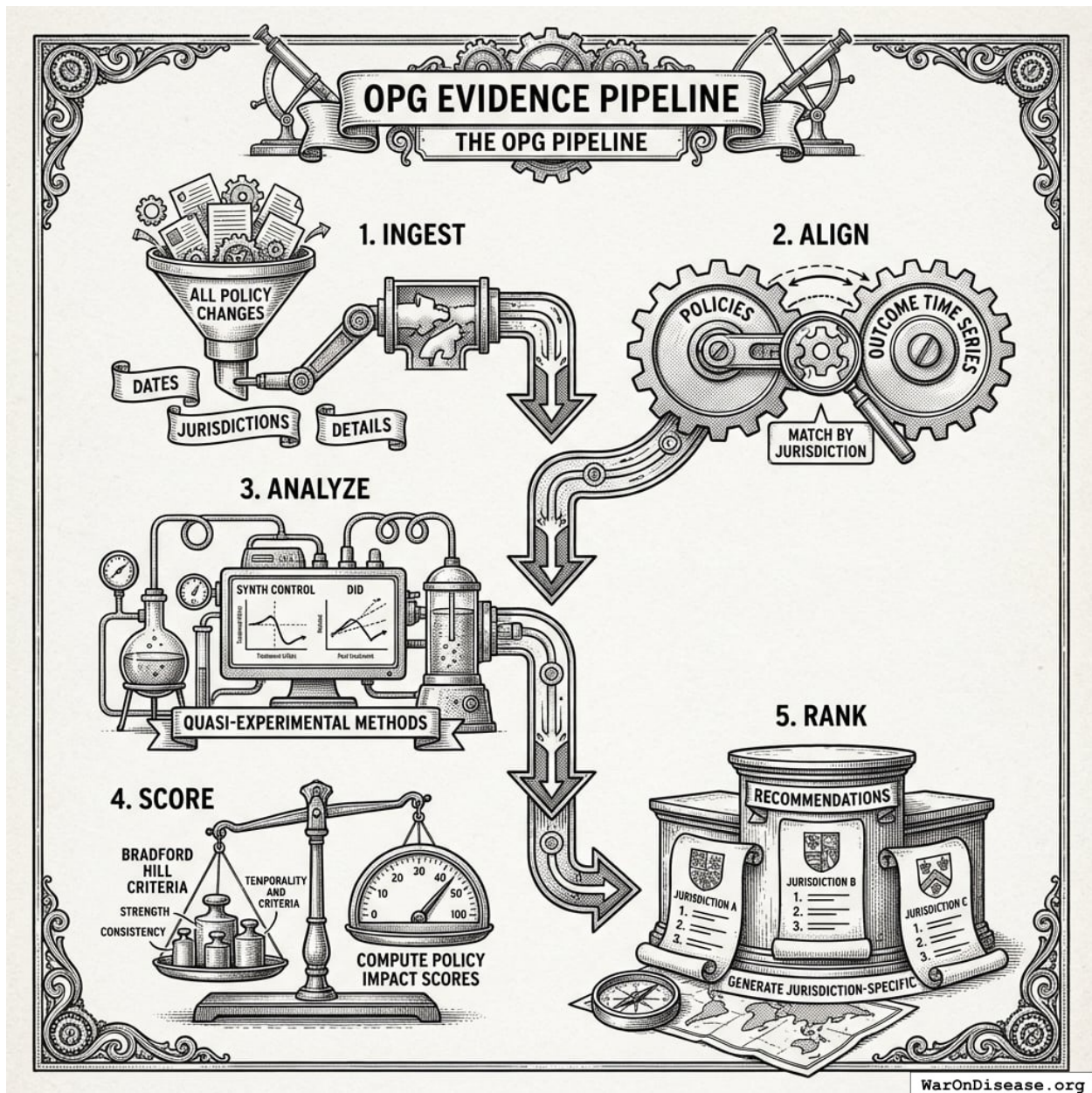


Figure 6: A five-stage data pipeline illustrating the transformation of policy data into ranked recommendations through ingestion, alignment, quasi-experimental analysis, and impact scoring.

OPG EVIDENCE PIPELINE

1. INGEST

All policy changes with dates, jurisdictions, details

2. ALIGN
Match policies to outcome time series by jurisdiction
3. ANALYZE
Apply quasi-experimental methods (synth control, DiD)
4. SCORE
Compute Policy Impact Scores using Bradford Hill
5. RANK
Generate jurisdiction-specific recommendations

2.3 Why This Hasn't Been Done Before

1. **Data fragmentation:** Policy records scattered across legislative databases, government archives, academic papers
2. **Computational limits:** Meta-analysis at this scale requires modern infrastructure
3. **Methodological advances:** Synthetic control (2003), modern DiD (2021) are recent
4. **Incentive structures:** No existing institution has mandate + capability + incentive

OPG aggregates fragmented evidence, applies modern causal inference at scale, and produces actionable output.

WHY THIS HASN'T BEEN DONE BEFORE & OPG'S APPROACH

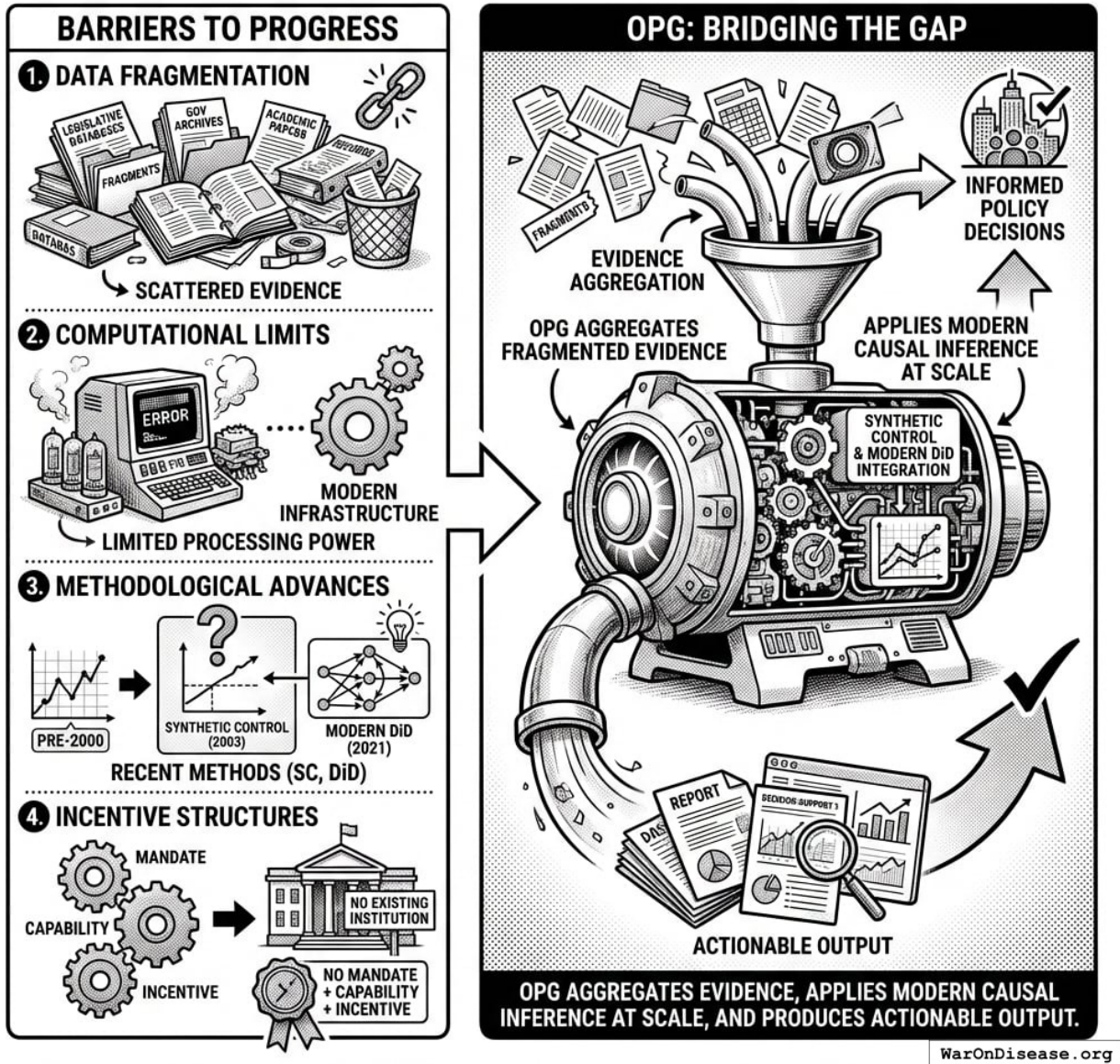


Figure 7: A diagram illustrating how four historical barriers (fragmented data, computational limits, older methodologies, and incentive gaps) converge into the OPG three-step workflow of aggregation, causal inference, and output.

3 System Overview

3.1 What Policymakers See

A jurisdiction-specific dashboard showing which policies to enact, replace, repeal, or maintain, ranked by expected welfare impact:

See [Appendix A](#) for a complete worked example showing jurisdiction-specific recommendations.

3.2 What Policy Analysts See

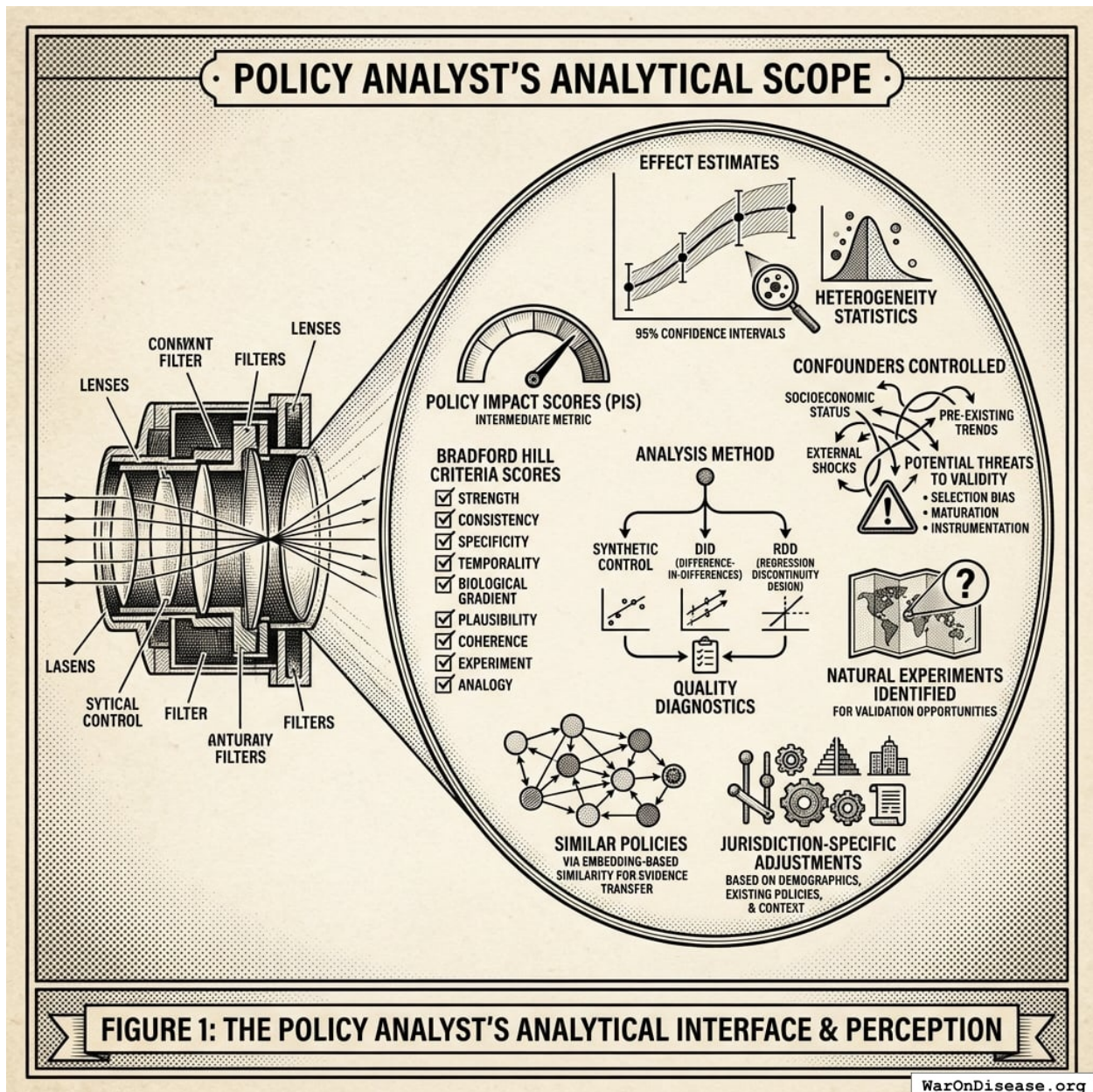


Figure 8: A conceptual framework illustrating the eight data components (ranging from statistical effect estimates to jurisdictional adjustments) that converge to form a comprehensive policy analysis profile.

- **Effect estimates** with standard errors, confidence intervals, and heterogeneity statistics
- **Policy Impact Scores (PIS)** for each policy-outcome relationship (intermediate metric)
- **Bradford Hill criteria scores** for causality assessment
- **Analysis method** used (synthetic control, DiD, RDD) with quality diagnostics
- **Confounders controlled** and potential threats to validity
- **Natural experiments identified** for validation opportunities

- **Similar policies** via embedding-based similarity for evidence transfer
- **Jurisdiction-specific adjustments** based on demographics, existing policies, and context

4 Introduction

4.1 Why Generic Policy Scores Are Not Enough

Previous approaches to evidence-based policy produce generic rankings:

“Tobacco taxes reduce smoking by 8.2 percentage points on average across 23 states...”

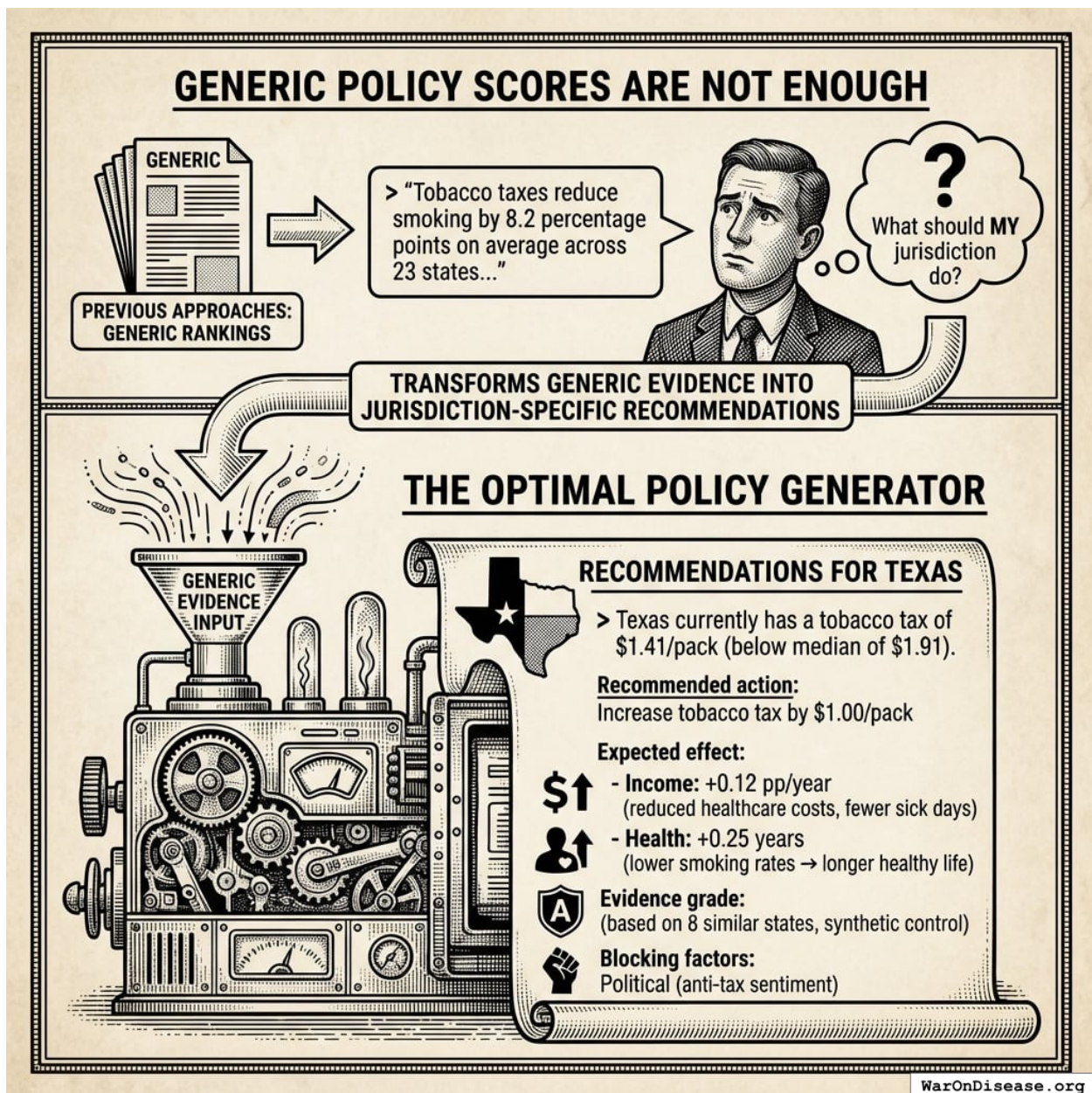


Figure 9: A comparative visualization showing the shift from a single generic national average to a localized dashboard of specific tax rates, health outcomes, and economic impacts for a specific jurisdiction.

This is useful background but doesn't answer the policymaker's question: **"What should MY jurisdiction do?"**

The Optimal Policy Generator transforms generic evidence into jurisdiction-specific recommendations:

Recommendations for Texas

Texas currently has a tobacco tax of \$1.41/pack (below median of \$1.91).

Recommended action: Increase tobacco tax by \$1.00/pack

Expected effect:

- **Income:** +0.12 pp/year (reduced healthcare costs, fewer sick days)
- **Health:** +0.25 years (lower smoking rates → longer healthy life)

Evidence grade: A (based on 8 similar states, synthetic control)

Blocking factors: Political (anti-tax sentiment)

4.2 Why Policy Ranking Fails Today

Current policy adoption follows a process dominated by political economy dynamics well-documented in the public choice literature^{140,141}:

1. **Lobbying intensity:** Policies that benefit concentrated interests (with resources to lobby) are adopted over policies that benefit diffuse majorities^{142,143}
2. **Ideological priors:** Policymakers filter evidence through pre-existing beliefs, accepting studies that confirm priors and rejecting those that don't
3. **Anecdote-driven reasoning:** Vivid individual cases drive policy more than systematic evidence ("If it saves one child...")
4. **Status quo bias:** Existing policies persist regardless of evidence because change requires political capital
5. **Salience heuristics:** Policies addressing visible problems (terrorism, rare diseases) receive disproportionate resources relative to invisible problems (air pollution, chronic disease)

The result: welfare losses from documented policy failures. Evidence-based policy movements have attempted to address these failures^{144,145}, but lack systematic, jurisdiction-specific recommendation generation.

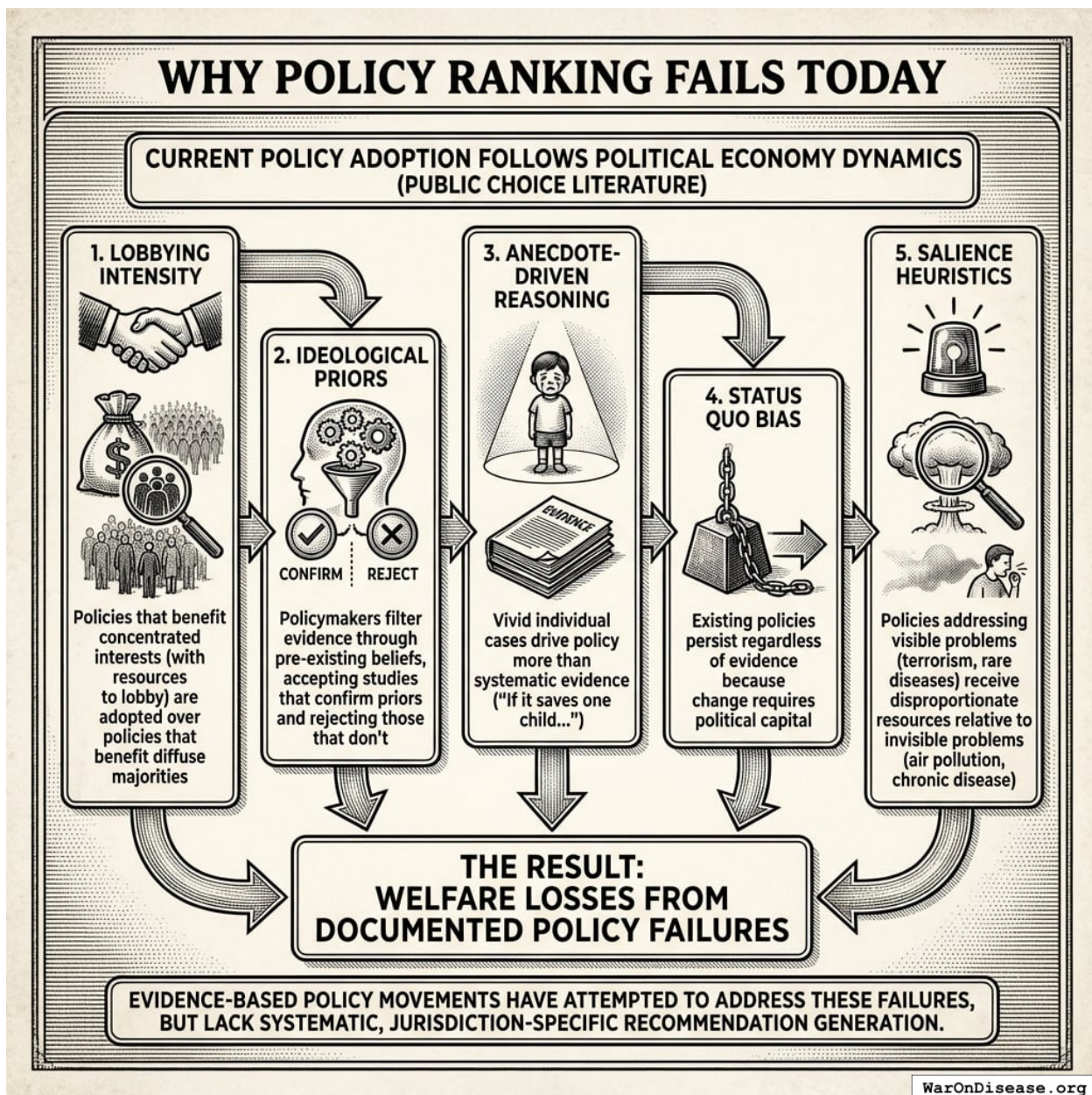


Figure 10: A conceptual diagram showing how political and psychological barriers like lobbying intensity, status quo bias, and salience heuristics filter evidence, leading to welfare losses in the policy adoption process.

4.3 Scale of Available Evidence

The evidence base comprises millions of policy-years of natural experiments across all jurisdictional levels (see Section 2 for detailed counts). Even with imperfect causal inference, systematically analyzing this data should improve on the current system of lobbying-driven, ideology-filtered policy adoption. How much improvement is an empirical question requiring validation (see Section 20).

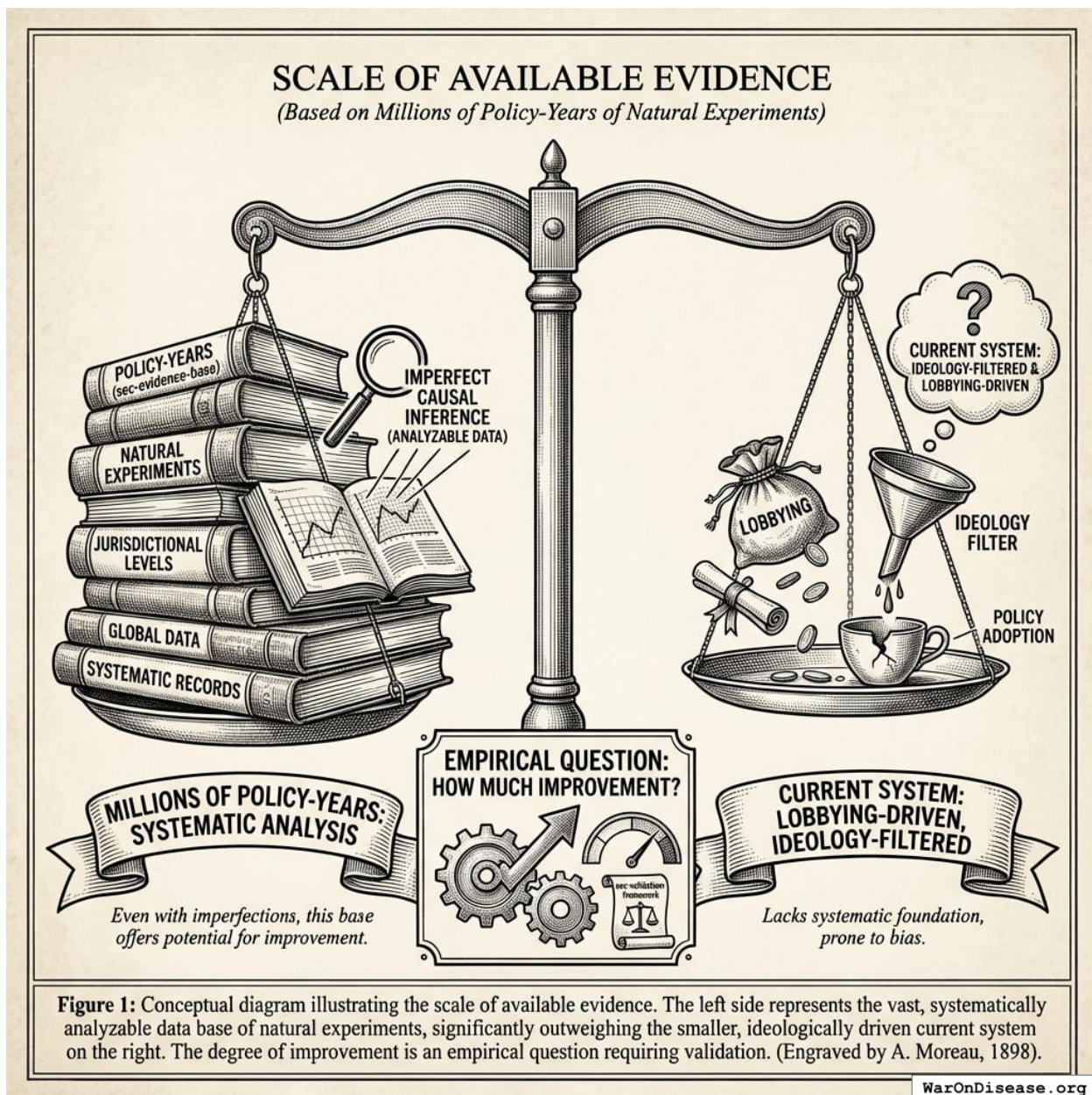


Figure 11: A conceptual comparison between the current ideology-driven policy system and a proposed evidence-based framework leveraging data from millions of policy-years across multiple jurisdictional levels.

4.4 Contributions

1. **Methodological:** A systematic framework for translating quasi-experimental evidence into jurisdiction-specific policy recommendations, extending beyond generic evidence ratings to actionable output in four categories (enact/replace/repeal/maintain).

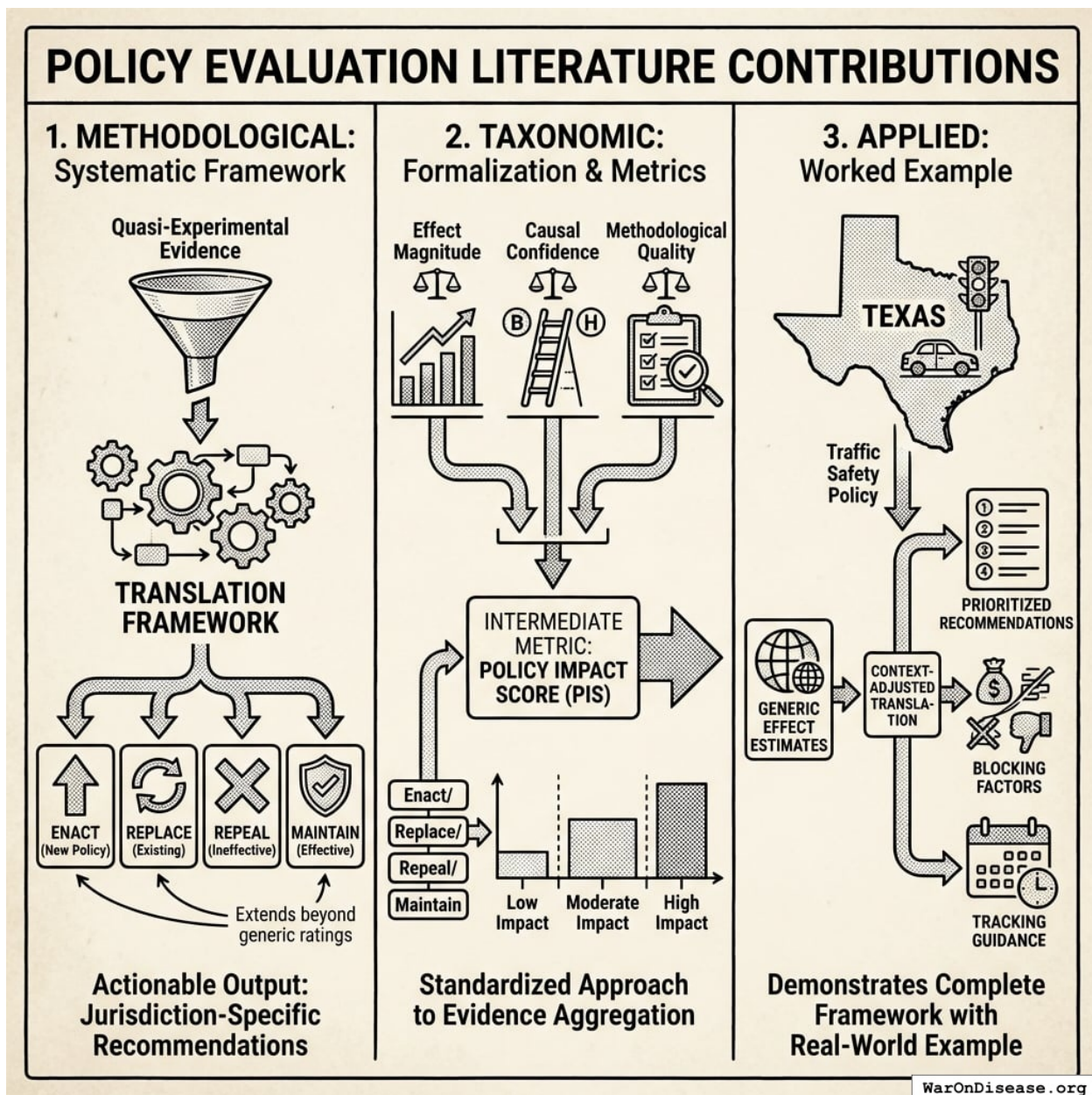


Figure 12: A workflow diagram showing the translation of evidence into a Policy Impact Score (PIS) and the subsequent categorization into four policy recommendation types: enact, replace, repeal, or maintain.

- 2. Taxonomic:** We formalize the four recommendation types and introduce the Policy Impact Score (PIS) as an intermediate metric combining effect magnitude, causal confidence (Bradford Hill criteria), and methodological quality. This provides a standardized approach to evidence aggregation.
- 3. Applied:** We demonstrate the complete framework with a worked example for Texas traffic safety policy, showing how generic effect estimates are translated into context-adjusted, prioritized recommendations with blocking factors and tracking guidance.

4.5 Validation Status

This specification describes a *proposed* framework. The methodology requires empirical validation before deployment. Specifically, retrospective studies should assess whether OPG-identified high-priority recommendations correlate with actual welfare improvements in jurisdictions that adopted them (see Section 20 for proposed validation design). Until such validation is completed, OPG outputs should inform but not replace expert judgment.

5 Related Work

5.1 Existing Policy Evaluation Frameworks

Regulatory Impact Analysis (RIA): Required by executive order for US federal regulations since 1981¹⁴⁶. RIA estimates costs and benefits of proposed rules but: (1) applies only to new regulations, not the existing policy stock; (2) lacks systematic cross-jurisdiction evidence aggregation; (3) does not produce jurisdiction-specific recommendations for subnational governments.

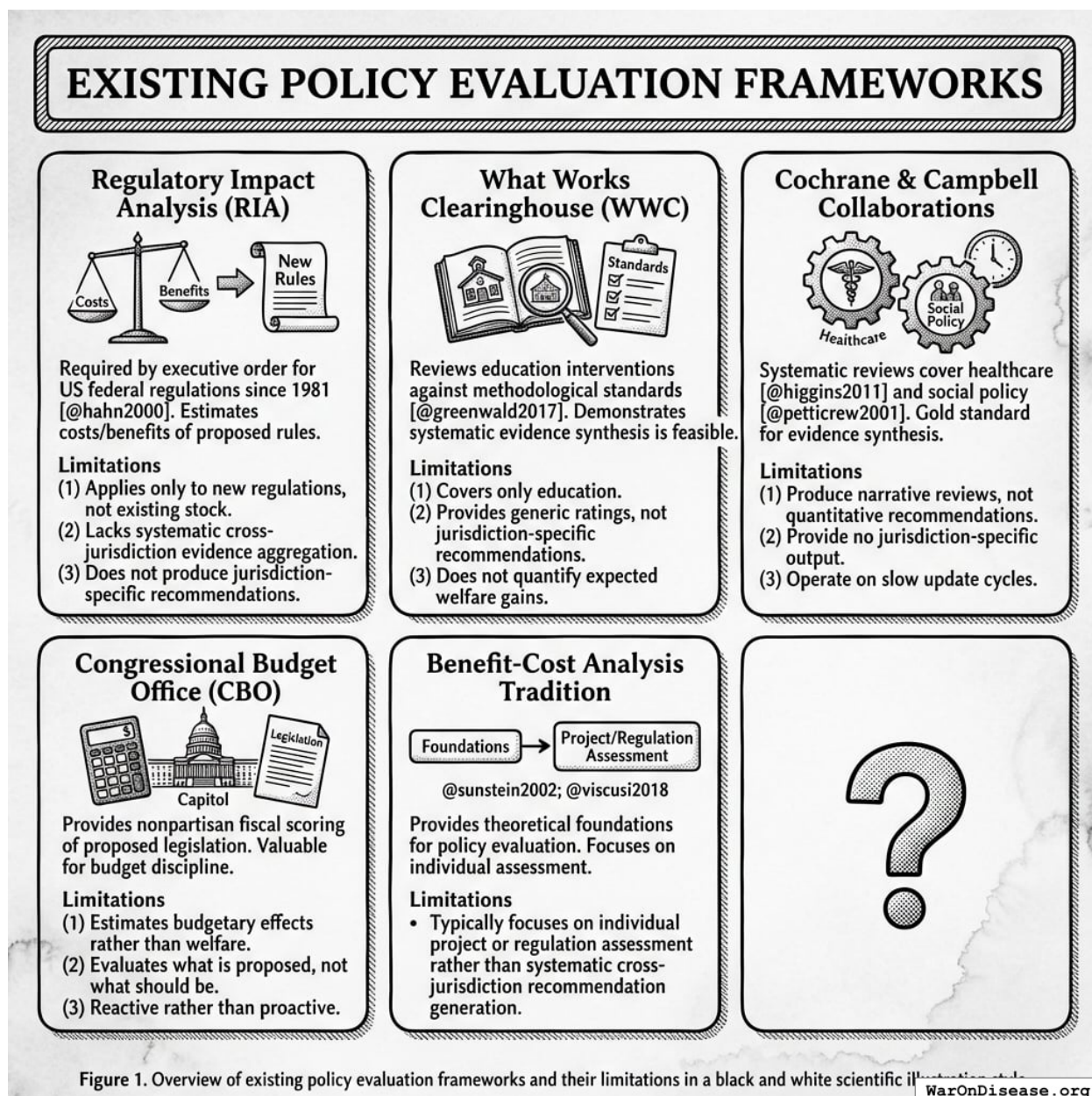


Figure 13: A comparison of major policy evaluation frameworks across key functional dimensions, highlighting the systematic gaps in jurisdiction-specific recommendations and evaluation of existing policy stocks.

What Works Clearinghouse (WWC): The Institute of Education Sciences operates WWC to review education interventions against methodological standards¹⁴⁷. WWC demonstrates that systematic evidence synthesis is feasible, but: (1) covers only education; (2) provides generic intervention ratings, not jurisdiction-specific recommendations; (3) does not quantify expected welfare gains.

Cochrane and Campbell Collaborations: These systematic review organizations cover healthcare¹⁴⁸ and social policy¹⁴⁹ respectively. They represent the gold standard for evidence

synthesis but: (1) produce narrative reviews rather than quantitative recommendations; (2) provide no jurisdiction-specific output; (3) operate on slow update cycles (years between reviews).

Congressional Budget Office (CBO): CBO provides nonpartisan fiscal scoring of proposed legislation. While valuable for budget discipline, CBO: (1) estimates budgetary effects rather than welfare; (2) evaluates what is proposed rather than what should be proposed; (3) is reactive rather than proactive.

Benefit-Cost Analysis Tradition: The broader benefit-cost literature^{150,151} provides theoretical foundations for policy evaluation but typically focuses on individual project or regulation assessment rather than systematic cross-jurisdiction recommendation generation.

5.2 This Framework's Contribution

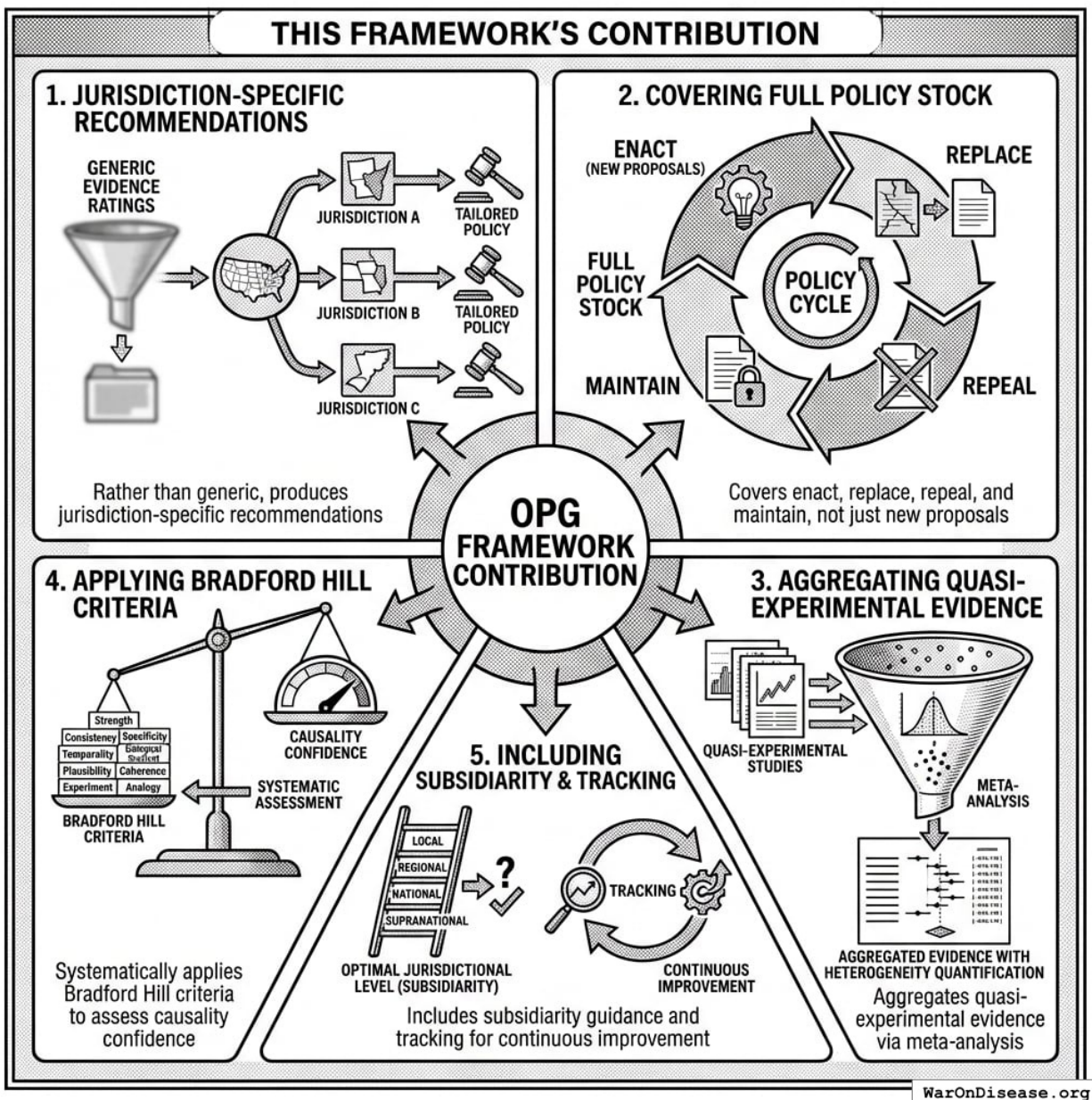


Figure 14: A comparative overview of the OPG framework's unique contributions, contrasting its jurisdiction-specific, comprehensive policy review process against traditional evaluation methods.

OPG differs from existing approaches by:

1. **Producing jurisdiction-specific recommendations** rather than generic evidence ratings
2. **Covering the full policy stock** (enact/replace/repeal/maintain) not just new proposals
3. **Aggregating quasi-experimental evidence** via meta-analysis with heterogeneity quantification
4. **Applying Bradford Hill criteria** systematically to assess causality confidence
5. **Including subsidiarity guidance** (optimal jurisdictional level) and tracking for continuous

improvement

6 Theoretical Framework

6.1 The Policy Optimization Problem

Let $\mathcal{P}$ denote the set of available policies. For jurisdiction j , let $P_j \subseteq \mathcal{P}$ denote the current policy bundle. Welfare under policy bundle P is defined using the two core metrics:

$$W_j(P) = \alpha \cdot \text{IncomeGrowth}_j(P) + (1 - \alpha) \cdot \text{HealthyYears}_j(P)$$

Where:

- $\text{IncomeGrowth}_j(P)$ = Real after-tax median income growth (pp/year)
- $\text{HealthyYears}_j(P)$ = Median healthy life expectancy (years)
- $\alpha = 0.5$ (default equal weighting; can be adjusted for jurisdiction priorities)

The social planner's problem:

$$P_j^* = \arg \max_{P \subseteq \mathcal{P}} W_j(P) \quad \text{subject to feasibility constraints}$$

Assumption 1 (Additive Separability): For tractability, assume each metric is approximately additively separable across policies:

$$\text{IncomeGrowth}_j(P) \approx \sum_{p \in P} \beta_{jp}^{\text{inc}} + \varepsilon_{\text{inc}}$$

$$\text{HealthyYears}_j(P) \approx \sum_{p \in P} \beta_{jp}^{\text{hlth}} + \varepsilon_{\text{hlth}}$$

where β_{jp}^{inc} and β_{jp}^{hlth} are the marginal effects of policy p on each metric in jurisdiction j , and interaction terms are assumed to be second-order.

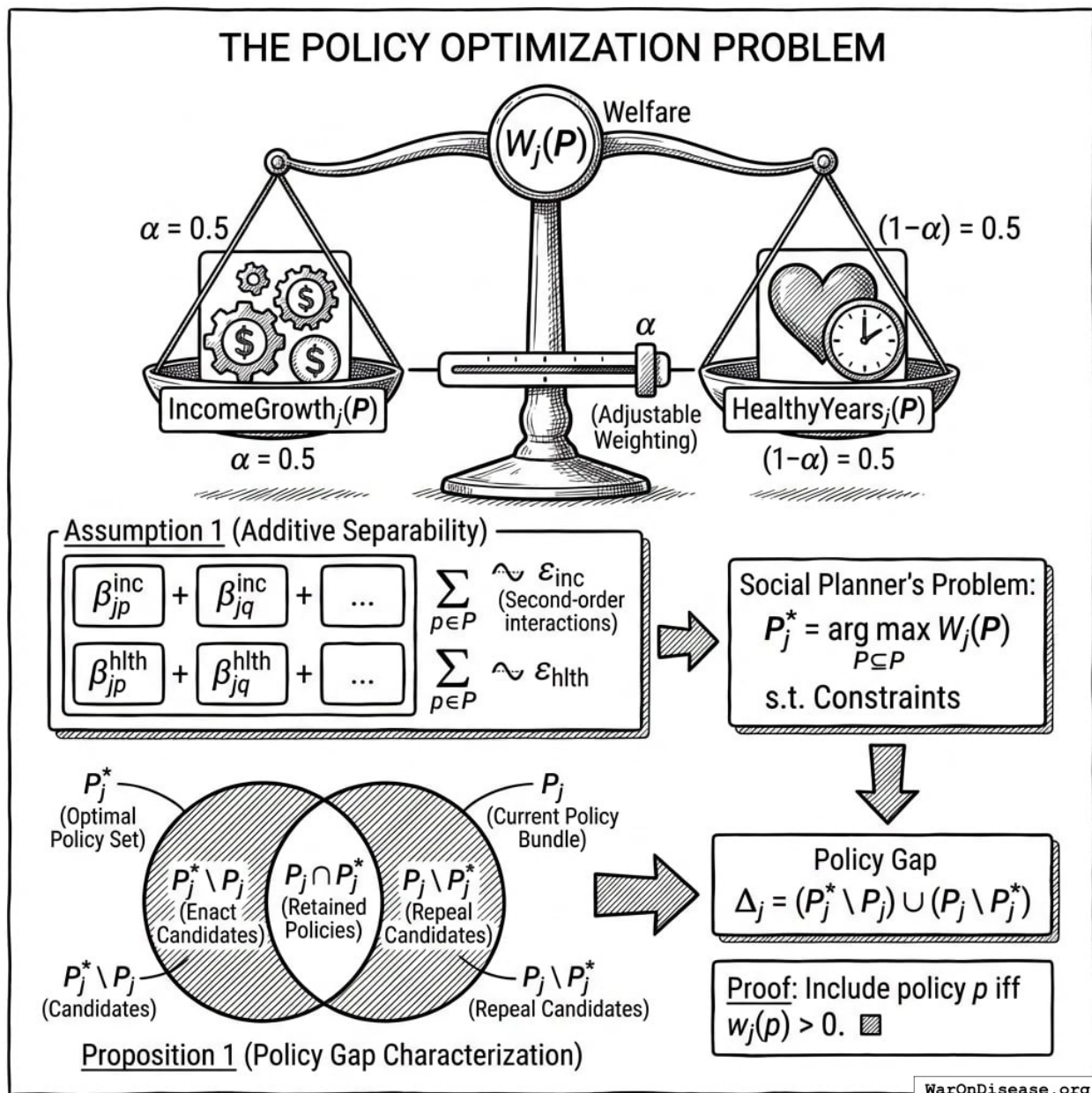


Figure 15: A Venn diagram illustrating the Policy Gap as the symmetric difference between a jurisdiction's current policy bundle and its welfare-optimal bundle, identifying specific areas for policy enactment and repeal.

Justification and limitations: Additive separability is a standard simplifying assumption in policy analysis (see¹⁴⁶ for regulatory impact analysis applications). This assumption is most valid when: (1) policies operate through distinct mechanisms, (2) jurisdictions have not reached saturation in any policy domain, and (3) policies do not create complementarities or substitution effects. When these conditions fail (for example, when a carbon tax interacts with renewable energy subsidies), the marginal effects may be mis-estimated.

Policy Interaction Detection:

OPG flags potential interaction effects using the following heuristics:

1. **Effect heterogeneity test:** If a policy’s effect varies significantly depending on whether another policy is present, flag the pair as potentially interacting.
2. **Known interaction database:** Documented policy complementarities and substitutes:

Policy A	Policy B	Interaction Type	Evidence
Seat belt law	Speed limit	Complementary	Both target crash fatalities
Nutrition labeling	School lunch programs	Complementary	Both improve dietary outcomes
Tobacco tax	Smoking ban	Complementary	Reinforce each other
Income tax cut	Sales tax increase	Substitutable	Offsetting fiscal effects

3. **Sensitivity analysis recommendation:** For high-priority recommendations, report: “How would this recommendation change if policies X and Y interact?” with bounds on combined effect.

Worked example (policy interaction):

Seat belt law + Speed limit reduction: - Seat belt alone: +0.15 years healthy life - Speed limit alone: +0.06 years healthy life - Additive prediction: +0.21 years - Observed when both present: +0.18 years (interaction term: -0.03 years)

The interaction term is negative because both policies target the same outcome (crash fatalities), creating partial overlap in benefits. OPG reports both the additive prediction and interaction-adjusted estimate when interaction data is available.

Proposition 1 (Policy Gap Characterization): Under Assumption 1, the welfare-optimal policy set satisfies:

$$P_j^* = \{p \in \mathcal{P} : w_j(p) > 0\}$$

and the policy gap for jurisdiction j is:

$$\Delta_j = (P_j^* \setminus P_j) \cup (P_j \setminus P_j^*)$$

where $(P_j^* \setminus P_j)$ represents beneficial policies the jurisdiction lacks (enact candidates) and $(P_j \setminus P_j^*)$ represents harmful policies the jurisdiction has (repeal candidates). See Section 9 for the operational implementation.

Proof: Direct consequence of additive separability. Include policy p if and only if $w_j(p) > 0$.

6.2 Evidence Aggregation Properties

Proposition 2 (PIS as Precision-Weighted Evidence): Under random-effects meta-analysis with between-jurisdiction variance τ^2 , the pooled effect estimate $\hat{\beta}_{\text{pooled}}$ is (see Section 13 for

implementation):

$$\hat{\beta}_{\text{pooled}} = \frac{\sum_j \frac{1}{\text{SE}_j^2 + \tau^2} \hat{\beta}_j}{\sum_j \frac{1}{\text{SE}_j^2 + \tau^2}}$$

with variance:

$$\text{Var}(\hat{\beta}_{\text{pooled}}) = \frac{1}{\sum_j \frac{1}{\text{SE}_j^2 + \tau^2}}$$

Proof: Standard random-effects meta-analysis derivation (DerSimonian-Laird).

Proposition 3 (Heterogeneity Bounds Transferability): When $I^2 > 75\%$ (high heterogeneity):

$$\text{Var}[\hat{\beta}_j | \hat{\beta}_{\text{pooled}}] > 0.75 \cdot \text{Var}[\hat{\beta}_j]$$

meaning the pooled estimate explains less than 25% of cross-jurisdiction variation. Context-specific estimates are required rather than direct application of the pooled effect. This constraint is operationalized in Section 13.4.

Proof: By definition, $I^2 = \frac{\tau^2}{\tau^2 + \bar{\sigma}^2}$ where $\bar{\sigma}^2$ is typical within-study variance. When $I^2 > 0.75$, between-study variance dominates, and the pooled estimate provides limited information about any individual jurisdiction’s true effect.

6.3 Information Value

Proposition 4 (Value of Additional Evidence): The expected value of information from an additional jurisdiction study is:

$$\text{VOI} = E\left[\max_{a \in \{\text{adopt}, \text{reject}\}} U(a | \text{new data})\right] - \max_a E[U(a | \text{current data})]$$

which is maximized when prior uncertainty is high and decision stakes are large.

Proof: Standard Bayesian decision theory¹⁴⁴.

Corollary 1 (Trial Prioritization): Policies with (1) high prior variance in effect estimates, (2) large potential welfare impact, and (3) low trial cost should be prioritized for experimental validation. See Section 17 for implementation.

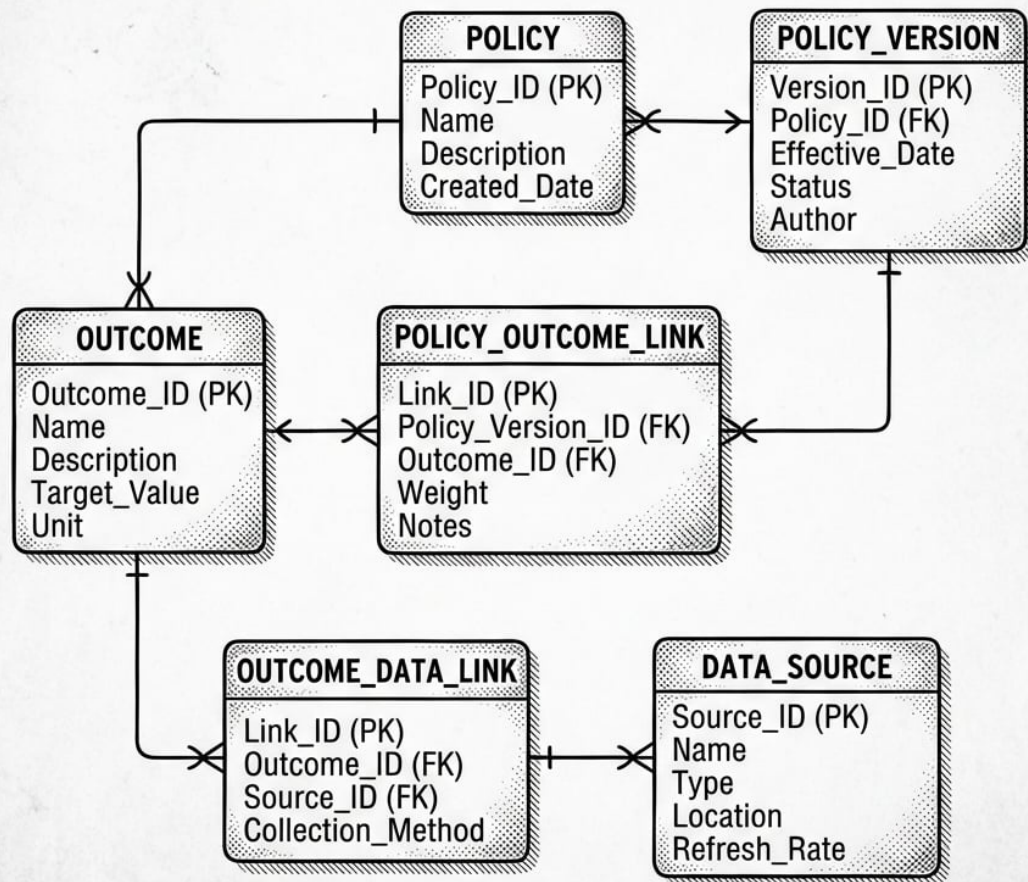
7 Core Methodology

7.1 Policy-Outcome Data Structure

The OPG system uses a relational database schema. The following is a **reference implementation** showing the conceptual data model; production deployments may vary.

OPG SYSTEM: POLICY-OUTCOME DATA STRUCTURE (CONCEPTUAL MODEL)

REFERENCE IMPLEMENTATION



WarOnDisease.org

Figure 16: An entity-relationship diagram illustrating the core tables and relational schema of the OPG system's policy-outcome data structure.

7.1.1 Core Tables

```

-- Hierarchical jurisdictions (country > state > county > city)
jurisdictions (
    id, name, jurisdiction_type, -- 'country', 'state', 'county', 'city'
    parent_id, -- FK to parent jurisdiction (e.g., Texas -> USA)
    iso_code, population, gdp_per_capita,
    constitution_type, -- constraints on policy space
    data_quality_score, -- how complete is our policy inventory?

```

```

    latitude, longitude, ...
)

-- Policy types (canonical definitions)
policy_types (
    id, name, policy_category_id, policy_type,
    is_continuous, typical_onset_delay_days,
    typical_duration_of_effect_years, canonical_text, ...
)

-- Current policy inventory by jurisdiction
jurisdiction_policies (
    jurisdiction_id, policy_type_id,
    has_policy BOOLEAN,
    policy_strength, -- e.g., tobacco tax amount, not just yes/no
    implementation_date,
    policy_details_json,
    data_source, last_verified
)

-- Two core welfare metrics (fixed schema)
outcome_metrics (
    id,
    metric_type ENUM('income', 'health'), -- Only two types
    jurisdiction_id,
    measurement_date,
    value, -- pp/year for income; years for health
    confidence_interval_low,
    confidence_interval_high,
    data_source -- Census/BLS for income; WHO/BRFSS for health
)

-- Policy recommendations (generated output)
policy_recommendations (
    jurisdiction_id, policy_type_id,
    recommendation_type, -- 'enact', 'replace', 'repeal', 'maintain'
    current_status, -- what they have now (NULL if nothing)
    recommended_target, -- what evidence suggests
    -- Two-metric effects
    income_effect_pp, -- Expected effect on median income growth (pp/year)
    income_effect_ci_low, income_effect_ci_high,
    health_effect_years, -- Expected effect on healthy life years
    health_effect_ci_low, health_effect_ci_high,
    evidence_grade, priority_score,
    blocking_factors, -- 'constitutional', 'federal_preemption', 'political', 'autonomy', etc.
    similar_jurisdictions,
    -- Jurisdictional level guidance

```

```

    minimum_effective_level, recommended_level,
    -- Tracking for feedback loop
    tracking_frequency, tracking_baseline_method,
    last_generated
)

```

7.1.2 Policy Types

Type	Description	Example	Measurement
law	Statutory law passed by legislature	Environmental regulation law	Binary (exists/not)
regulation	Administrative rule by agency	Agency emission standards	Continuous (stringency)
tax_policy	Tax rate, bracket, credit, deduction	Investment income tax rate	Continuous (rate)
budget_allocation	Spending decision	Education spending per pupil	Continuous (\$/capita)
executive_order	Executive action	Enforcement priority directive	Binary
court_ruling	Judicial precedent	Constitutional interpretation	Binary
treaty	International agreement	Multilateral cooperation treaty	Binary
local_ordinance	Municipal rule	Land use restrictions	Categorical

7.2 Analysis Methods

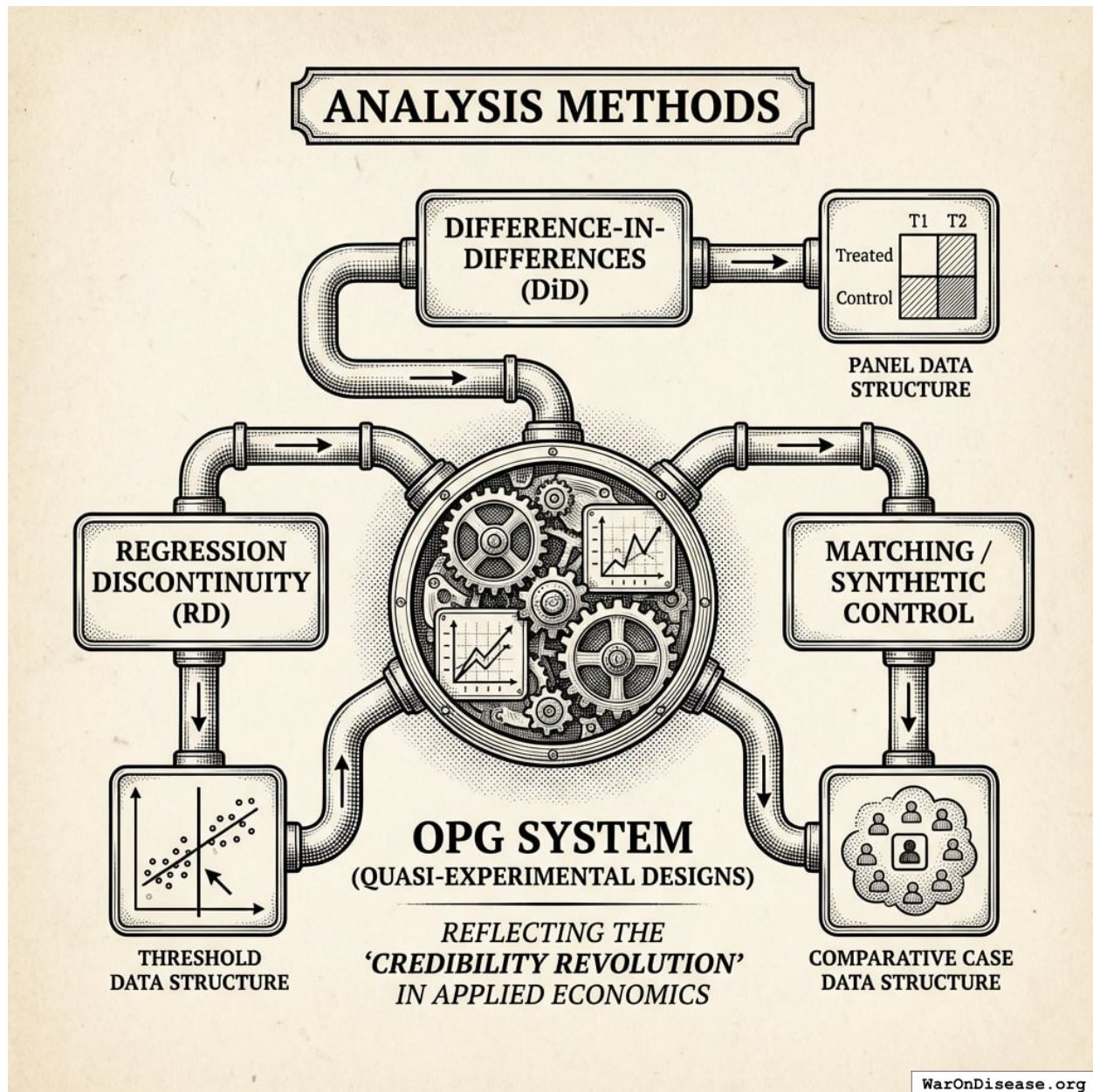


Figure 17: A conceptual mapping of quasi-experimental research designs supported by the OPG system, illustrating how different methods align with specific data structures.

The OPG system supports multiple quasi-experimental designs, reflecting the “credibility revolution” in applied economics¹⁵². Each method is appropriate for different data structures¹⁵³:

7.2.1 Synthetic Control Method

Use case: Single treated jurisdiction, good donor pool of similar untreated jurisdictions.

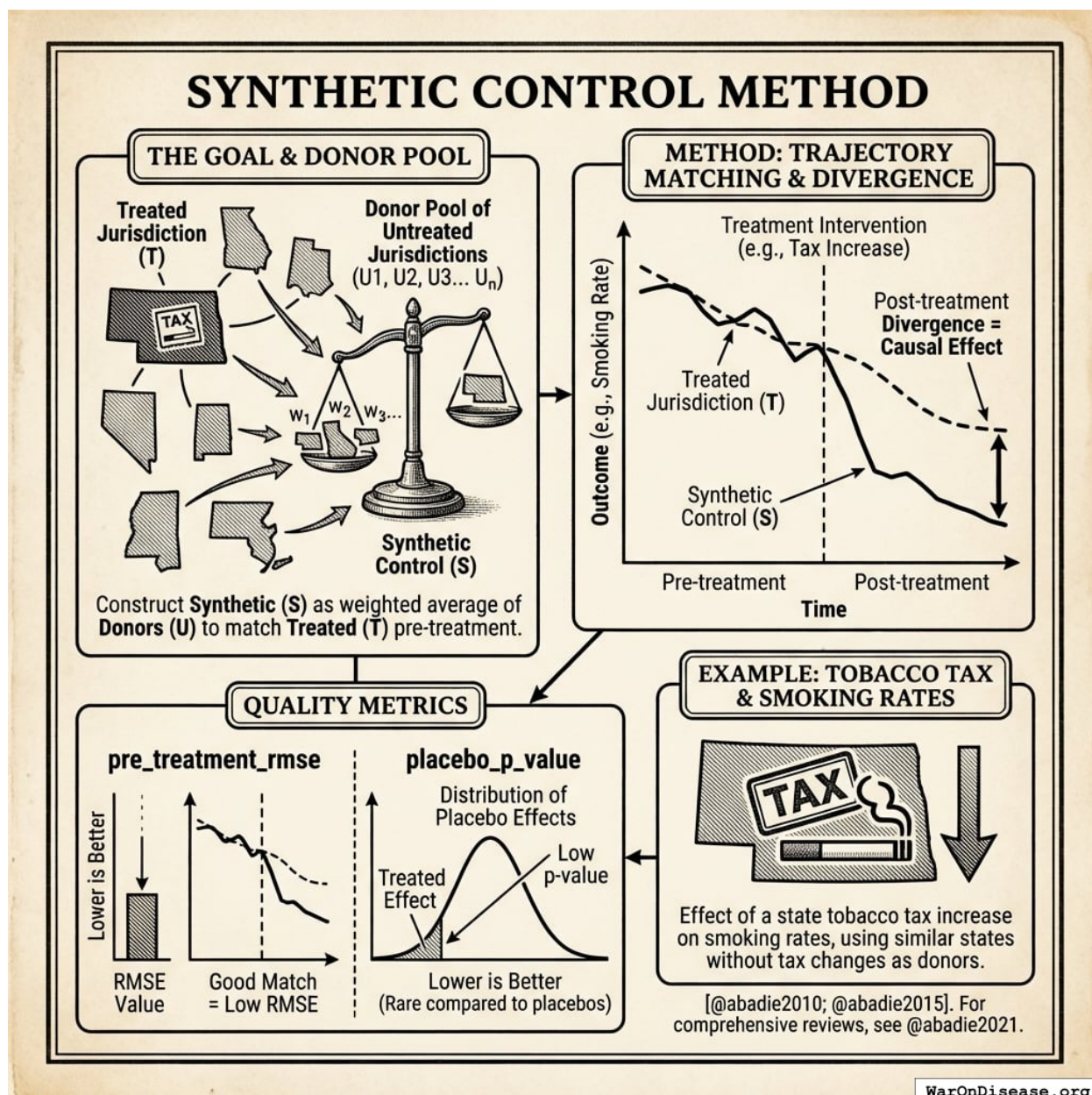


Figure 18: A time-series line chart illustrating the synthetic control method, showing the treated jurisdiction and synthetic control units tracking together pre-treatment and diverging after the intervention.

Method: Construct a “synthetic” control as a weighted average of untreated jurisdictions that matches the treated jurisdiction’s pre-treatment outcome trajectory. Post-treatment divergence estimates the causal effect.

Quality metrics: - `pre_treatment_rmse`: How well does synthetic control match pre-treatment? (Lower is better) - `placebo_p_value`: Permutation test comparing treated effect to placebo effects (Lower is better)

Example: Effect of a state tobacco tax increase on smoking rates, using similar states without tax

changes as donors^{154,155}. For comprehensive reviews of the synthetic control method, see¹⁵⁶.

7.2.2 Difference-in-Differences (DiD)

Use case: Multiple treated jurisdictions, staggered adoption timing, parallel trends assumption plausible.

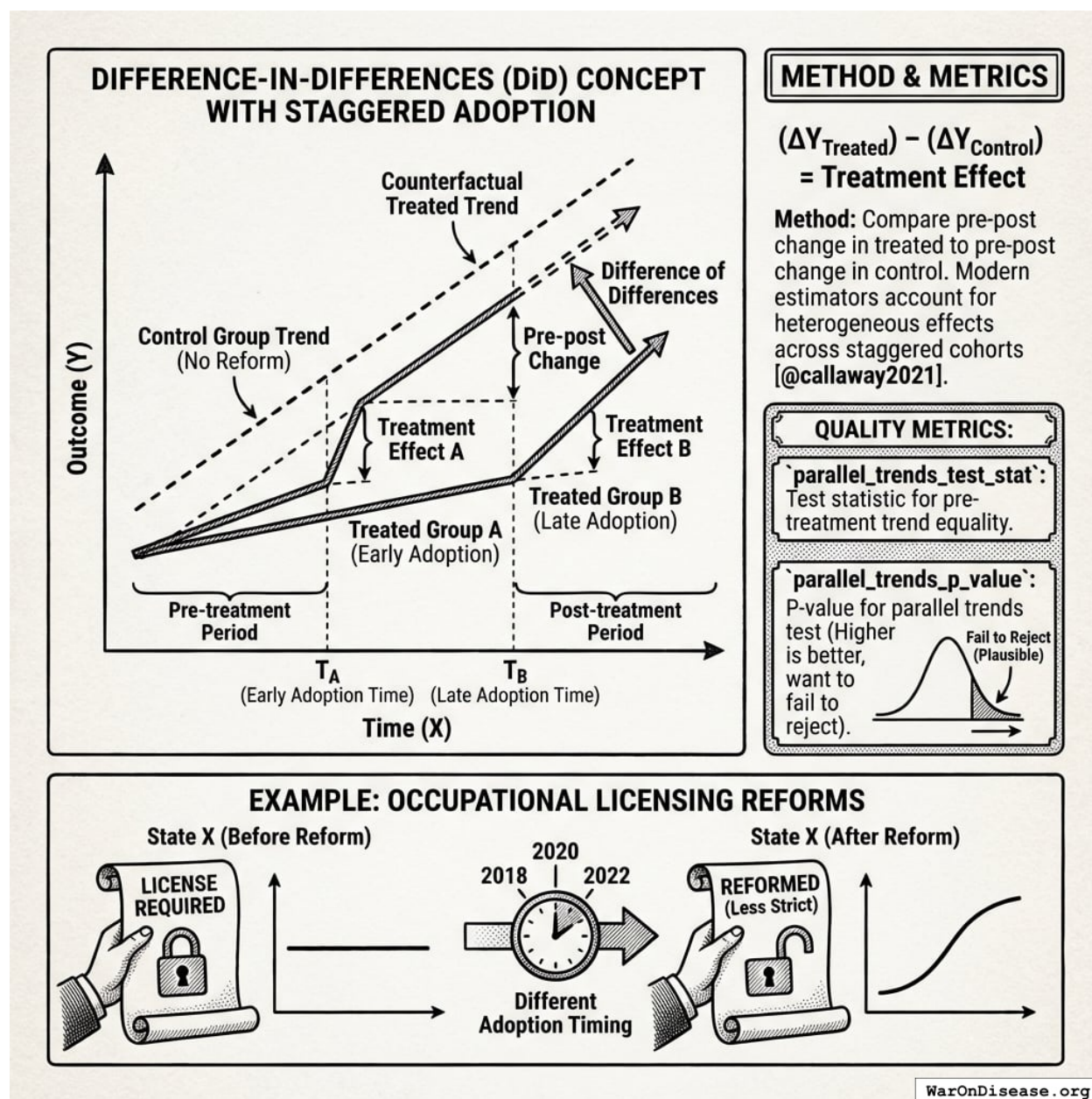


Figure 19: A line chart illustrating the Difference-in-Differences framework, showing the parallel trajectories of treatment and control groups and the divergence used to calculate the treatment effect.

Method: Compare pre-post change in treated jurisdictions to pre-post change in control jurisdictions.

Difference of differences estimates treatment effect. For settings with staggered adoption, modern estimators account for heterogeneous treatment effects across cohorts¹⁵⁷.

Quality metrics: - `parallel_trends_test_stat`: Test statistic for pre-treatment trend equality - `parallel_trends_p_value`: P-value for parallel trends test (Higher is better, want to fail to reject)

Example: Effect of occupational licensing reforms across states with different adoption timing.

7.2.3 Regression Discontinuity Design (RDD)

Use case: Sharp eligibility threshold determines treatment assignment.

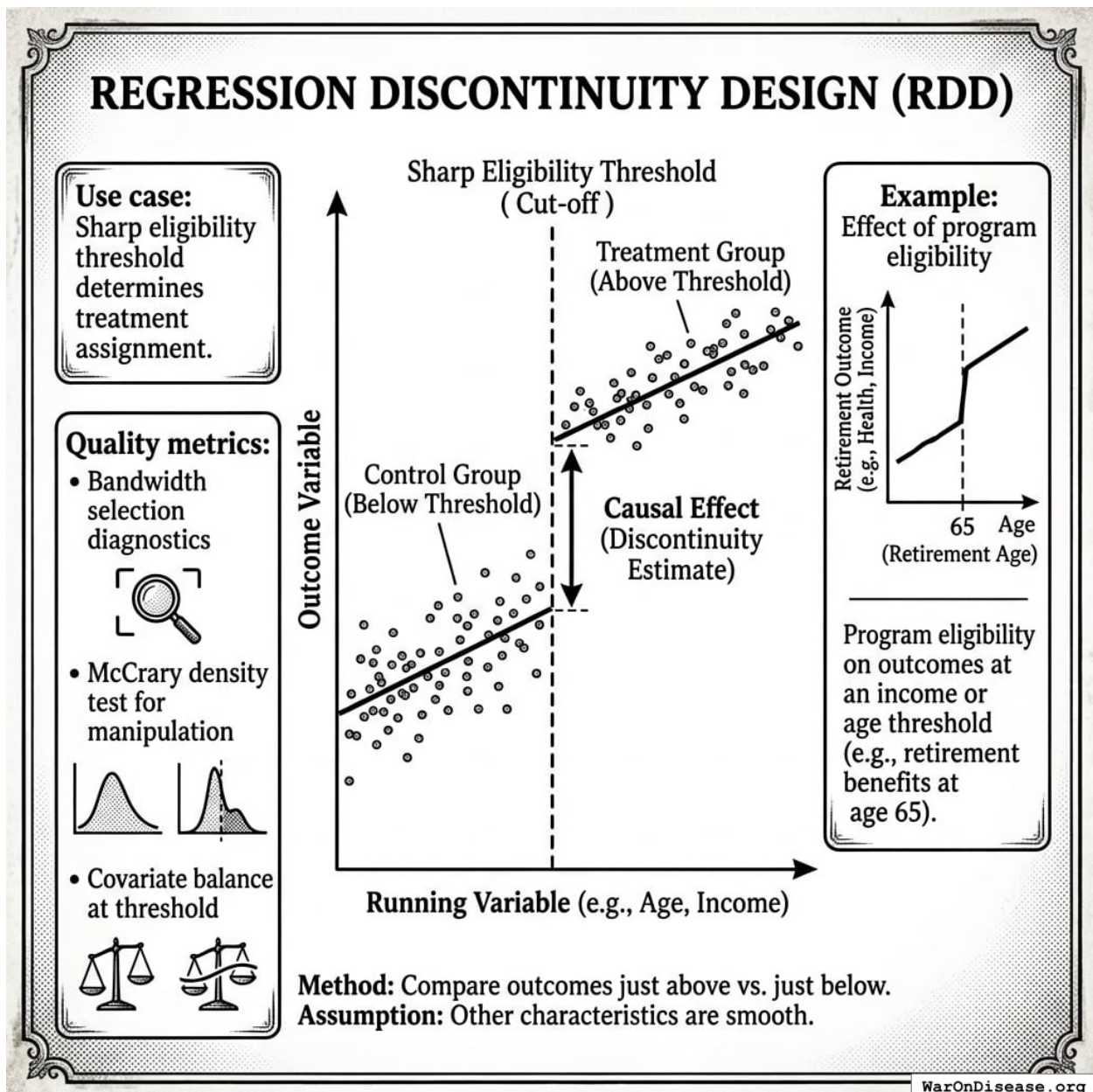


Figure 20: A scatter plot illustrating a regression discontinuity design, showing a jump in the outcome variable at a sharp vertical threshold line with regression curves on either side.

Method: Compare outcomes just above vs. just below the threshold. If other characteristics are smooth across the threshold, the discontinuity in outcomes estimates the causal effect.

Quality metrics: - Bandwidth selection diagnostics - McCrary density test for manipulation - Covariate balance at threshold

Example: Effect of program eligibility on outcomes at an income or age threshold (e.g., retirement benefits at age 65).

7.2.4 Event Study / Interrupted Time Series

Use case: Need to visualize pre-trends and dynamic treatment effects.

Event Study / Interrupted Time Series Analysis: Dynamic Treatment Effects & Pre-Trends

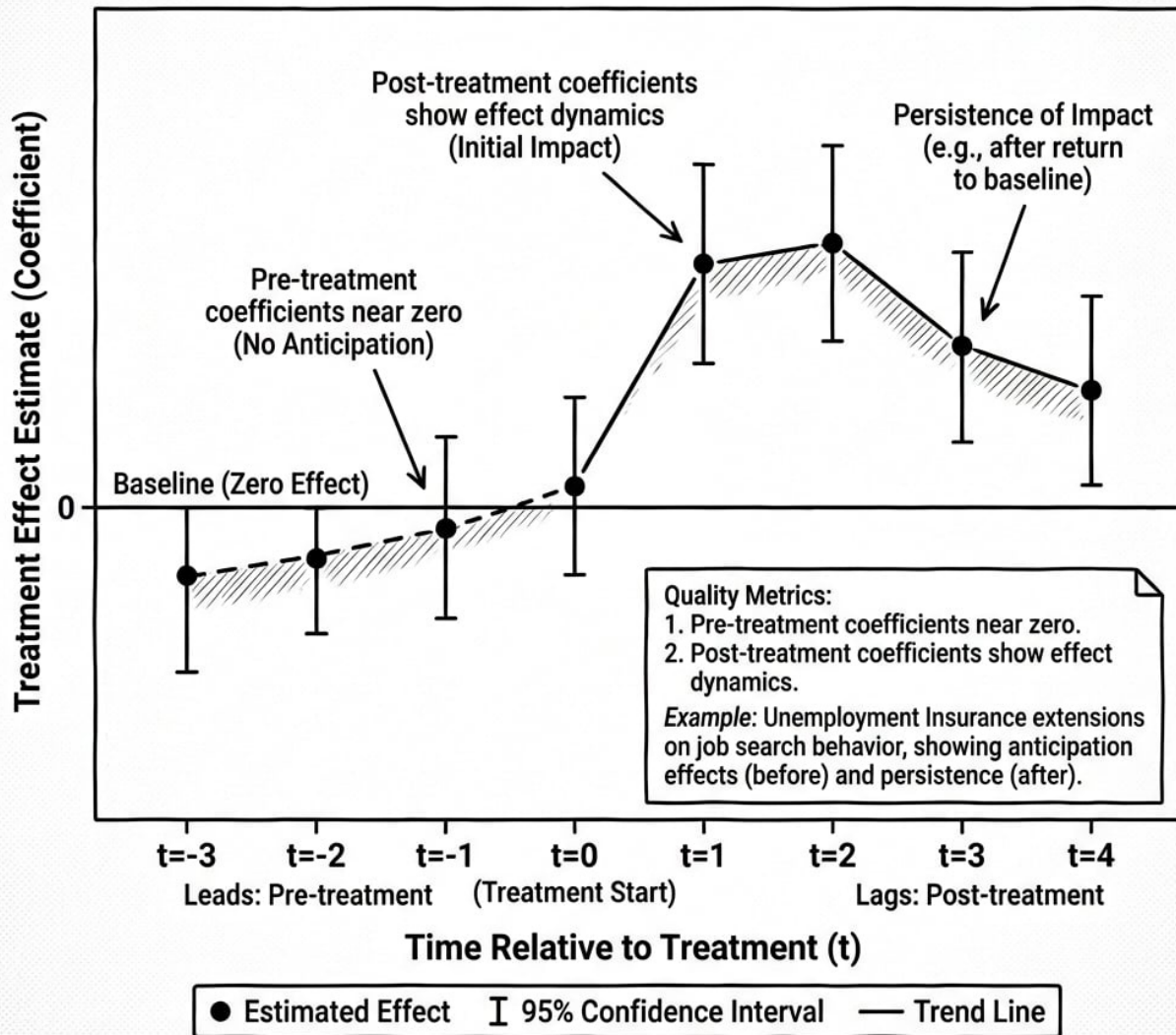


Figure 21: A coefficient plot representing an event study with pre-treatment estimates near zero and post-treatment estimates showing the dynamic evolution of treatment effects.

Method: Estimate treatment effects at each time period relative to treatment, including leads (pre-treatment) and lags (post-treatment).

Quality metrics: - Pre-treatment coefficients should be near zero (no anticipation) - Post-treatment coefficients show effect dynamics

Example: Effect of unemployment insurance extensions on job search behavior, showing both anticipation effects (before benefits expire) and persistence of impact (after return to baseline).

7.2.5 Confidence Weighting by Method

The following weights reflect proposed defaults based on the methodological rigor hierarchy in applied economics. These weights have not been empirically calibrated and should be treated as starting points for implementation.

Method	Base Confidence Weight	Rationale
Randomized experiment	1.00	Gold standard; rare for policies
Regression discontinuity	0.90	Local randomization at threshold
Synthetic control	0.85	Good pre-treatment fit implies validity
Difference-in-differences	0.80	Requires untestable parallel trends
Event study	0.75	Descriptive of dynamics; less rigorous
Interrupted time series	0.65	Single-unit; history threats
Simple before-after	0.40	No control group; confounding likely
Cross-sectional	0.25	Snapshot; severe confounding

7.3 Bradford Hill Criteria Scoring Functions

Bradford Hill's criteria for causality¹⁵⁸, originally developed for epidemiology, are operationalized here as explicit scoring functions. Each criterion maps to a saturation function that produces a score in $[0, 1]$.

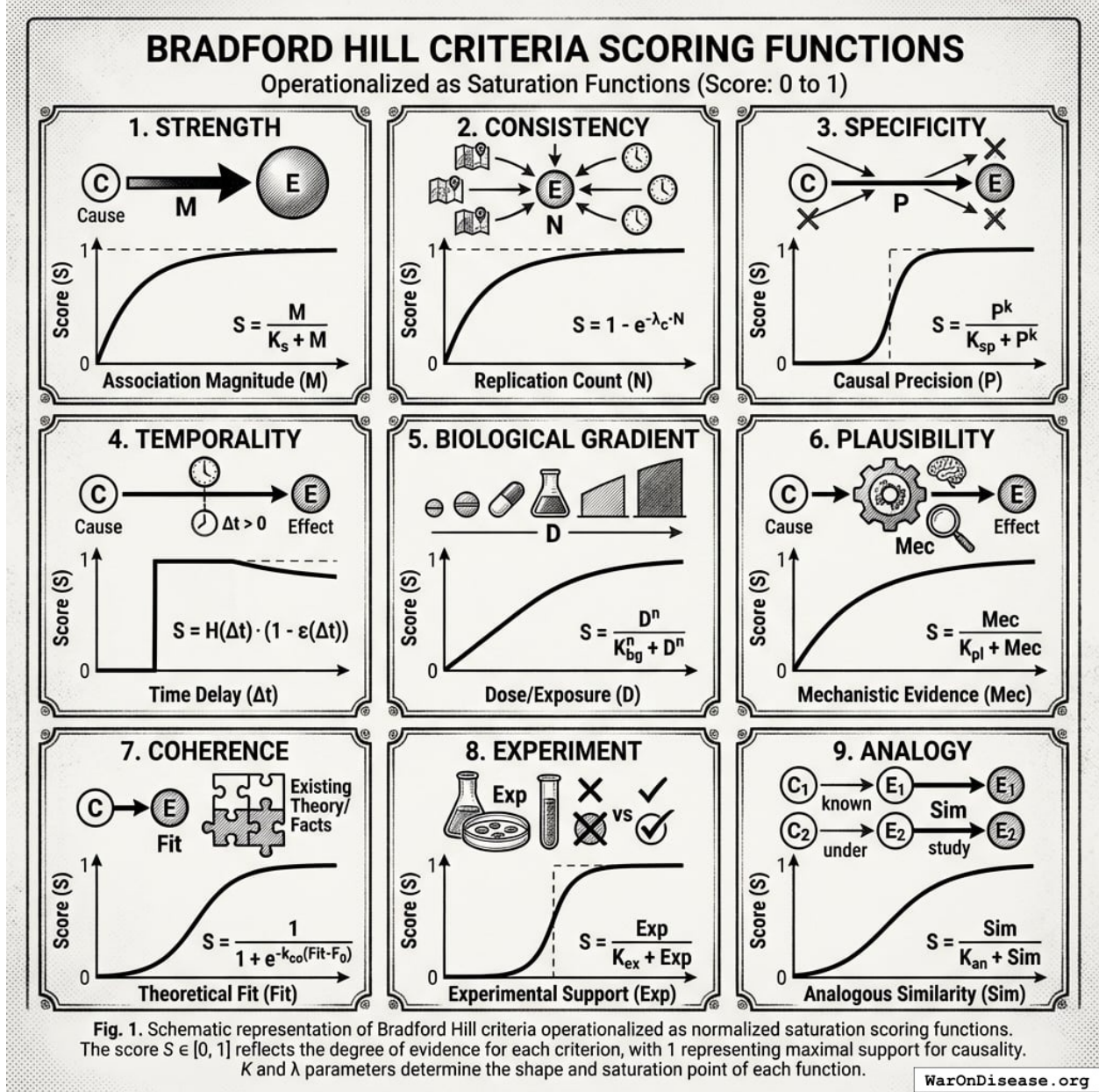


Figure 22: A conceptual diagram showing the mapping process where individual Bradford Hill criteria are processed through saturation functions to yield normalized scores between 0 and 1.

7.3.1 Strength of Association

Larger effect estimates provide stronger evidence. We use an exponential saturation function:

$$S_{\text{strength}} = 1 - e^{-|\hat{\beta}_{\text{std}}|/\beta_{\text{sig}}}$$

Where $|\hat{\beta}_{\text{std}}|$ is the absolute standardized effect size and $\beta_{\text{sig}} = 0.3$ is the saturation parameter.

Parameter justification: The threshold $\beta_{\text{sig}} = 0.3$ corresponds to Cohen's convention for a

“medium” effect size in social science¹⁵⁹. This is a starting point; sensitivity analysis shows PIS changes by $\pm 15\%$ when β_{sig} varies from 0.2 to 0.4. A standardized effect of 0.3 yields $S_{\text{strength}} \approx 0.63$; effects of 0.6+ yield scores > 0.86 .

7.3.2 Consistency Across Jurisdictions

Replication across contexts provides stronger evidence. Scored by number of independent jurisdiction studies:

$$S_{\text{consistency}} = 1 - e^{-N_j/N_{\text{sig}}}$$

Where N_j is the number of jurisdictions with concordant effect direction and $N_{\text{sig}} = 10$ is the saturation parameter.

Parameter justification: The threshold $N_{\text{sig}} = 10$ reflects that replication across 10+ independent jurisdictions provides strong evidence against idiosyncratic local effects. This aligns with meta-analytic conventions where 10+ studies enable reliable heterogeneity estimation¹⁶⁰. Sensitivity analysis shows PIS varies by $\pm 12\%$ when N_{sig} ranges from 7 to 15. Five concordant jurisdictions yield $S_{\text{consistency}} \approx 0.39$; ten yield ≈ 0.63 .

7.3.3 Temporality (Required)

Policy adoption must precede outcome change. This is binary (either satisfied or not):

$$S_{\text{temporality}} = \begin{cases} 1.0 & \text{if } \delta > 0 \\ 0.0 & \text{otherwise} \end{cases}$$

Where δ is the lag between policy implementation and outcome measurement. **If temporality is violated, the overall CCS is zeroed regardless of other criteria.**

7.3.4 Dose-Response Gradient

For continuous policies (tax rates, spending levels), dose-response strengthens causal inference:

$$S_{\text{gradient}} = \frac{r_{\text{dose}}^2}{r_{\text{dose}}^2 + r_{\text{sig}}^2}$$

Where r_{dose} is the correlation between policy intensity and outcome magnitude, and $r_{\text{sig}} = 0.5$ is the saturation parameter.

Parameter justification: The threshold $r_{\text{sig}} = 0.5$ reflects that a correlation of 0.5 between policy intensity and outcome represents moderate dose-response evidence. This is analogous to toxicological dose-response standards where monotonic relationships strengthen causal inference¹⁶¹. Sensitivity analysis shows PIS varies by $\pm 8\%$ when r_{sig} ranges from 0.3 to 0.7. A dose-response correlation of 0.5 yields $S_{\text{gradient}} = 0.5$; correlation of 0.7 yields ≈ 0.66 .

Binary policies: For binary (yes/no) policies, dose-response cannot be assessed. Rather than defaulting to a neutral score of 0.5, binary policies are marked as “N/A” for gradient and this criterion is **excluded from the CCS calculation** (weights are renormalized across remaining

criteria). This prevents binary policies from being systematically penalized relative to continuous policies.

7.3.5 Experiment Quality

Quality of the quasi-experimental design, weighted by validity diagnostic violations:

$$S_{\text{experiment}} = w_{\text{method}} \times (1 - v_{\text{violations}})$$

Where w_{method} is the base method weight and $v_{\text{violations}} \in [0, 1]$ is the proportion of validity checks failed (parallel trends, pre-treatment fit, placebo tests).

7.3.6 Plausibility (Mechanistic)

Economic or behavioral mechanism linking policy to outcome. Scored by expert-validated mechanism database:

$$S_{\text{plausibility}} = \frac{\sum_i w_i \cdot m_i}{\sum_i w_i}$$

Where $m_i \in \{0, 1\}$ indicates whether mechanism component i is satisfied and w_i are component weights.

Mechanism component checklist:

Component	Weight	Assessment Criterion
Economic theory predicts direction	0.30	Peer-reviewed theory paper supports predicted sign
Behavioral response documented	0.25	Empirical evidence of behavioral change in response to similar policies
No implausible required assumptions	0.20	Mechanism doesn't require assumptions contradicted by evidence
Timing consistent with mechanism	0.15	Effect onset matches expected mechanism timeline
Magnitude plausible	0.10	Effect size within range predicted by mechanism

Scoring procedure: Each component is scored binary (0 or 1) by literature review. The weighted sum yields $S_{\text{plausibility}} \in [0, 1]$. When expert-validated mechanism assessments are unavailable, this score defaults to 0.5 with a note that mechanism plausibility is unassessed.

7.3.7 Coherence with Literature

Consistency with broader economic and social science evidence:

$$S_{\text{coherence}} = 1 - e^{-N_{\text{studies}}/N_{\text{sig}}}$$

Where N_{studies} is the count of supporting studies in the literature and $N_{\text{sig}} = 5$. Three supporting studies yield $S_{\text{coherence}} \approx 0.45$; ten yield ≈ 0.86 .

7.3.8 Analogy to Similar Policies

Evidence transfer from similar policies via embedding similarity:

$$S_{\text{analogy}} = \max_{p' \neq p} [\text{sim}(\mathbf{e}_p, \mathbf{e}_{p'}) \times \text{PIS}_{p'}]$$

Where $\mathbf{e}_p$ is the policy embedding vector, $\text{sim}(\cdot)$ is cosine similarity, and $\text{PIS}_{p'}$ is the Policy Impact Score of similar policy p' . Strong evidence for a similar policy provides partial evidence for the focal policy.

7.3.9 Specificity

Whether the policy affects specific outcomes rather than everything:

$$S_{\text{specificity}} = \frac{1}{1 + \log(1 + N_{\text{outcomes}})}$$

Where N_{outcomes} is the number of outcome categories with significant effects. A policy affecting 1-2 outcomes has $S_{\text{specificity}} > 0.7$; a policy affecting 10+ outcomes has $S_{\text{specificity}} < 0.3$. Lower specificity suggests confounding or measurement artifact.

7.4 Causal Confidence Score (CCS) Calculation

The aggregate CCS combines the eight non-temporality criteria with explicit weights, gated by temporality:

$$\text{CCS} = S_{\text{temporality}} \times \frac{\sum_{k \neq \text{temp}} w_k \cdot S_k}{\sum_{k \neq \text{temp}} w_k}$$

$S_{\text{temporality}}$ acts as a binary gate: if temporality fails (policy doesn't precede outcome), the entire CCS is zero regardless of other criteria scores.

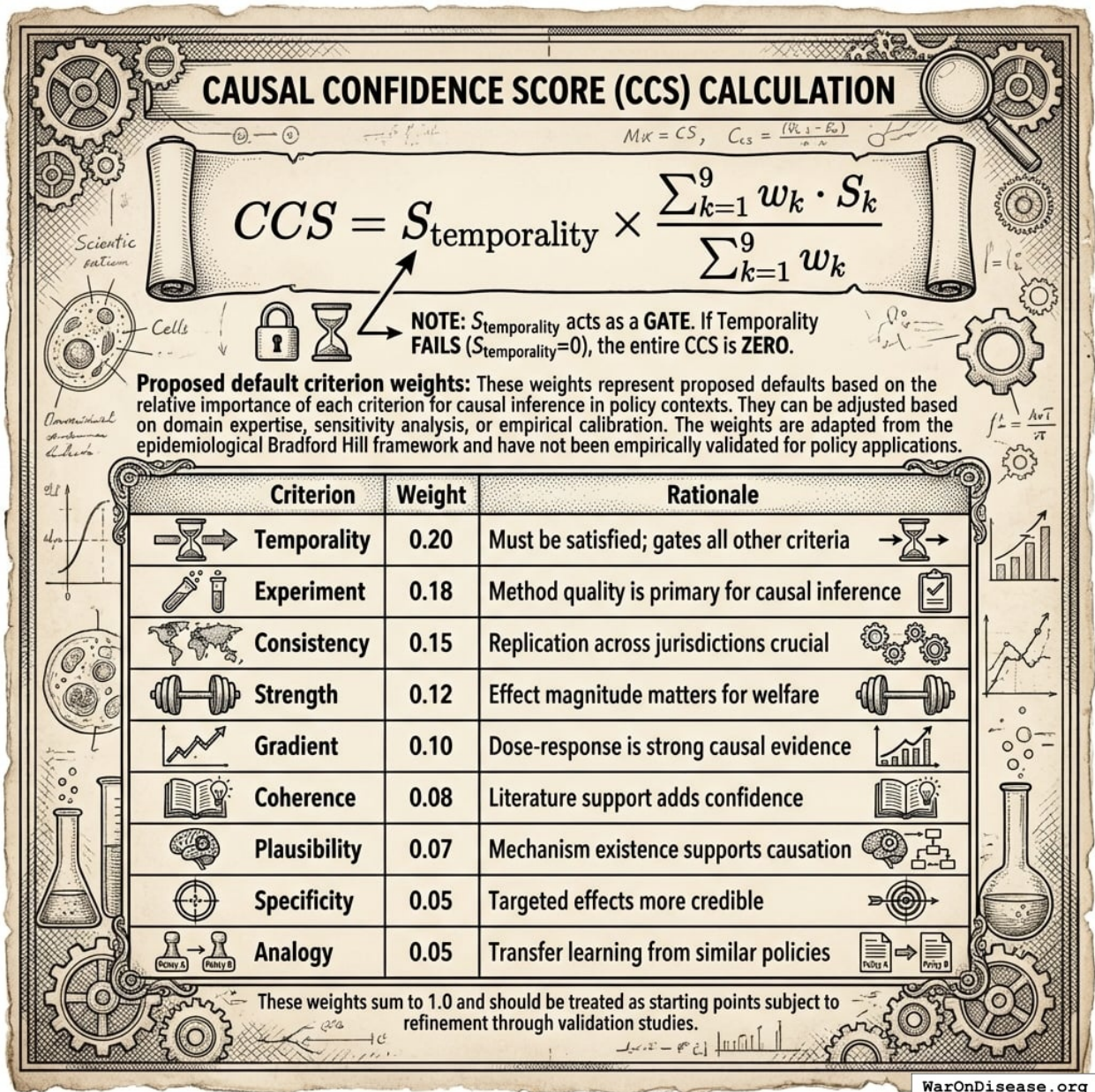


Figure 23: A horizontal bar chart illustrating the nine criteria weights for the Causal Confidence Score, ranking them from highest impact to lowest impact.

Proposed default criterion weights:

These weights represent proposed defaults based on the relative importance of each criterion for causal inference in policy contexts. They can be adjusted based on domain expertise, sensitivity analysis, or empirical calibration. The weights are adapted from the epidemiological Bradford Hill framework and have not been empirically validated for policy applications.

Criterion	Weight	Role
Temporality	Gate	Binary prerequisite (must be 1.0 to proceed)
Experiment	0.225	Method quality is primary for causal inference
Consistency	0.19	Replication across jurisdictions crucial
Strength	0.15	Effect magnitude matters for welfare
Gradient	0.125	Dose-response is strong causal evidence
Coherence	0.10	Literature support adds confidence
Plausibility	0.09	Mechanism existence supports causation
Specificity	0.06	Targeted effects more credible
Analogy	0.06	Transfer learning from similar policies

Weights for the eight scored criteria sum to 1.0. Temporality is not weighted because it is a binary gate, not a continuous score.

8 Jurisdiction Policy Inventory

8.1 Tracking Current Policies by Jurisdiction

Before generating recommendations, OPG must know what policies each jurisdiction currently has. The `jurisdiction_policies` table tracks:

Field	Description	Example
<code>has_policy</code>	Whether jurisdiction has this policy type	TRUE/FALSE
<code>policy_strength</code>	For continuous policies, the current level	\$1.41/pack (tobacco tax)
<code>implementation_date</code>	When current policy took effect	2009-01-01
<code>policy_details_json</code>	Structured details about implementation	{"primary_enforcement": false}
<code>data_source</code>	Where this information came from	"Texas Tax Code §154.021"
<code>last_verified</code>	When this was last confirmed accurate	2024-06-15

8.2 Data Sources for Policy Status

Jurisdiction Level	Primary Sources	Update Frequency
Country	WTO, OECD, IMF policy databases	Annual
US State	NCSL, state legislative databases, LexisNexis	Continuous
EU Member	EUR-Lex, national legal databases	Continuous
US City/County	Municipal code databases, Municode	Varies
Other Subnational	National statistics offices, academic datasets	Varies

8.3 Handling Missing Data

Data completeness varies by jurisdiction and policy type:

Data Quality Score	Interpretation	Recommendation Confidence
> 0.9	Comprehensive inventory	Full confidence
0.7 - 0.9	Most major policies tracked	High confidence
0.5 - 0.7	Significant gaps	Medium confidence; flag gaps
< 0.5	Sparse data	Low confidence; prioritize data collection

Recommendations are only generated when policy status is known with reasonable confidence.

9 Policy Gap Analysis

9.1 Comparing Current to Optimal

For each jurisdiction j , the policy gap for policy type p is:

$$\text{Gap}_{jp} = \text{Evidence-Supported}_p - \text{Current}_{jp}$$

Where:

- **Evidence-Supported:** What the evidence suggests the jurisdiction should have
- **Current:** What the jurisdiction actually has

9.2 Gap Types

Gap Type	Definition	Example
Missing policy	Jurisdiction lacks a policy with strong positive evidence	Texas lacks primary seat belt enforcement
Harmful policy	Jurisdiction has a policy with strong negative evidence	Jurisdiction has policy X shown to increase mortality
Suboptimal strength	Continuous policy set below evidence-supported level	Tobacco tax at \$1.41 vs. optimal ~\$2.50
Excessive strength	Continuous policy set above evidence-supported level	Speed limit at 85 mph vs. optimal ~70 mph

9.3 Priority Scoring

Recommendations are ranked by priority score, which combines gap magnitude, evidence quality, and expected welfare impact:

$$\text{Priority}_{jp} = |\text{Gap}_{jp}| \times \text{PIS}_p \times M_{jp}$$

Where:

- $|\text{Gap}_{jp}|$ = Absolute difference between evidence-supported and current policy level (normalized to $[0, 1]$)
- PIS_p = Policy Impact Score (see Section 12), capturing effect magnitude and causal confidence
- M_{jp} = Monetized annual welfare impact, adjusted for jurisdiction j 's population and context

Priority tiers:

Tier	Priority Score	Interpretation
Critical	≥ 0.80	Immediate action recommended
High	$[0.50, 0.80)$	Strong candidate for adoption
Medium	$[0.25, 0.50)$	Consider if political capital available
Low	< 0.25	Monitor for better evidence

High-priority recommendations have: 1. Large gap between current and optimal 2. Strong evidence (Grade A or B; high PIS) 3. Large expected welfare impact (high M)

9.4 Context Adjustment

Effect estimates are adjusted for jurisdiction characteristics:

Adjustment Factor	Description	Example
Demographics	Age structure, income distribution	Tobacco tax effect varies by income
Existing policies	Interaction with current policy bundle	Effect depends on what else is in place
Institutional capacity	Enforcement capability	Weak institutions $\rightarrow$ smaller effects
Cultural factors	Compliance norms	Varies by society

Context Adjustment Algorithm:

The context adjustment multiplier is computed as:

$$\text{Context Adjustment}_j = \prod_{k \in \{D, P, I, C\}} (1 + \delta_k \cdot d_{jk})$$

Where: - d_{jk} = Standardized distance between jurisdiction j and the evidence-weighted mean on factor k - δ_k = Sensitivity coefficient for factor k (estimated from heterogeneity analysis)

Factor	δ_k Default	Quantification Method
Demographics (D)	0.15	Mahalanobis distance on age/income/education
Existing policies (P)	0.10	Policy bundle cosine distance
Institutional (I)	0.20	World Bank Governance Indicators

Factor	δ_k Default	Quantification Method
Cultural (C)	0.10	Hofstede dimensions + compliance indices

Uncertainty widening:

When jurisdiction j differs substantially from the evidence base (context adjustment < 0.7 or > 1.3), confidence intervals are widened:

$$CI_{\text{adjusted}} = CI_{\text{pooled}} \times (1 + 0.5 \times |1 - \text{Context Adjustment}_j|)$$

This reflects increased uncertainty when extrapolating beyond the observed evidence distribution.

Worked example:

Texas tobacco tax recommendation: - Evidence base: Primarily northeast/west coast states (higher income, lower smoking rates) - Texas demographics: Lower median income, higher baseline smoking - Context adjustment: 0.85 (effect likely 15% smaller in Texas context) - CI widening: $1.075 \times$ (modest extrapolation)

$$\text{Expected Effect}_{\text{Texas}} = 0.30 \times 0.85 = 0.25 \text{ healthy years}$$

10 Recommendation Generation

10.1 Recommendation Types

Type	Question	When to Use	Example
Enact	“Add this?”	New policy the jurisdiction doesn’t have	“ENACT primary seat belt law”
Replace	“Change this?”	Modify existing policy level or approach	“REPLACE tobacco tax: \$1.41 $\rightarrow$ \$2.50”
Repeal	“Remove this?”	Remove policy with negative evidence	“REPEAL [harmful policy]”
Maintain	“Keep this?”	Current policy is evidence-supported	“MAINTAIN DUI threshold at 0.08 BAC”

For continuous policies (taxes, spending levels), **Replace** specifies the change from current to optimal level. **Enact** is reserved for truly new policies that don’t exist in the jurisdiction.

10.2 Blocking Factors

Recommendations flag constraints that may impede adoption:

Blocking Factor	Severity	Description	Example
constitutional	Hard	Requires constitutional amendment	Takings Clause limits on land use regulations
federal_preemption	Hard	Federal law prevents state/local action	Federal minimum wage floor
treaty_obligation	Hard	International agreement constrains policy	WTO rules on tariffs
autonomy_concern	Soft	Restricts individual freedom/choice	Mandatory helmet laws
political_feasibility	Soft	Strong organized opposition	Industry lobbying
implementation_cost	Soft	High fixed costs to implement	New regulatory agency needed

Design rationale: Why blocking factors are metadata only

OPG produces **evidence-based rankings**, not political forecasts. Blocking factors are flagged but do not affect algorithmic priority scores, for three reasons:

1. **Political feasibility shifts over time.** A policy “impossible” in 2020 may be mainstream by 2025. Filtering by current political feasibility would lock in the status quo and fail to surface the evidence-supported set.
2. **Politicians know their context.** An elected official in Texas understands local political dynamics better than any algorithm. OPG provides the evidence; filtering is left to policymaker judgment.
3. **Autonomy tradeoffs require human judgment.** A universal helmet law may save lives but restrict freedom. This is a value judgment, not an evidence question. OPG surfaces the health/income effects; the autonomy tradeoff is for democratic deliberation.

Hard vs. Soft blocking factors:

- **Hard blockers** (constitutional, preemption, treaty): These represent legal impossibility at the current jurisdictional level. Recommendations with hard blockers are marked distinctly but still shown, as they may inform advocacy for constitutional change or higher-level policy.
- **Soft blockers** (political, cost, autonomy): These represent practical difficulty, not impossibility. Many transformative policies faced “impossible” political opposition before adoption.

Important: The full evidence-supported recommendation set is always shown. Users can filter by blocking factor severity if desired, but the default view shows all recommendations ranked by expected welfare impact.

10.3 Similar Jurisdictions

For each recommendation, OPG identifies jurisdictions that: 1. Had similar characteristics to the target jurisdiction 2. Adopted the recommended policy 3. Experienced the predicted effects

This provides concrete examples for policymakers: “Vermont (similar demographics, adopted this in 2015, saw -7.1 pp smoking reduction).”

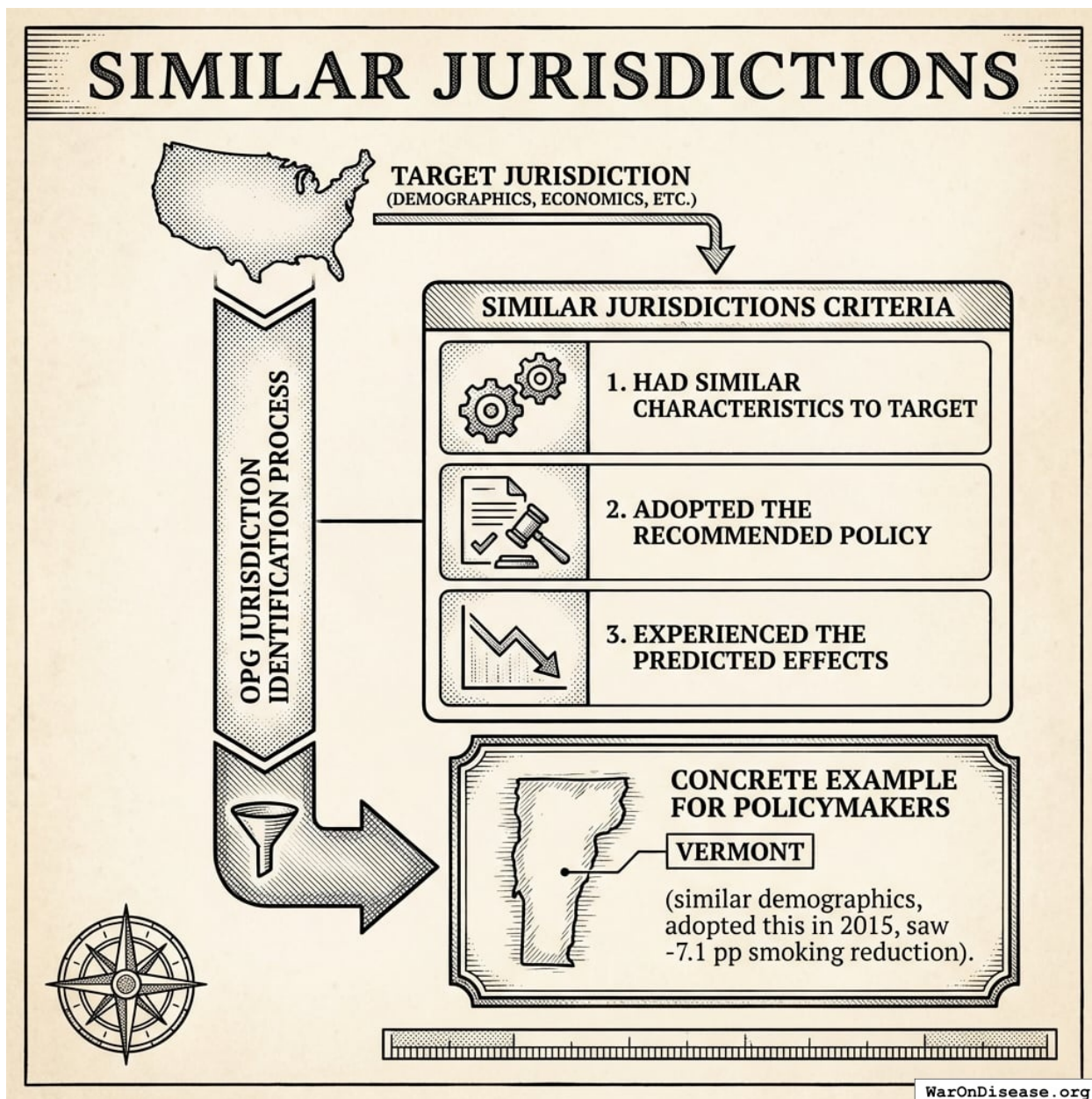


Figure 24: A visual breakdown of the three criteria used to select peer jurisdictions: matching characteristics, policy adoption, and documented positive outcomes.

10.3.1 Computing Jurisdiction Similarity

Similarity between jurisdictions j_1 and j_2 is computed as a weighted sum across three dimensions:

$$\text{sim}(j_1, j_2) = w_D \cdot \text{sim}_D(j_1, j_2) + w_P \cdot \text{sim}_P(j_1, j_2) + w_I \cdot \text{sim}_I(j_1, j_2)$$

Where default weights are $w_D = 0.4$, $w_P = 0.3$, $w_I = 0.3$.

Demographic Similarity (sim_D):

Variable	Weight	Normalization
Log GDP per capita	0.25	By cross-jurisdiction SD
Population log	0.15	By cross-jurisdiction SD
Median age	0.20	By cross-jurisdiction SD
Urban population %	0.15	By cross-jurisdiction SD
Education (years)	0.15	By cross-jurisdiction SD
Gini coefficient	0.10	By cross-jurisdiction SD

$$\text{sim}_D = 1 - \frac{\sum_k w_k |z_{j_1,k} - z_{j_2,k}|}{\sum_k w_k \cdot 4}$$

Where z values are z-scores and the denominator normalizes to [0,1] (4 SD maximum difference).

Policy Similarity (sim_P): Cosine similarity of policy embedding vectors, where each jurisdiction’s policy bundle is embedded via weighted sum of individual policy embeddings.

Institutional Similarity (sim_I):

Feature	Comparison
Federal vs. unitary	Binary match (1.0 if same, 0.5 if different)
Legal tradition	Common law, civil law, mixed (1.0/0.5/0.0)
Enforcement capacity	World Bank governance indicator proximity
Corruption level	Transparency International CPI proximity

Usage: Jurisdictions with $\text{sim}(j_1, j_2) > 0.7$ are considered “similar” for evidence transfer purposes. Effect estimates from similar jurisdictions receive higher weight in context adjustment.

10.4 Recommended Tracking (for OPG Feedback)

Each recommendation includes minimal tracking guidance to enable continuous OPG improvement:

Field	Description	Example
Primary metric	The outcome variable to track	Traffic deaths per 100K
Data source	Where to get it	State vital statistics
Measurement frequency	How often	Annual
Comparison baseline	What to compare against	Pre-implementation 3-year average

This creates a learning loop: OPG recommends → jurisdiction implements → reports outcomes → OPG improves future recommendations.

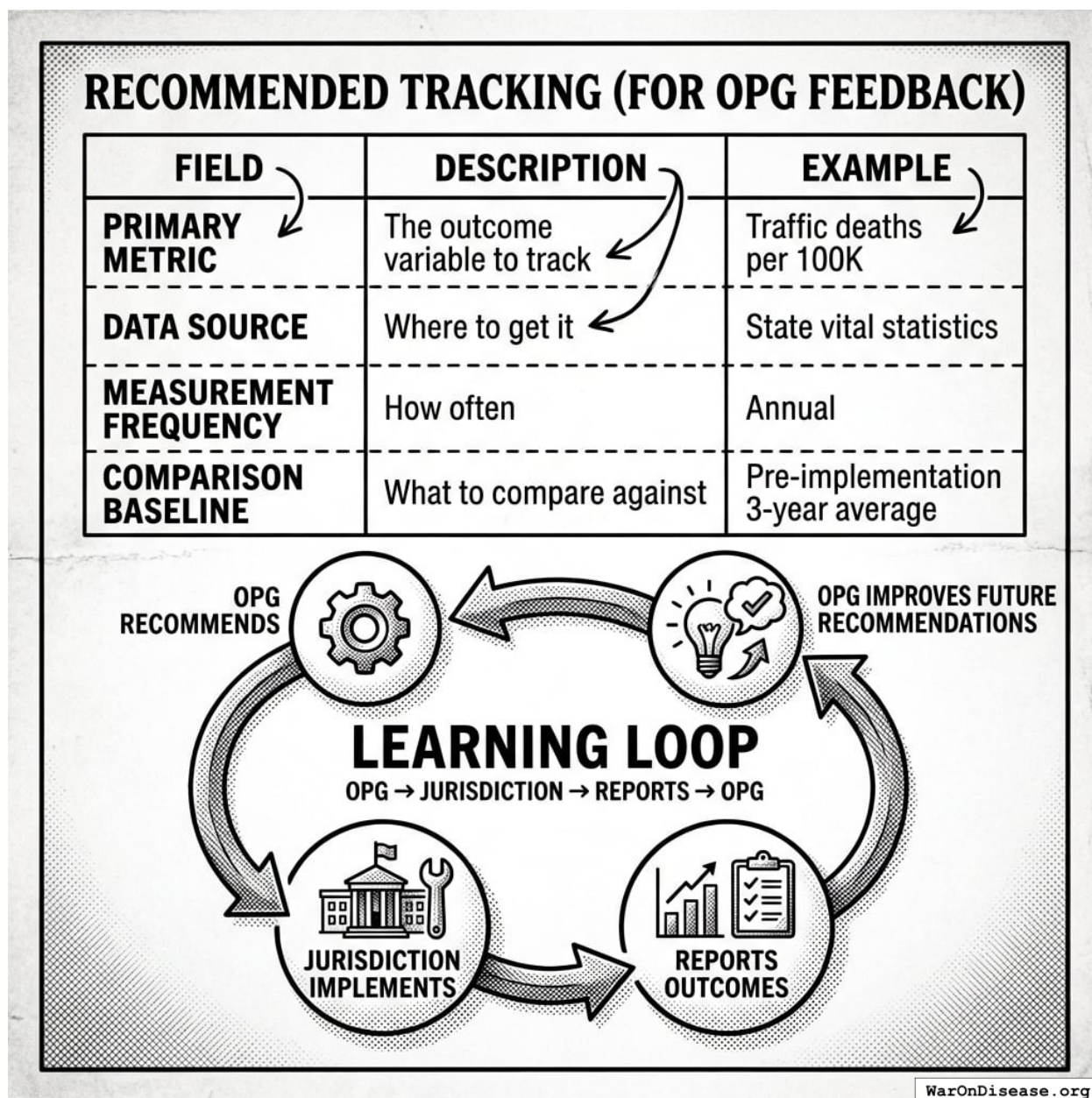


Figure 25: A circular flowchart illustrating the four-step learning loop: OPG recommendation, jurisdictional implementation, outcome reporting, and continuous improvement of future guidance.

11 Optimal Jurisdictional Level for Policy Implementation

11.1 The Subsidiarity Principle for Evidence Generation

OPG recommends policies be implemented at the **lowest jurisdictional level where the policy can be effective**, for two reasons:

1. **Maximize experimental data**: 50 states experimenting > 1 federal policy. 3,000+ counties > 50 states. More jurisdictions = more natural experiments = faster evidence accumulation.

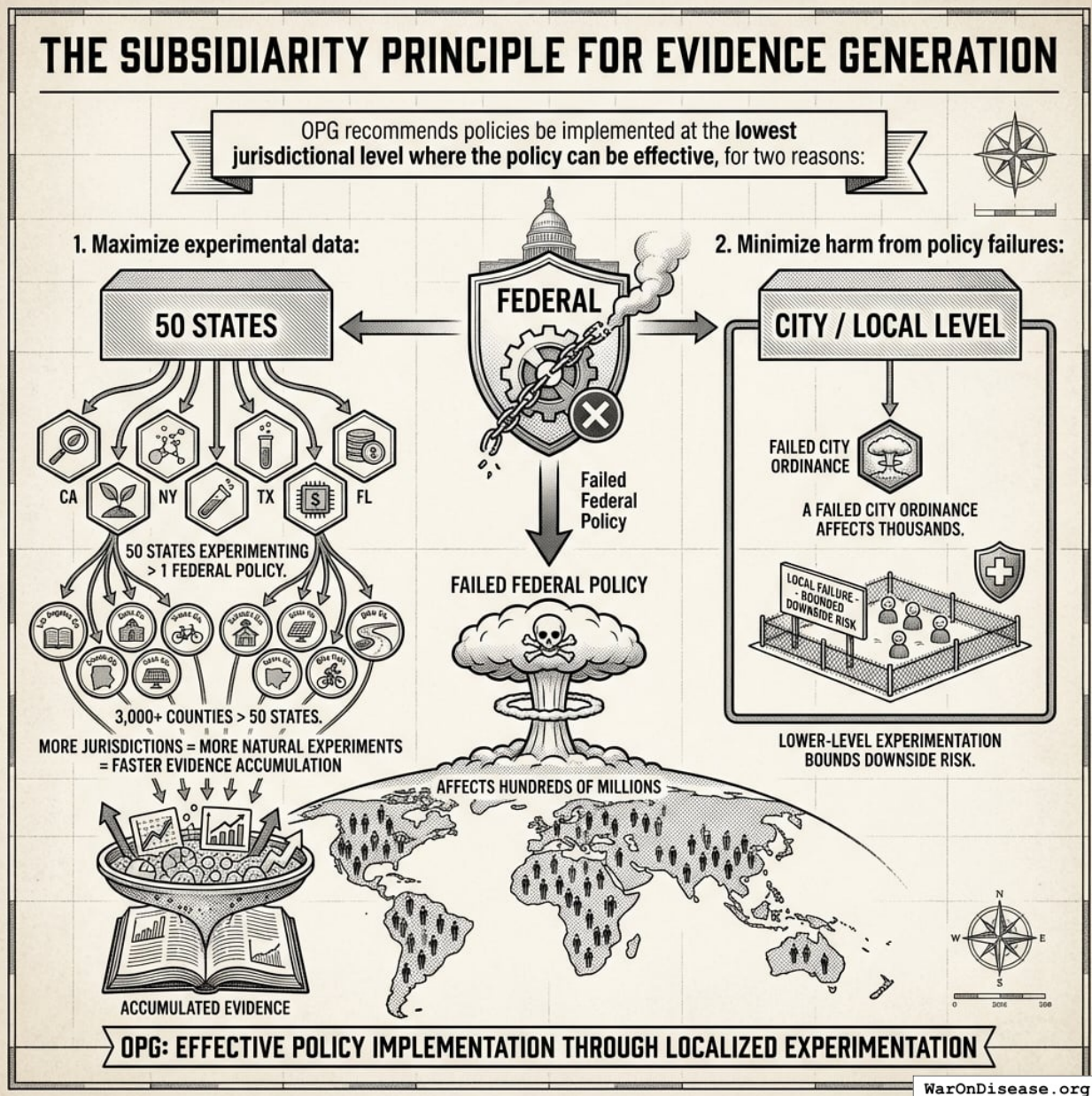


Figure 26: A hierarchical visualization comparing the volume of data generation and risk potential across federal, state, and county levels.

2. **Minimize harm from policy failures:** A failed city ordinance affects thousands; a failed federal policy affects hundreds of millions. Lower-level experimentation bounds downside risk.

11.2 When Higher Levels Are Necessary

Some policies require higher jurisdictional levels:

Reason	Example	Recommendation
Externalities	Pollution crosses borders	State or federal

Reason	Example	Recommendation
Race-to-bottom risk	Labor standards, tax competition	Federal floor, state variation above
Network effects	Infrastructure standards	Federal coordination
Economies of scale	Defense, diplomacy	National

11.3 Jurisdictional Level in Recommendations

For each policy recommendation, OPG specifies:

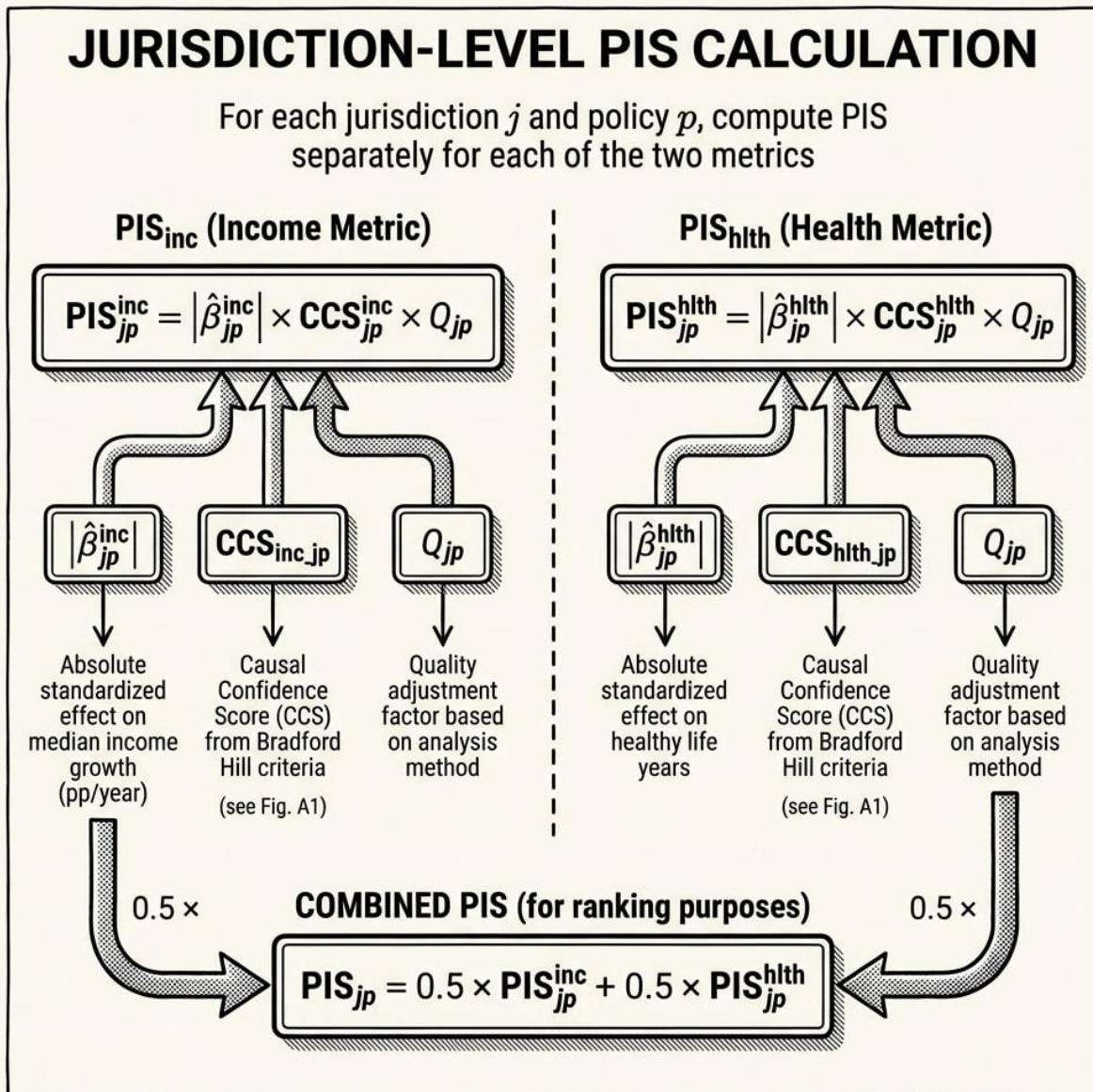
Field	Example
Minimum effective level	“City or higher”
Recommended level	“City (maximize data collection)”
Current adoption	“12 states, 47 cities have this”
Level constraints	“Federal preemption prevents city-level”

12 Policy Impact Score (Intermediate Metric)

12.1 Overview

The Policy Impact Score (PIS) is the intermediate metric used to generate recommendations. It quantifies the strength of evidence that a policy affects an outcome, combining effect magnitude, causal confidence, and analysis quality into a single score.

12.2 Jurisdiction-Level PIS Calculation



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Figure 27: A flowchart illustrating the logic for calculating the combined Policy Impact Score (PIS) by aggregating standardized effect sizes, causal confidence scores, and quality adjustments for both income and health metrics.

For each jurisdiction j and policy p , compute PIS separately for each of the two metrics:

$$\text{PIS}_{jp}^{\text{inc}} = |\hat{\beta}_{jp}^{\text{inc}}| \times \text{CCS}_{jp}^{\text{inc}} \times Q_{jp}$$

$$\text{PIS}_{jp}^{\text{hlth}} = |\hat{\beta}_{jp}^{\text{hlth}}| \times \text{CCS}_{jp}^{\text{hlth}} \times Q_{jp}$$

Where:

- $|\hat{\beta}_{jp}^{\text{inc}}|$ = Absolute standardized effect on median income growth (pp/year)
- $|\hat{\beta}_{jp}^{\text{hlth}}|$ = Absolute standardized effect on healthy life years
- CCS = Causal Confidence Score from Bradford Hill criteria (see Section 7.3)
- Q_{jp} = Quality adjustment factor based on analysis method

The **combined PIS** for ranking purposes is:

$$\text{PIS}_{jp} = 0.5 \times \text{PIS}_{jp}^{\text{inc}} + 0.5 \times \text{PIS}_{jp}^{\text{hlth}}$$

! Always Report Both Metrics Separately

While combined PIS is useful for ranking, recommendations should always display effects on both metrics:

Policy	Income Effect	Health Effect	CCS	Grade
Tobacco tax +\$1	-0.02 pp/yr	+0.25 years	0.85	A
Primary seat belt	+0.02 pp/yr	+0.15 years	0.81	A

This prevents hiding tradeoffs. For example, the tobacco tax has a negative income effect (direct tax burden) but positive health effect (reduced smoking). The two-metric format makes this tradeoff explicit.

12.3 Effect Estimate Standardization

Each metric is standardized using cross-jurisdictional standard deviations:

$$\hat{\beta}_{\text{std}}^{\text{inc}} = \frac{\hat{\beta}_{\text{raw}}^{\text{inc}}}{\sigma_{\text{income}}} \quad \hat{\beta}_{\text{std}}^{\text{hlth}} = \frac{\hat{\beta}_{\text{raw}}^{\text{hlth}}}{\sigma_{\text{health}}}$$

Where σ_{income} is the cross-jurisdictional SD of median income growth (typically ~1.5 pp/year) and σ_{health} is the cross-jurisdictional SD of healthy life expectancy (typically ~3-5 years).

12.4 Quality Adjustment Factor

$$Q = w_{\text{method}} \cdot (1 - \text{violations})$$

Where: - w_{method} = Method confidence weight (see table above) - violations = Proportion of validity checks failed (parallel trends, pre-treatment fit, etc.)

12.5 Confounder Adjustment

For each analysis, we track which confounders were controlled:

```
{
  "confounders_controlled": ["gdp_growth", "unemployment", "population_age_structure"],
  "confounders_not_controlled": ["neighboring_policy_spillovers", "measurement_error"],
```

```

"confounder_sensitivity": 0.85
}

```

The `confounder_sensitivity` field estimates how much the effect estimate might change if uncontrolled confounders were addressed (Oster's delta,¹⁶²).

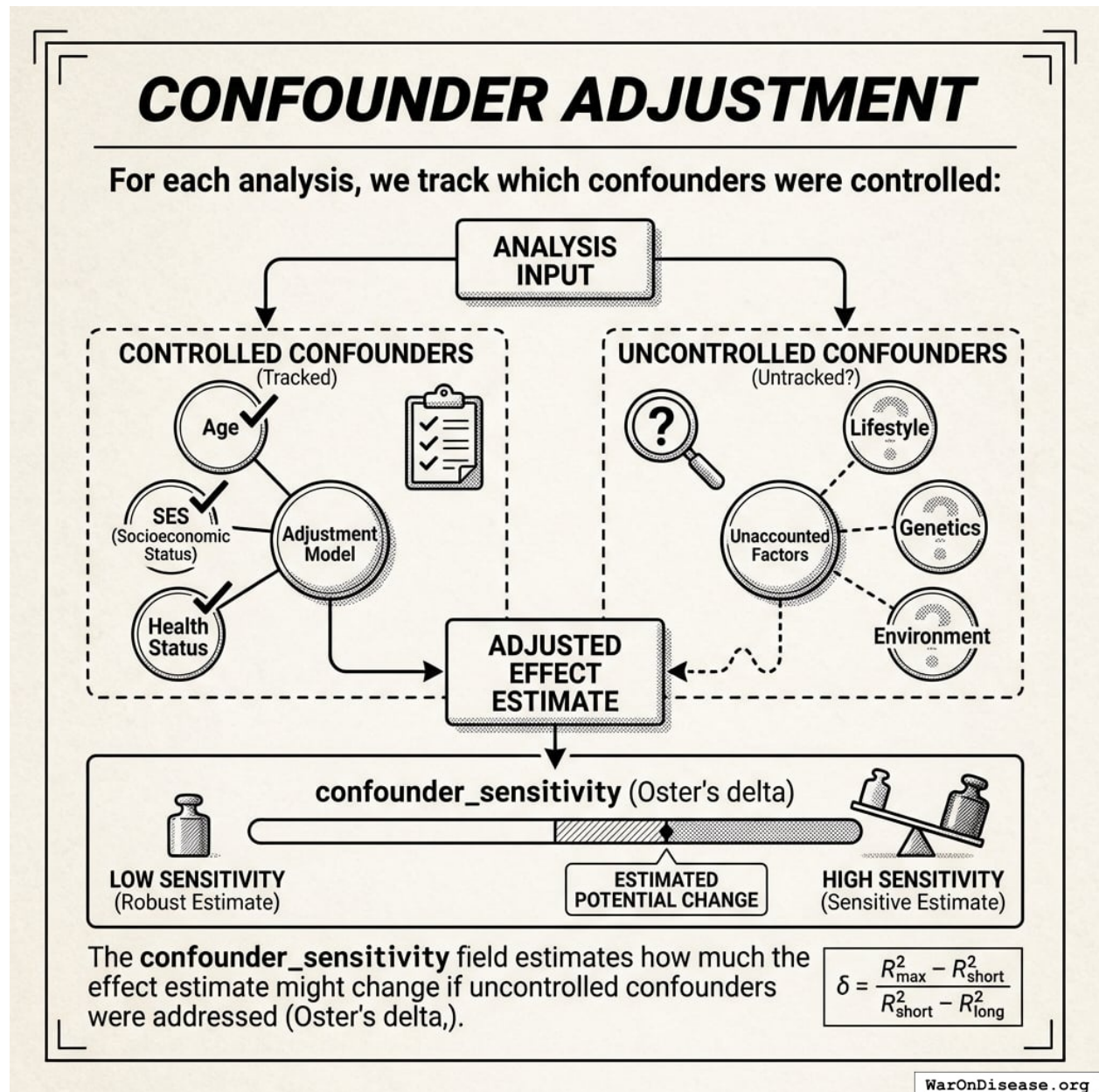


Figure 28: A conceptual diagram showing the relationship between a treatment and outcome, distinguishing between controlled confounders and the potential bias introduced by uncontrolled factors.

13 Global (Aggregate) PIS Calculation

Aggregate estimates combine jurisdiction-level analyses via random-effects meta-analysis.

13.1 Pooled Effect Estimate

$$\hat{\beta}_{\text{pooled}} = \frac{\sum_j w_j \hat{\beta}_j}{\sum_j w_j}$$

Where weights incorporate both within-study variance and between-study heterogeneity:

$$w_j = \frac{1}{\text{SE}_j^2 + \tau^2}$$

13.2 Pooled PIS Across Jurisdictions

The aggregate PIS for policy p and outcome o is:

$$\text{PIS}_{\text{pooled}} = \frac{\sum_j w_j \cdot \text{PIS}_j}{\sum_j w_j}$$

This precision-weighted average gives more influence to high-precision estimates (low SE) while accounting for true heterogeneity (τ^2).

13.3 Heterogeneity Statistics

Following standard meta-analysis conventions¹⁴⁸:

- **I²**: Percentage of variance due to heterogeneity (vs. sampling error)
 - $I^2 < 25\%$: Low heterogeneity
 - $25\% \leq I^2 < 75\%$: Moderate heterogeneity
 - $I^2 \geq 75\%$: High heterogeneity (effects vary substantially across jurisdictions)
- **τ^2** : Estimated between-study variance
- **Q statistic**: Cochran’s test for heterogeneity

High heterogeneity suggests moderators (policy effects vary by context) rather than a single true effect.

13.4 Evidence Grading

Evidence grades are assigned using explicit thresholds on PIS, heterogeneity (I^2), and jurisdiction count (N_j):

$$\text{Grade} = \begin{cases} A & \text{if PIS} \geq 0.80 \text{ AND } I^2 < 0.50 \text{ AND } N_j \geq 5 \\ B & \text{if PIS} \geq 0.60 \text{ AND } I^2 < 0.50 \text{ AND } N_j \geq 3 \\ C & \text{if PIS} \geq 0.40 \text{ AND } I^2 < 0.75 \text{ AND } N_j \geq 2 \\ D & \text{if PIS} \geq 0.20 \\ F & \text{otherwise} \end{cases}$$

Grade interpretation:

Grade	PIS Threshold	Heterogeneity	Jurisdictions	Interpretation
A	≥ 0.80	$I^2 < 50\%$	≥ 5	Strong evidence; ready for implementation
B	≥ 0.60	$I^2 < 50\%$	≥ 3	Good evidence; consider piloting
C	≥ 0.40	$I^2 < 75\%$	≥ 2	Suggestive evidence; needs validation
D	≥ 0.20	Any	Any	Weak evidence; exploratory only
F	< 0.20	Any	Any	Insufficient evidence

Threshold calibration methodology:

These thresholds are proposed defaults requiring retrospective calibration. The calibration procedure:

1. **Historical validation:** Apply OPG grading to policies adopted 10+ years ago with known outcomes
2. **Target validation rates:** Grade A recommendations should validate at 70%+ rate; Grade B at 50%+
3. **Threshold adjustment:** If observed validation rates differ from targets, adjust PIS and I^2 thresholds

Heterogeneity threshold rationale:

The $I^2 < 50\%$ threshold for Grades A and B follows Cochrane Collaboration guidance that heterogeneity above 50% indicates “substantial” variability across studies¹⁴⁸. Grade C allows heterogeneity up to 75% (the “high” threshold) with explicit acknowledgment that effects are context-dependent. Above 75%, pooled estimates provide limited guidance for any specific jurisdiction.

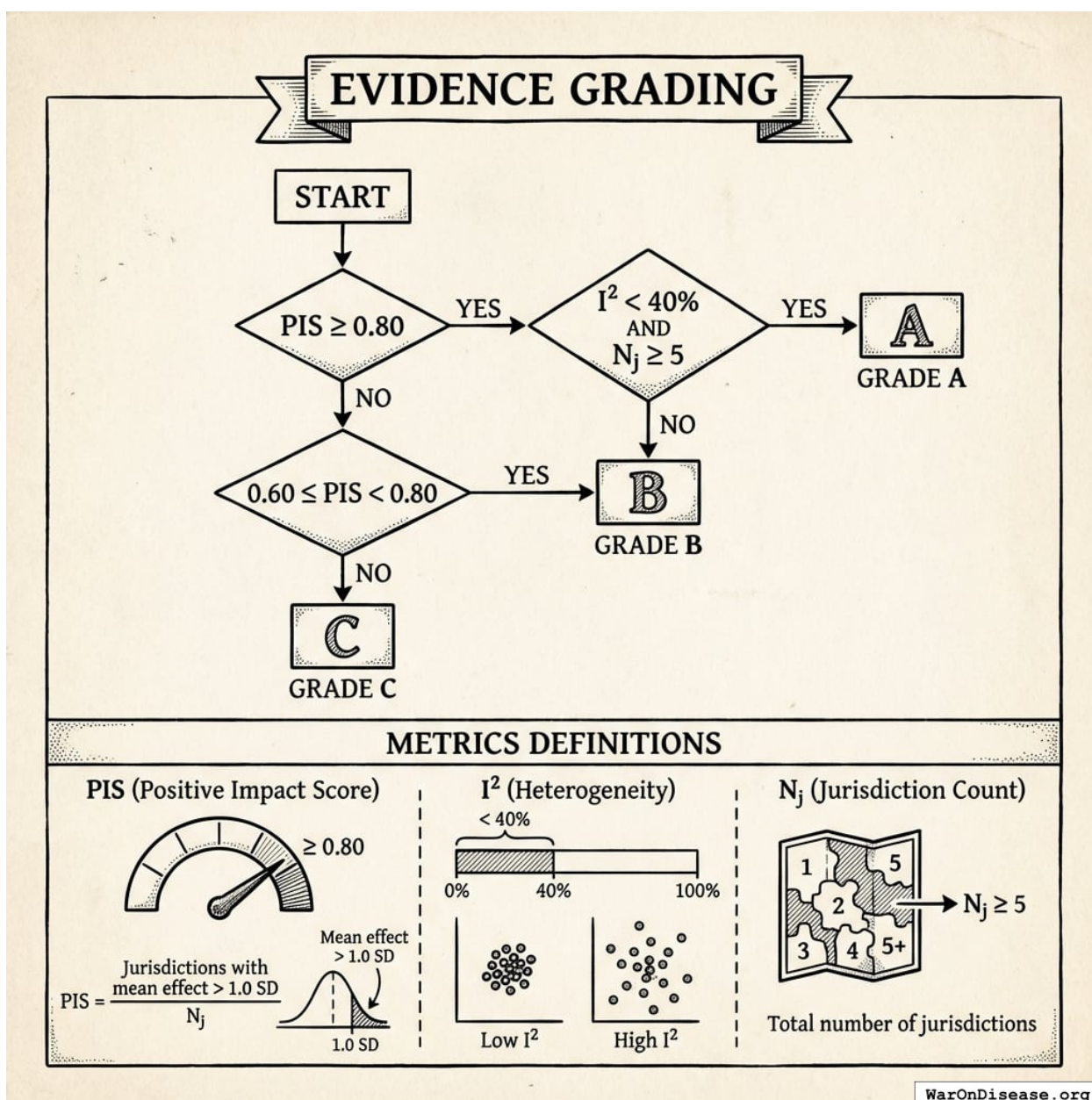


Figure 29: A decision tree illustrating the evidence grading logic based on PIS, heterogeneity, and jurisdiction count, including adjustment modifiers.

Additional grade modifiers:

- **Conflicting evidence:** Downgrade by 1 letter if direction of effect differs across high-quality studies
- **High-quality RCT:** Automatic Grade A if RCT with low risk of bias, regardless of other criteria
- **Single jurisdiction:** Maximum Grade C unless effect is extraordinarily large ($|\hat{\beta}| > 1.0 \text{ SD}$)

13.5 Context-Specific Confidence

Effects may vary by jurisdiction characteristics. We report confidence separately for:

Context	Description	Example Modifier
High-income countries	OECD members, GDP/capita > \$30K	Tax policy effects
Low-income countries	GDP/capita < \$5K	Different institutional capacity
Federal systems	Policy set at national level	vs. subnational variation
Subnational	States, provinces, cities	Local policy autonomy

14 Quality Requirements & Validation

14.1 Minimum Thresholds for Inclusion

Criterion	Minimum	Rationale
Pre-treatment periods	4	Need to assess pre-trends
Post-treatment periods	2	Need to observe effect
Outcome observations	20	Statistical power
Control jurisdictions (for DiD)	5	Donor pool size
Pre-treatment RMSE (synthetic control)	< 2 SD	Acceptable pre-treatment fit

14.2 Parallel Trends Testing (DiD)

For difference-in-differences analyses, we test whether treated and control jurisdictions had parallel outcome trends before treatment:

1. Estimate event study with pre-treatment leads
2. Test joint significance of pre-treatment coefficients
3. If $p < 0.10$, flag as potential parallel trends violation
4. Report sensitivity: how different would trends need to be to explain away the effect?

Parallel Trends Testing (DiD)

Event Study: Pre-treatment Leads & Trends

Legend:

- Treated (T=1)
- Control (T=0)

Y-axis: Outcome (Y)

X-axis: Time relative to Treatment (t)

Vertical dashed line: TREATMENT

2. Test Joint Significance

$$H_0: \beta_{-3} = \beta_{-2} = \beta_{-1} = 0$$

$$\Rightarrow F = \frac{\sum (3 - \beta_{-1})}{n^2}$$

$$\Rightarrow \text{P-value} = 0.35 \text{ (e.g.)}$$

3. If $p < 0.10$, Flag as Potential VIOLATION

$p < 0.10?$
FLAG!

Check Robustness!

1. Estimate Event Study

$$Y_{it} = \alpha + \sum_j \beta_j \cdot \text{Lead}_{j,it} + \gamma \cdot \text{Post}_{it} + \delta \cdot \text{Treated}_i + \varepsilon_{it}$$

Lead -3

Lead -2

Lead -1

4. Report Sensitivity: Explain Away Effect?

Plausible Alternative Trends

How different would trends need to be to explain away the effect? Calculate and report.

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Figure 30: A four-step workflow illustrating the parallel trends testing procedure, including the statistical significance check and subsequent sensitivity reporting.

14.3 Pre-Treatment Fit (Synthetic Control)

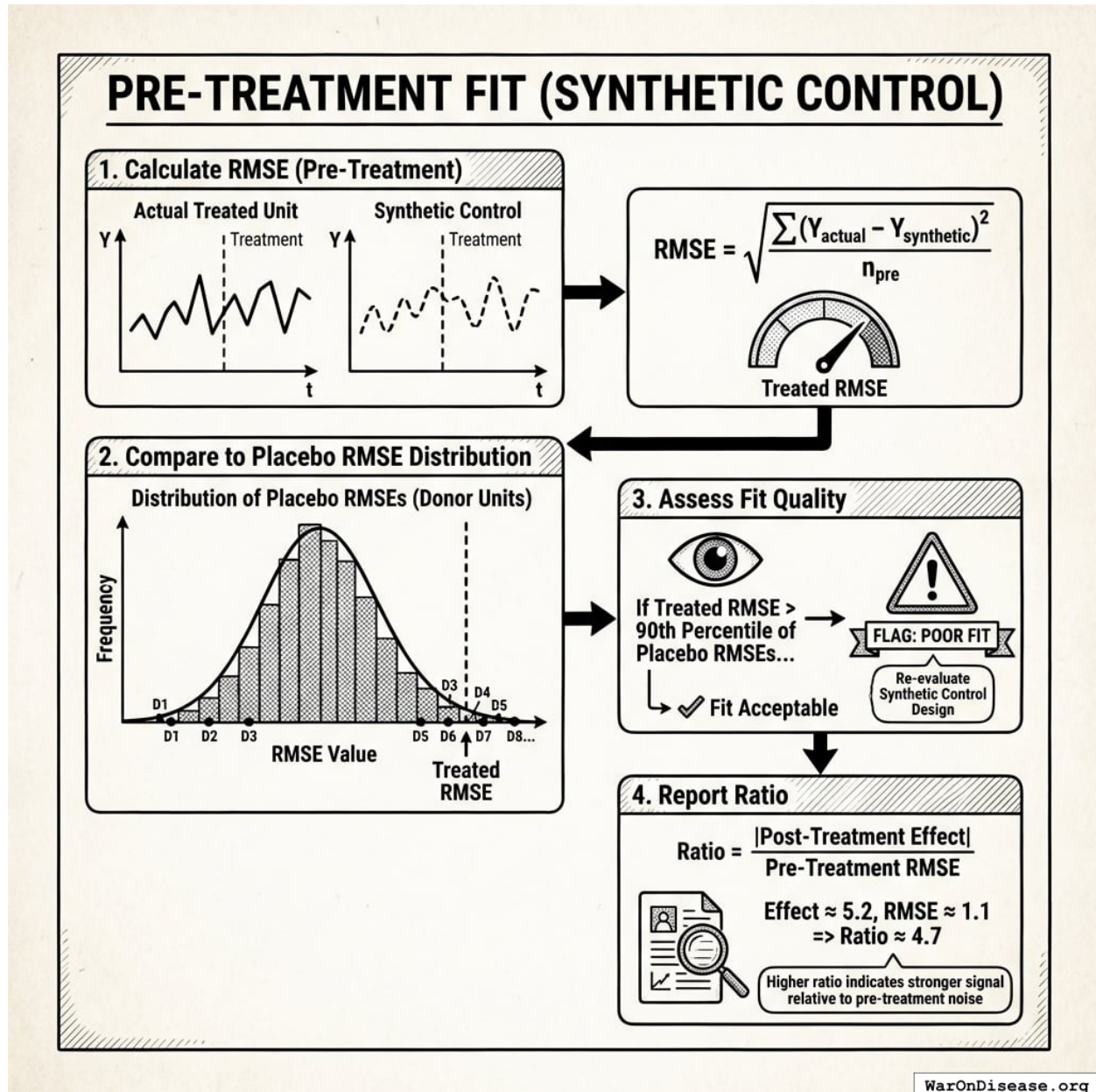


Figure 31: A workflow diagram outlining the four steps for validating synthetic control fit, including RMSE calculation, placebo comparison, and the conditional logic for flagging poor fits.

For synthetic control analyses:

1. Calculate RMSE of synthetic vs. actual treated unit pre-treatment
2. Compare to distribution of placebo RMSEs (treating each donor as “treated”)
3. If treated RMSE is in top 10% of placebo RMSEs, flag as poor fit
4. Report ratio of post-treatment effect to pre-treatment RMSE

14.4 Placebo and Robustness Tests

Test	Purpose	Implementation
In-time placebo	Does “treatment” show effect before it happened?	Assign fake treatment date before actual
In-space placebo	Do untreated units show similar effects?	Apply analysis to control jurisdictions
Leave-one-out	Is result driven by single jurisdiction?	Re-estimate dropping each jurisdiction
Bandwidth sensitivity	(For RDD) Is result robust to bandwidth choice?	Estimate with multiple bandwidths
Covariate adjustment	Does controlling for confounders change result?	Add covariates, compare estimates

15 Interpreting Recommendations

15.1 Priority Tiers

Tier	Criteria	Action
Quick Wins	High impact, low blocking factors, Grade A evidence	Immediate adoption recommended
Major Reforms	High impact, significant blocking factors	Requires political capital; strategic timing
Long-Term	Moderate impact, constitutional or treaty constraints	Requires structural change
Monitor	Moderate impact, Grade C/D evidence	Watch for better evidence

15.2 Political Feasibility Notes

While OPG does not filter by political feasibility, it provides context:

- **Organized opposition:** Industries or groups likely to lobby against
- **Public opinion:** Polling data on similar policies where available
- **Adjacent jurisdictions:** Whether neighbors have adopted (diffusion effects)
- **Historical attempts:** Previous failed attempts and why

15.3 Sequencing Guidance

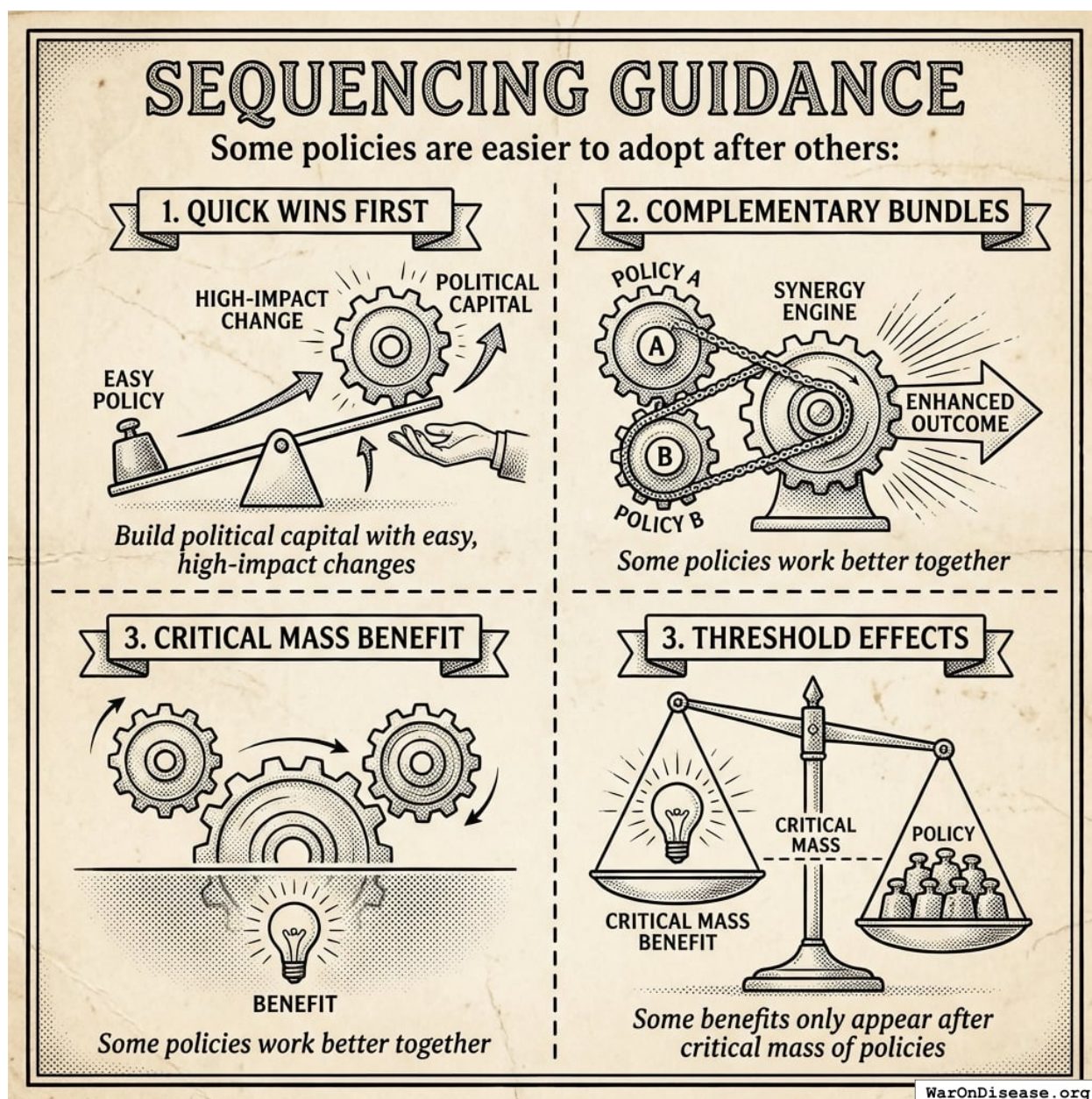


Figure 32: A strategic roadmap showing the progression of policy adoption from immediate quick wins to bundled implementations and reaching critical mass thresholds.

Some policies are easier to adopt after others:

1. **Quick wins first:** Build political capital with easy, high-impact changes
2. **Complementary bundles:** Some policies work better together
3. **Threshold effects:** Some benefits only appear after critical mass of policies

16 Effect Size Benchmarks

Effect sizes are calibrated to cross-jurisdictional variation to aid interpretation:

Size	Income (pp/year)	Health (years)	Example
Small	< 0.05	< 0.1	Minor regulatory changes
Medium	0.05 - 0.15	0.1 - 0.3	Typical tax policy effects
Large	0.15 - 0.30	0.3 - 0.5	Major reform programs
Very Large	> 0.30	> 0.5	Transformative policies (rare)

Calibration basis: US states vary by ~1.5 pp/year in median income growth and ~3-5 years in healthy life expectancy. A “medium” effect represents ~10% of cross-state variation.

Confidence interval interpretation:

- **Narrow** (< 25% of effect): Precise estimate; high confidence
- **Moderate** (25-50% of effect): Reasonable precision
- **Wide** (> 50% of effect): Imprecise; low confidence

For the complete two-metric framework definition, see Section 1.

17 Trial Prioritization

17.1 Value of Information Calculation

The expected value of running a randomized trial on policy p is:

$$VOI_p = P(\text{adopt}|\text{trial}) \cdot E[\text{benefit}|\text{trial}] - P(\text{adopt}|\text{no trial}) \cdot E[\text{benefit}|\text{no trial}] - \text{Cost}_{\text{trial}}$$

Policies with high VOI have: - **High prior uncertainty:** Current evidence is inconclusive - **High potential impact:** If the policy works, benefits are large - **Low trial cost:** Policy can be randomized in small jurisdictions cheaply - **Decision relevance:** Trial result would change adoption decision

17.2 Natural Experiment Identification

The system automatically identifies potential natural experiments:

Type	Identification Method	Example
Border discontinuity	Adjacent jurisdictions with different policies	Minimum wage differences at state borders
Temporal discontinuity	Abrupt policy change	Court ruling invalidating previous policy

Type	Identification Method	Example
Eligibility threshold	Sharp cutoff for policy application	Income threshold for benefit eligibility
Staggered adoption	Different jurisdictions adopting at different times	Unemployment insurance extensions by state
Lottery	Random assignment (rare)	Charter school lotteries
Court mandate	Externally imposed change	Desegregation orders

Identified natural experiments are stored in `natural_experiments` table for validation.

17.3 Recommended Pilot Jurisdictions

For policies with PIS in the “pilot” range (0.10-0.25), we recommend jurisdictions based on:

1. **Variation feasibility:** Jurisdiction has autonomy to adopt the policy
2. **Data quality:** Good administrative data for outcome measurement
3. **Donor pool:** Similar jurisdictions available as controls
4. **Political openness:** Leadership interested in evidence-based pilots
5. **Scalability:** Results can inform larger-scale adoption

18 Data Sources

18.1 Primary Policy Databases

Database	Coverage	URL	Use Case
V-Dem	202 countries, 1789-present	v-dem.net	Democracy indices, political institutions
Polity V	167 countries, 1800-present	systemic-peace.org	Regime type, political stability
CPDS	36 OECD, 1960-present	cpds-data.org	Economic policy, welfare state
OECD iLibrary	OECD members	oecd-ilibrary.org	Tax, labor, education policy
Congress.gov	US federal, 1973-present	congress.gov	US federal legislation
EUR-Lex	EU, 1951-present	eur-lex.europa.eu	EU legislation and regulations

18.2 Primary Outcome Databases

Database	Coverage	URL	Use Case
World Bank WDI	217 countries, 1960-present	data.world-bank.org	GDP, poverty, education, health

Database	Coverage	URL	Use Case
Our World in Data	Global, varies	ourworldindata.org	Curated outcome metrics
WHO GHO	Global	who.int/data/globe	Health outcomes
Penn World Tables	183 countries, 1950-present	ggdc.net/pwt	GDP, productivity, prices
SIPRI	Global, 1949-present	sipri.org	Military spending
IMF	190 countries	imf.org/data	Fiscal, monetary indicators

18.3 Subnational Data

Country	Source	Coverage
United States	Census Bureau, BLS, state agencies	50 states + territories
European Union	Eurostat regional database	~300 NUTS-2 regions
India	CMIE, NSS, state data portals	28 states + territories
China	National Bureau of Statistics	31 provinces
Brazil	IBGE	27 states

18.4 Jurisdiction Policy Inventory Sources

Level	Source	Coverage
US States	NCSL State Legislation Database	All 50 states, continuous updates
US States	State government websites	Primary verification
US Cities	Municode, American Legal Publishing	Major cities
Countries	OECD Government at a Glance	OECD members
Countries	World Bank Doing Business (archived)	190 economies
EU	EUR-Lex	All member states

19 Limitations

19.1 Oracle Capture Risk

The measurement process itself can be captured:

1. **Outcome measurement:** Agencies reporting outcomes have incentives to manipulate
2. **Policy implementation dates:** Recording when policies “really” took effect is subjective
3. **Confounder selection:** Which confounders to control affects estimates

Mitigation: Multiple independent data sources, pre-registered analysis protocols, adversarial audits.

19.2 Confounding Severity

Policy effects face more confounding than drug trials:

Confounder Type	Example	Mitigation
Economic cycles	Recession coincides with policy	Control for GDP growth, unemployment
Secular trends	Improving health over time	Include time trends, compare to controls
Selection	Jurisdictions adopting policies differ	Matching, synthetic control
Spillovers	Neighboring policies affect outcomes	Spatial controls, SUTVA violations noted
Reverse causality	Outcomes drive policy adoption	Instruments, timing-based identification

19.3 Heterogeneous Effects

Policy effects vary by: - Jurisdiction characteristics (income, institutions, culture) - Implementation fidelity - Complementary policies - Time period

High heterogeneity ($I^2 > 75\%$) suggests context-dependence rather than universal effects.

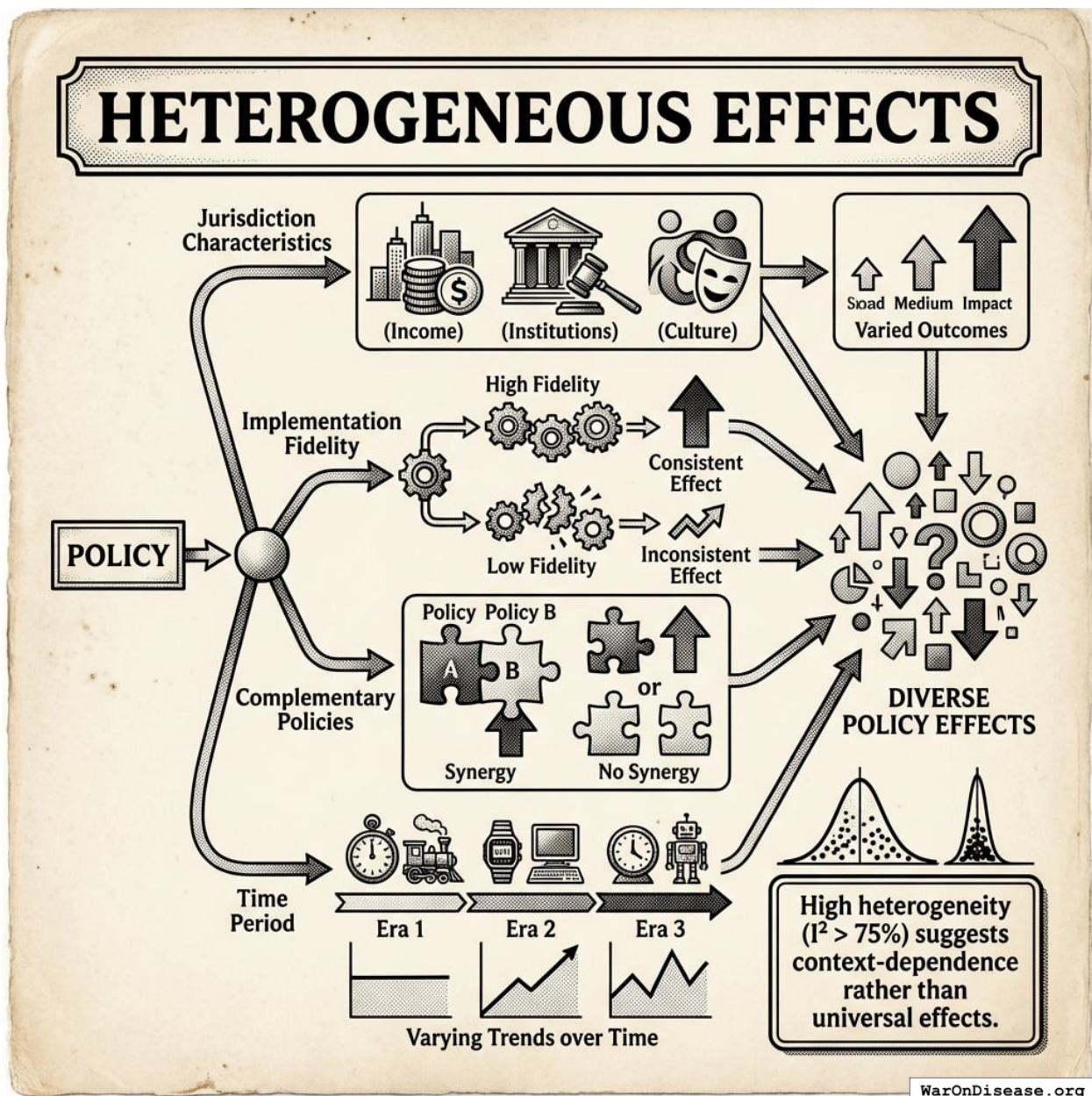


Figure 33: A visualization showing how a single policy can result in diverse outcomes across different jurisdictions based on factors like culture, implementation quality, and timing.

19.4 Jurisdiction-Specific Caveats

Caveat	Description	Mitigation
Data completeness	Policy inventory may be incomplete	Flag data quality; recommend verification
Context transfer	Effect in State A may not transfer to State B	Adjust for observable differences; widen CIs

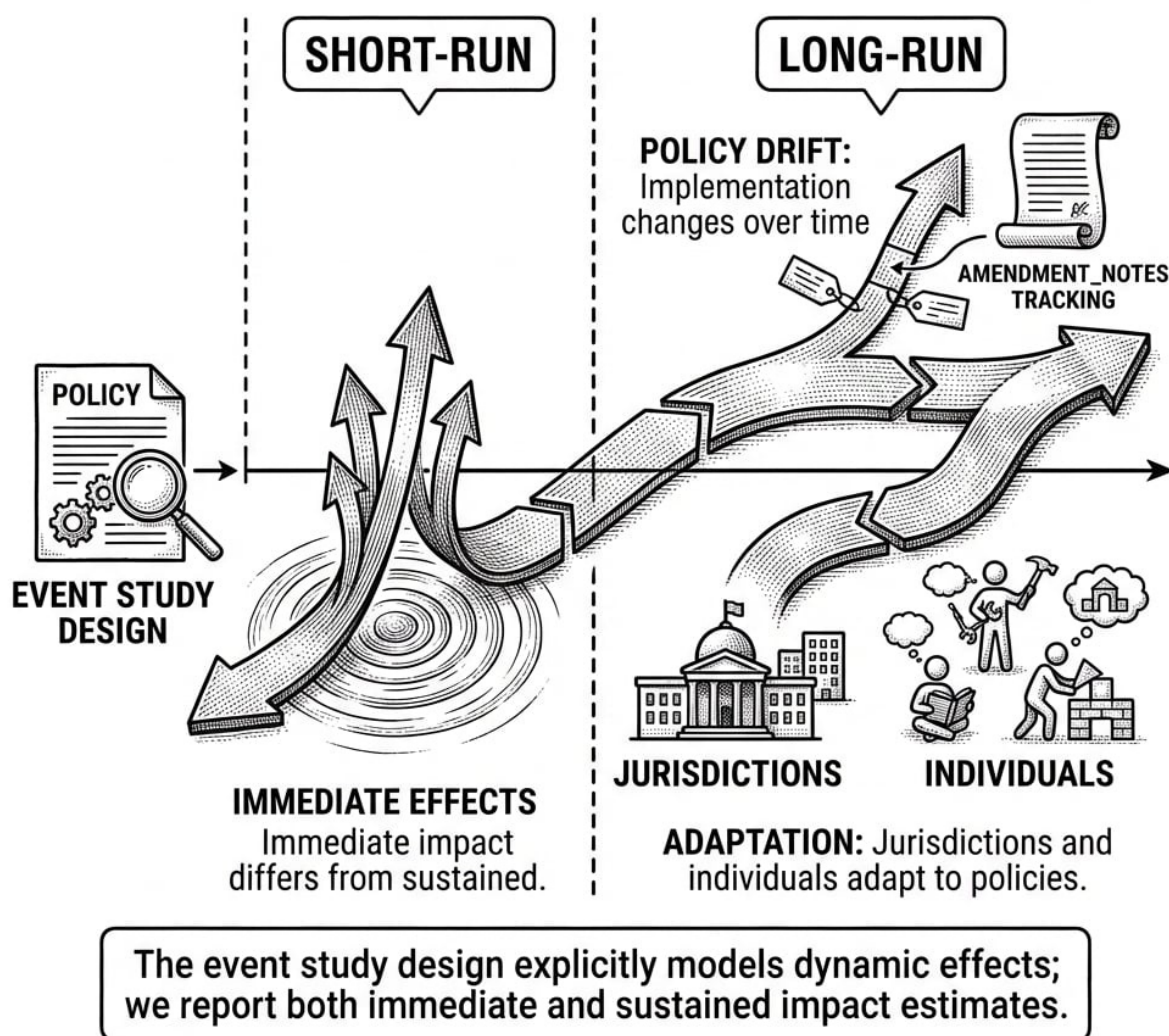
Caveat	Description	Mitigation
Implementation variation	Same policy, different enforcement	Track implementation quality where possible
Interaction effects	Effect depends on other policies in place	Model policy bundles, not just single policies

19.5 Time-Varying Effects

- **Short-run vs. long-run:** Immediate effects may differ from sustained effects
- **Policy drift:** Implementation changes over time (amendment_notes tracking)
- **Adaptation:** Jurisdictions and individuals adapt to policies

The event study design explicitly models dynamic effects; we report both immediate and sustained impact estimates.

TIME-VARYING EFFECTS



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Figure 34: A dynamic impact chart illustrating the progression from short-run immediate effects to long-run sustained effects, accounting for policy drift and adaptation over time.

19.6 Publication Bias

The policy evaluation literature suffers from systematic publication bias:

1. **Null effects underreported:** Studies finding “no significant effect” are less likely to be published
2. **Positive framing:** Researchers may frame results to emphasize statistically significant findings
3. **File drawer problem:** Failed replications rarely published
4. **Jurisdiction selection:** Jurisdictions with cleaner natural experiments are overrepresented

Mitigation strategies:

- Weight by inverse probability of publication (using funnel plot asymmetry tests)
- Require pre-registration of analysis protocols before data access
- Include unpublished working papers and government reports
- Apply trim-and-fill or PET-PEESE corrections for funnel plot asymmetry
- Report null findings prominently in the database

19.7 Epistemic Limitations

OPG provides *evidence-weighted recommendations*, not causal proof:

What OPG Can Do	What OPG Cannot Do
Rank policies by strength of quasi-experimental evidence	Prove any policy causes an outcome
Generate jurisdiction-specific recommendations	Guarantee effects transfer to new contexts
Identify promising candidates for randomized pilots	Replace randomized policy experiments
Quantify uncertainty and heterogeneity	Eliminate unmeasured confounding
Flag potential harms with moderate confidence	Guarantee a policy is safe
Transfer evidence across similar jurisdictions	Account for all local factors

Important: The quasi-experimental methods used provide evidence *consistent with* causation under assumptions that are often untestable. Synthetic control assumes the donor pool adequately represents the counterfactual; difference-in-differences assumes parallel trends would have continued; regression discontinuity assumes no manipulation around the threshold. These assumptions cannot be verified from data alone.

20 Validation Framework

20.1 The Critical Question

The ultimate test of OPG validity: **Do jurisdictions that adopt high-priority OPG recommendations see better outcomes than those that don't?**

20.2 Addressing Adoption Bias

A naive retrospective comparison suffers from **adoption bias**: jurisdictions that voluntarily adopt policies may differ systematically from those that don't. States adopting tobacco tax increases may already have anti-smoking momentum, overstating the causal effect of the tax itself.

Instrumental variable approach:

To address adoption bias, validation should exploit **exogenous shocks to adoption**:

Exogenous Shock	Example	Rationale
Court rulings	State court strikes down previous policy	Adoption forced by legal ruling, not political choice
Federal mandates	Clean Air Act state implementation	Compliance driven by federal law, not state preference
Close electoral outcomes	Ballot measure passes 51-49%	Near-randomization around threshold
Leadership turnover	New governor from different party	Adoption reflects leadership change, not underlying trends

These quasi-random adoption events provide cleaner tests of OPG predictions than voluntary adoption comparisons.

20.3 Proposed Validation Study

Design: Retrospective prediction using instrumental variable identification.

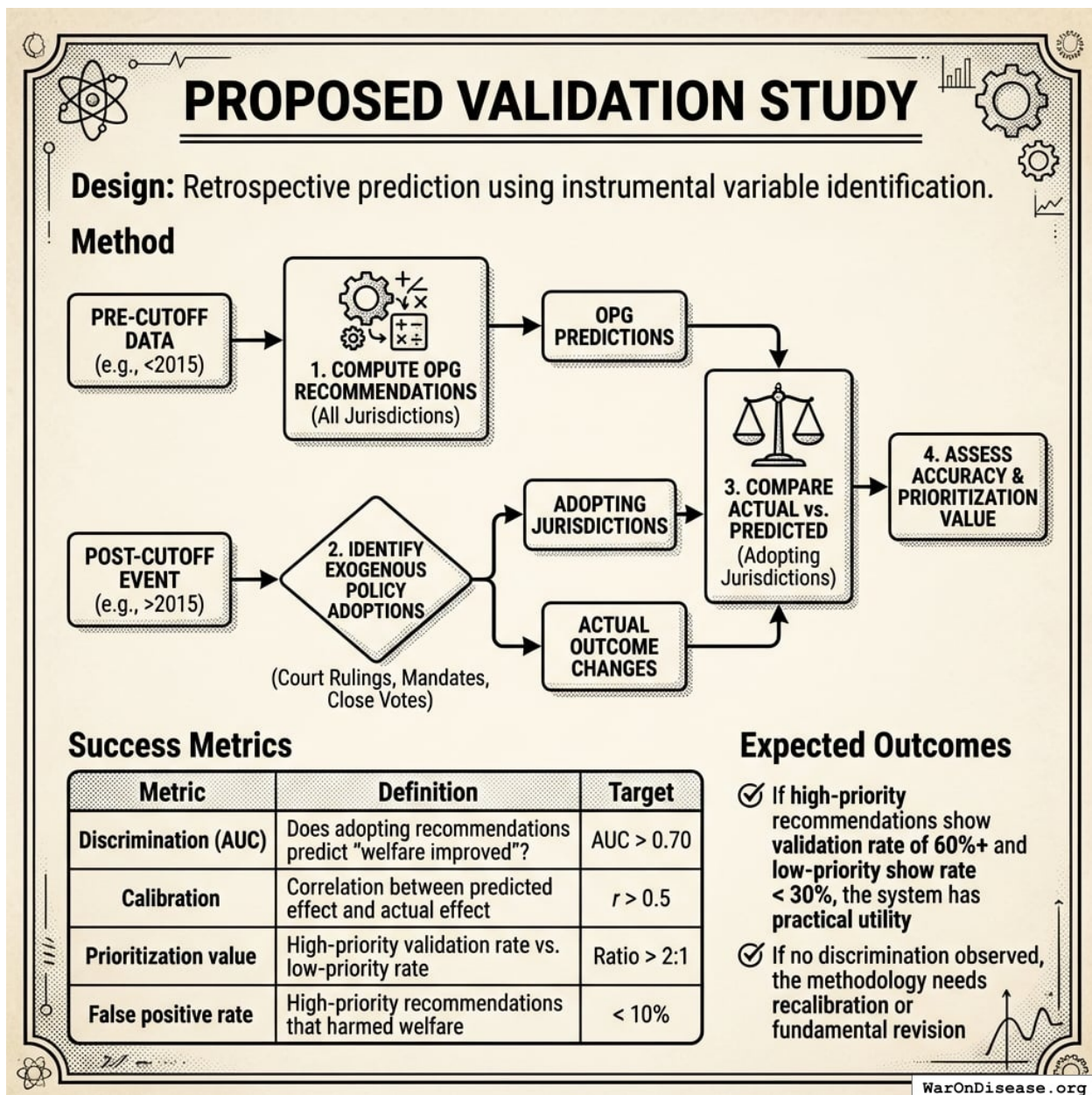


Figure 35: A flowchart illustrating the four-step retrospective validation methodology, from initial data computation through policy identification to final performance assessment.

Method:

1. Compute OPG recommendations for all jurisdictions using only data available before a cutoff date (e.g., 2015)
2. Identify **exogenously-induced** policy adoptions (court rulings, mandates, close votes) after the cutoff
3. Compare actual outcome changes in adopting jurisdictions to OPG predictions
4. Assess prediction accuracy and prioritization value

Success Metrics (strengthened from initial draft):

Metric	Definition	Target
Discrimination (AUC)	Does adopting recommendations predict “welfare improved”?	$AUC > 0.70$
Calibration	Correlation between predicted effect and actual effect	$r > 0.5$
Prioritization value	High-priority validation rate vs. low-priority rate	Ratio $> 2:1$
False positive rate	High-priority recommendations that harmed welfare	$< 10\%$

Expected Outcomes:

- If high-priority recommendations show validation rate of 60%+ and low-priority show rate $< 30\%$, the system has practical utility
- If no discrimination observed, the methodology needs recalibration or fundamental revision

20.4 Prospective Pre-Registration

To prevent hindsight bias, OPG should **publish recommendations before adoption decisions are made**:

1. Quarterly publication of jurisdiction-specific recommendations with timestamps
2. Public pre-commitment to methodology (no post-hoc adjustments)
3. Tracking of which recommendations were subsequently adopted
4. Comparison of pre-registered predictions to actual outcomes

This creates an auditable record that prevents retrofitting methodology to match observed outcomes.



Figure 36: A workflow illustrating the four-step pre-registration cycle from methodology commitment to outcome comparison to ensure an auditable record.

20.5 Known Limitations Requiring Validation

1. **Context adjustment accuracy:** Do jurisdiction-specific adjustments improve prediction?
2. **Blocking factor impact:** Are recommendations with blocking factors less likely to be adopted?
3. **Evidence grade thresholds:** Are the A-F grade cutoffs appropriately calibrated?
4. **Heterogeneity interpretation:** Does high I^2 actually indicate context-dependence vs. measurement noise?
5. **Translation pipeline accuracy:** Do surrogate→terminal metric conversions introduce

systematic bias?

20.6 Continuous Improvement via Adoption Feedback

OPG improves through a learning loop:

1. **OPG generates recommendation** with expected effect $\pm$ uncertainty
2. **Jurisdiction adopts** policy at recommended level
3. **Jurisdiction tracks** primary metric per tracking guidance
4. **Jurisdiction reports** outcomes to OPG feedback system
5. **OPG incorporates** new data point into meta-analysis
6. **Future recommendations** reflect updated evidence

This transforms OPG from a static evidence aggregator into a self-improving system where every adoption strengthens the evidence base. The tracking guidance included with each recommendation standardizes what data jurisdictions should collect and report.

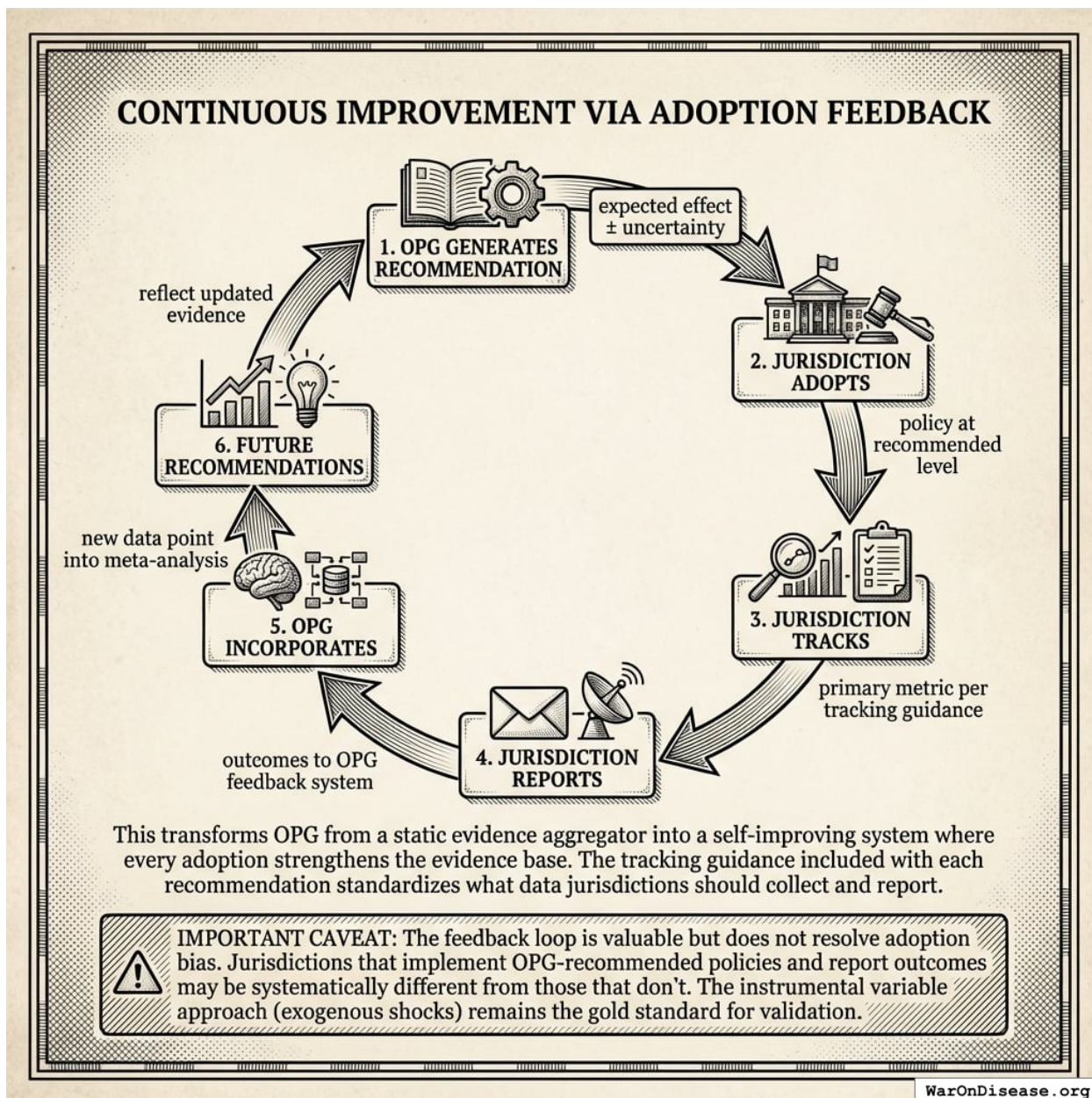


Figure 37: A circular flowchart depicting the six-step learning loop where OPG recommendations lead to jurisdictional adoption and data reporting, which then feeds back into OPG's meta-analysis to refine future policy guidance.

Important caveat: The feedback loop is valuable but does not resolve adoption bias. Jurisdictions that implement OPG-recommended policies and report outcomes may be systematically different from those that don't. The instrumental variable approach (exogenous shocks) remains the gold standard for validation.

21 Future Directions

21.1 Methodological Improvements

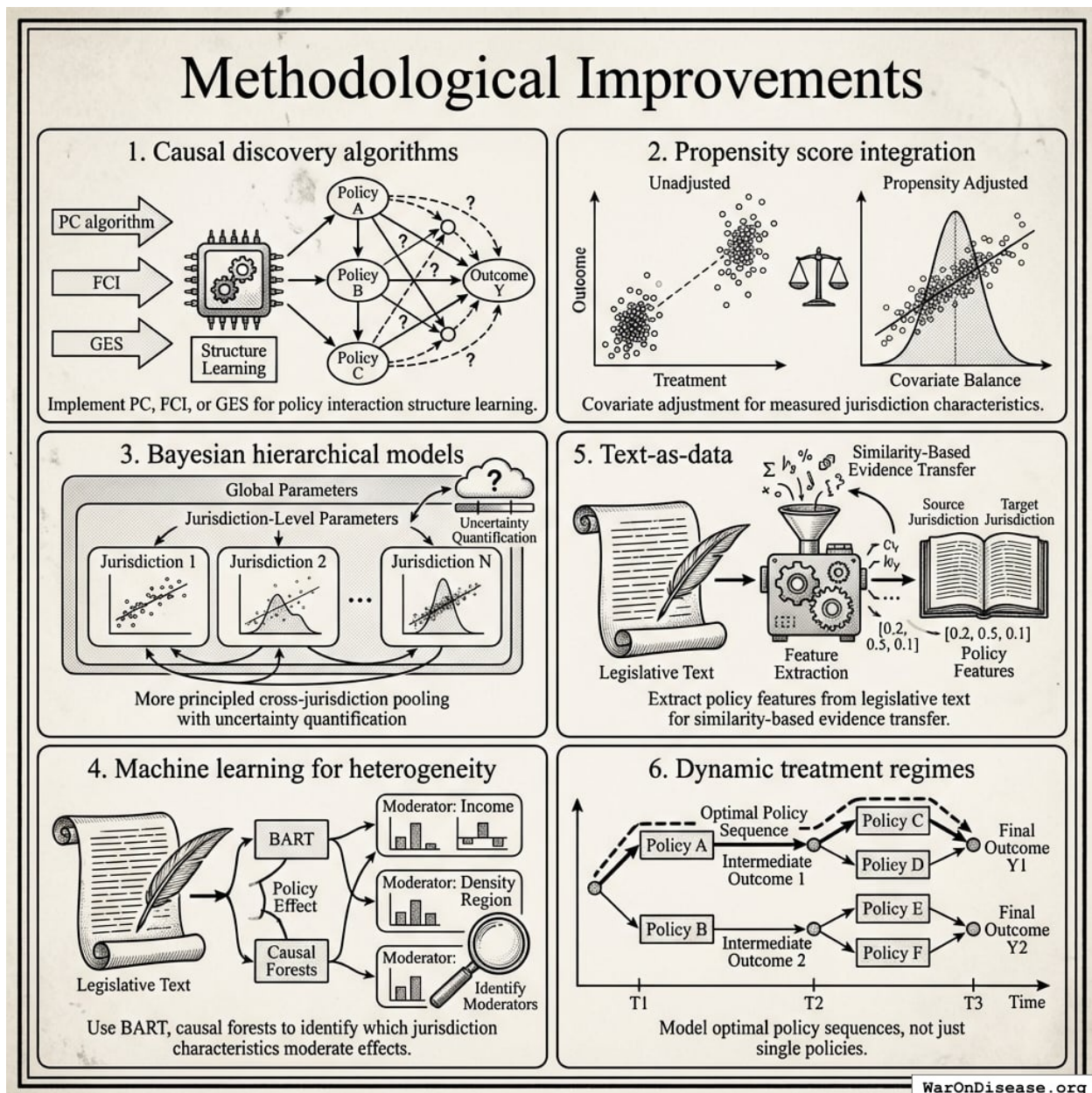


Figure 38: A conceptual framework illustrating how advanced methodological components like causal discovery, text-as-data, and machine learning integrate into a comprehensive policy evaluation pipeline.

- 1. Causal discovery algorithms:** Implement PC algorithm, FCI, or GES for policy interaction structure learning
- 2. Propensity score integration:** Covariate adjustment for measured jurisdiction characteristics
- 3. Bayesian hierarchical models:** More principled cross-jurisdiction pooling with uncertainty

quantification

4. **Machine learning for heterogeneity:** Use BART, causal forests to identify which jurisdiction characteristics moderate effects¹⁶³
5. **Text-as-data:** Extract policy features from legislative text for similarity-based evidence transfer
6. **Dynamic treatment regimes:** Model optimal policy sequences, not just single policies

21.2 Validation Priorities

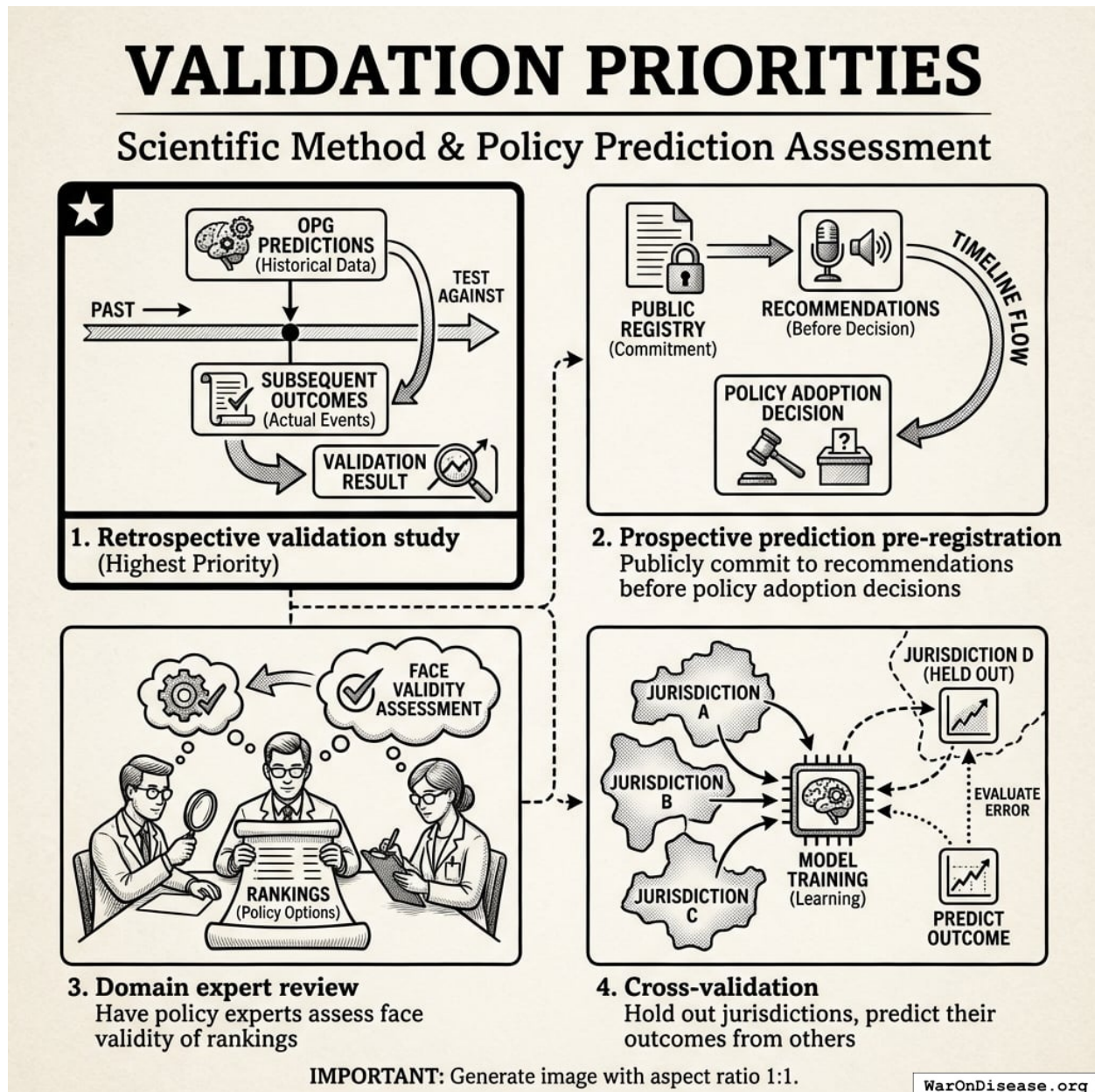


Figure 39: A strategic framework illustrating the four validation priorities for OPG predictions, highlighting the hierarchy of methods from retrospective studies to expert reviews.

1. **Retrospective validation study** (highest priority): Test OPG predictions against subsequent outcomes
2. **Prospective prediction pre-registration**: Publicly commit to recommendations before policy adoption decisions
3. **Domain expert review**: Have policy experts assess face validity of rankings
4. **Cross-validation**: Hold out jurisdictions, predict their outcomes from others

21.3 Data Infrastructure

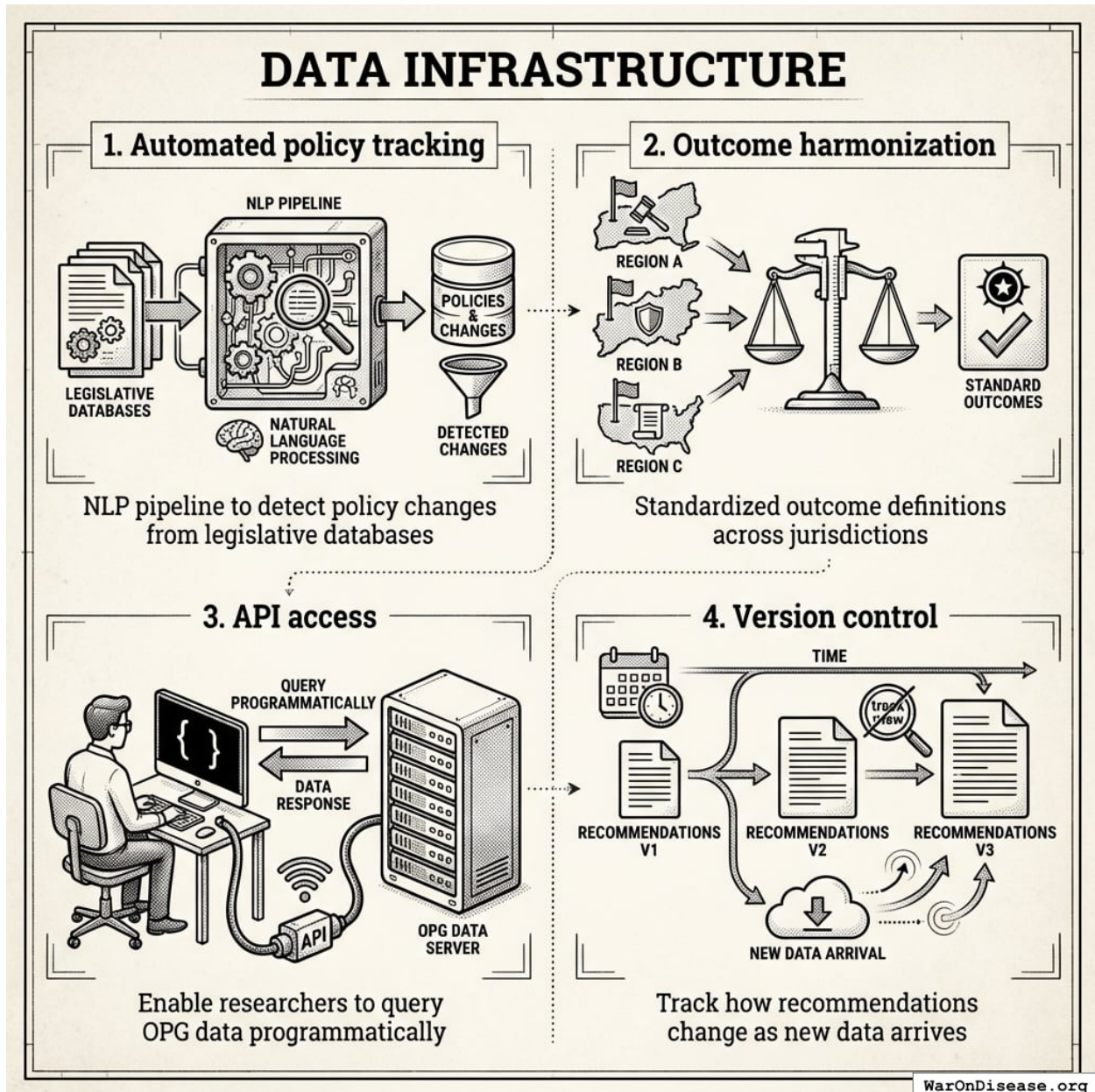


Figure 40: A data infrastructure pipeline illustrating the flow from legislative data collection and NLP processing to standardized outcome harmonization and researcher API access.

1. **Automated policy tracking:** NLP pipeline to detect policy changes from legislative databases
2. **Outcome harmonization:** Standardized outcome definitions across jurisdictions
3. **API access:** Enable researchers to query OPG data programmatically
4. **Version control:** Track how recommendations change as new data arrives

21.4 Integration with Decision-Making

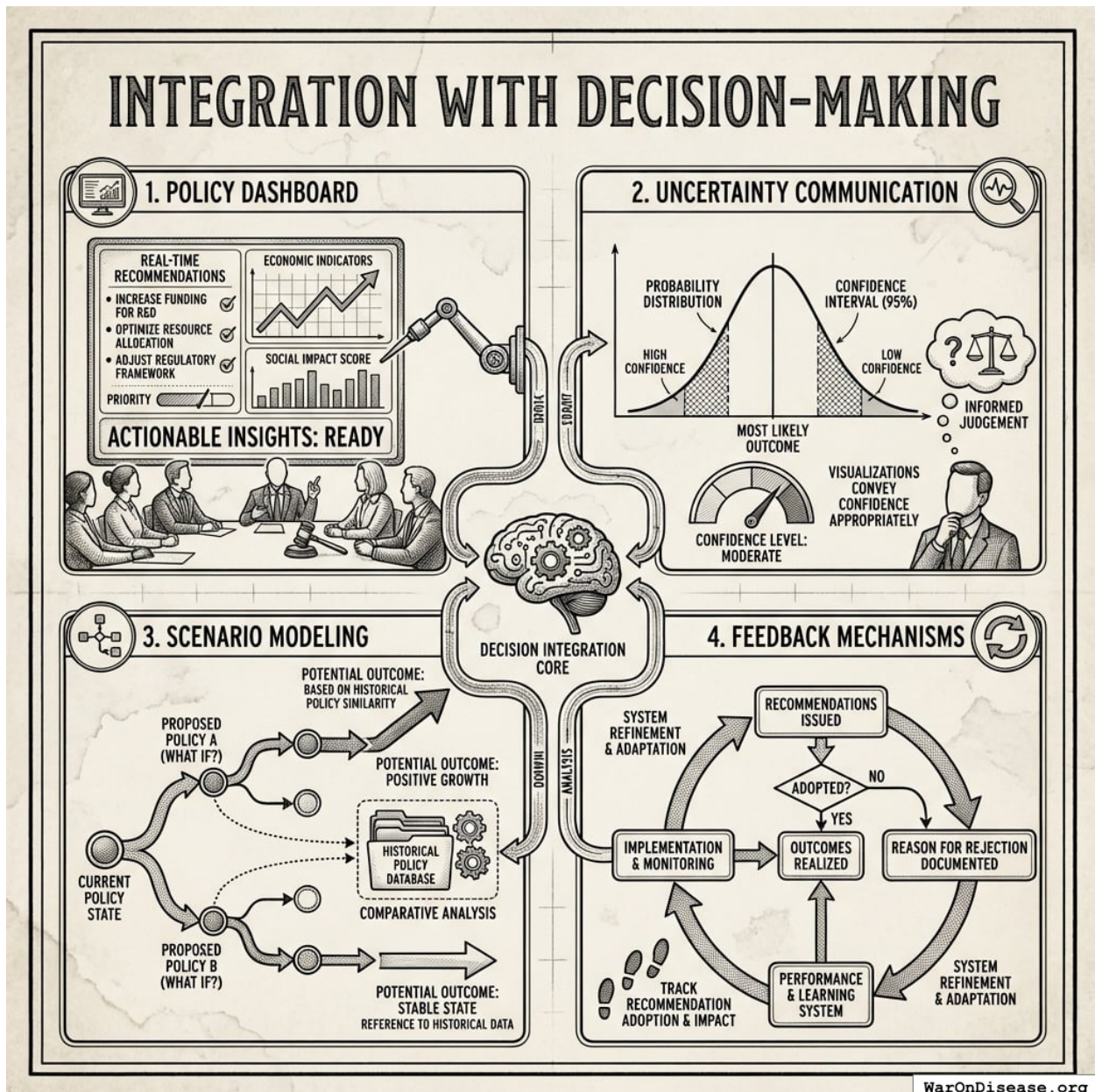


Figure 41: A circular process diagram showing the iterative loop between policy dashboards, uncertainty communication, scenario modeling, and feedback mechanisms.

1. **Policy dashboard:** Real-time recommendations for policymakers

2. **Uncertainty communication:** Visualizations that convey confidence appropriately
3. **Scenario modeling:** “What if” analysis for proposed policies based on similar historical policies
4. **Feedback mechanisms:** Track whether recommendations were actually adopted and outcomes realized

22 Conclusion

The Optimal Policy Generator provides a systematic framework for translating policy-outcome evidence into jurisdiction-specific recommendations. By comparing each jurisdiction’s current policy inventory to the evidence-supported set, OPG produces actionable recommendations in four categories (enact/replace/repeal/maintain) ranked by expected welfare impact. The framework transforms scattered natural experimental evidence into actionable, jurisdiction-specific guidance.

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24 **Appendix A: Worked Example - Texas Policy Recommendations**

 **SYNTHETIC DATA - NOT EMPIRICAL FINDINGS**

All numbers in this appendix are fabricated for illustration. The effect sizes (+0.15 years, +0.02 pp/year, etc.), confidence intervals, and Bradford Hill scores are synthetic placeholders demonstrating the OPG framework’s output format. They were not derived from actual data analysis.

Do not cite these numbers as empirical findings. Actual policy effects would require jurisdiction-specific evidence analysis using real data from the sources described in this specification.

24.1 **Overview**

This worked example demonstrates the complete OPG output for a specific jurisdiction: Texas. It shows how generic policy evidence is translated into jurisdiction-specific recommendations.

24.2 **Texas Policy Inventory (Sample)**

Policy	Status	Income Effect	Health Effect	Recommendation	Grade
Primary seat belt	Missing	+0.02 pp/yr	+0.15 years	ENACT	A
Motorcycle helmet (all ages)	Partial	+0.01 pp/yr	+0.08 years	ENACT	A
Tobacco tax (\$1.41→\$2.50)	Suboptimal	-0.02 pp/yr	+0.25 years	REPLACE	A
Speed limit (85→70 mph)	Excessive	+0.01 pp/yr	+0.06 years	REPLACE	B
DUI threshold (0.08 BAC)	Optimal	N/A	N/A	MAINTAIN	A
Graduated licensing	Optimal	N/A	N/A	MAINTAIN	A

24.3 Step 1: Calculate Policy Impact Scores

Example: Primary Seat Belt Law

From meta-analysis of 47 US states (2000-2020):

Metric	Effect	SE	I ²	Grade
Income (pp/yr)	+0.025	0.008	28%	A
Health (years)	+0.18	0.04	28%	A

Income effect derives from reduced healthcare costs and fewer disability-related productivity losses. Health effect converts mortality reduction to median healthy life years.

Bradford Hill Criteria Scores (applies to both metrics):

Criterion	Score	Rationale
Strength	0.75	Moderate standardized effects on both metrics
Consistency	0.82	I ² = 28%, consistent across states
Temporality	0.95	Clear temporal ordering
Plausibility	0.90	Clear mechanism (increased compliance)
Experiment	0.85	Multiple synthetic control studies

CCS = 0.81 → Grade A

24.4 Step 2: Apply Context Adjustment for Texas

Factor	Texas Value	Adjustment
Current seat belt use	91.5%	Effect may be smaller (already high)
Rural driving proportion	High	Effect may be larger (more severe crashes)
Population	29.5M	Scale up total state-level impact

Adjusted expected effects for Texas:

- Income: +0.02 pp/year (slightly smaller due to already-high compliance)
- Health: +0.15 years

24.5 Step 3: Generate Recommendations

OPG Recommendations for Texas

💡 ENACT (New Policies to Adopt)

1. Primary Seat Belt Enforcement Law (Current: secondary enforcement only)

Metric	Expected Effect	95% CI
Income	+0.02 pp/year	[+0.01, +0.03]
Health	+0.15 years	[+0.10, +0.20]

- **Evidence grade:** A
- **Priority:** High
- **Blocking factors:** None identified
- **Similar jurisdictions:** Florida adopted 2009; saw +0.18 years health effect

2. Universal Motorcycle Helmet Requirement (Current: partial - under 21 only)

Metric	Expected Effect	95% CI
Income	+0.01 pp/year	[+0.005, +0.015]
Health	+0.08 years	[+0.05, +0.12]

- **Evidence grade:** A
- **Priority:** Medium
- **Blocking factors:** Political (autonomy concerns from rider groups)
- **Similar jurisdictions:** California (all ages since 1992)

⚠️ REPLACE (Policies to Modify)

3. Tobacco Tax: \$1.41 → \$2.50/pack

Metric	Expected Effect	95% CI
Income	-0.02 pp/year	[-0.04, +0.01]
Health	+0.25 years	[+0.18, +0.32]

- **Current level:** \$1.41/pack (illustrative; below median \$1.91)
- **Recommended level:** \$2.50/pack (illustrative target)
- **Evidence grade:** A
- **Priority:** High
- **Blocking factors:** Political (anti-tax sentiment)

Tradeoff Demonstration

This recommendation illustrates how the two-metric framework surfaces genuine tradeoffs:

- **Income effect is negative:** The tax directly reduces disposable income for smokers. Productivity gains (fewer sick days) partially offset this, but the net income effect is small and negative.
- **Health effect is strongly positive:** Reduced smoking leads to longer healthy life expectancy.
- **Combined welfare is positive:** At equal weighting ($\alpha=0.5$), the health gain outweighs the income cost.

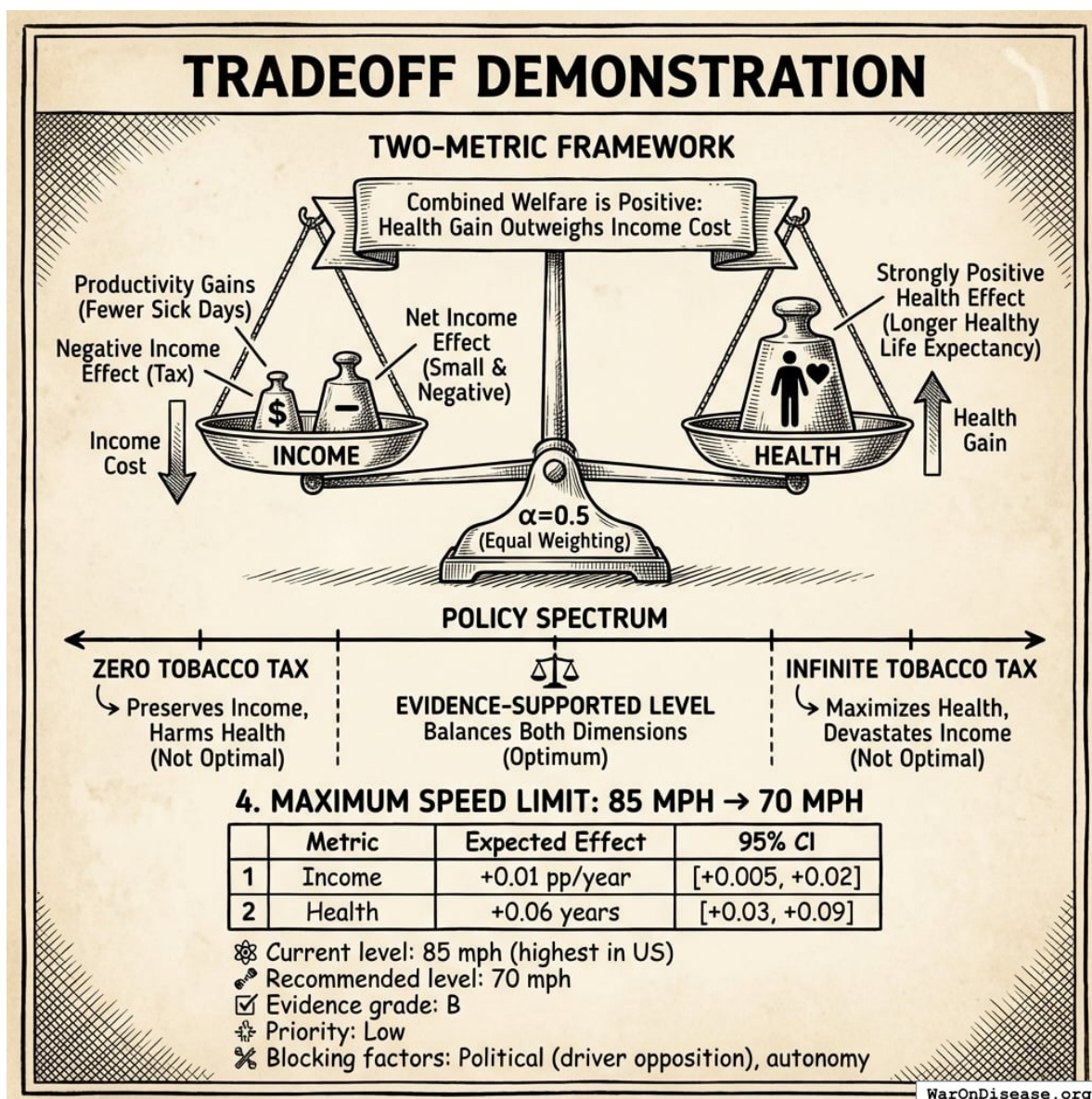


Figure 42: A two-axis tradeoff visualization plotting income impact against health gains, illustrating the optimal policy point between the extremes of infinite and zero tobacco taxes.

This demonstrates why the two-metric framework prevents corner solutions: - An **infinite tobacco tax** would maximize health but devastate income → not optimal
- A **zero tobacco tax** would preserve income but harm health → not optimal
- The **evidence-supported level** balances both dimensions

The recommendation shows where evidence places the optimum, while making the tradeoff explicit for democratic deliberation. Jurisdictions that weight income more heavily may choose a lower tax; those prioritizing health may choose higher.

4. Maximum Speed Limit: 85 mph → 70 mph

Metric	Expected Effect	95% CI
Income	+0.01 pp/year	[+0.005, +0.02]
Health	+0.06 years	[+0.03, +0.09]

- **Current level:** 85 mph (highest in US)
- **Recommended level:** 70 mph
- **Evidence grade:** B
- **Priority:** Low
- **Blocking factors:** Political (driver opposition), autonomy

REPEAL (Policies to Remove)

*No high-priority repeal recommendations for Texas at this time.
(Example format: If Texas had a policy shown to cause net harm, it would appear here with expected welfare gain from removal.)*

MAINTAIN (No Change Needed)

5. DUI Threshold at 0.08 BAC

- **Current level:** 0.08 BAC (national standard)
- **Evidence:** Aligned with evidence-supported level
- **Status:** Continue current policy

6. Graduated Driver Licensing Program

- **Current level:** Three-stage system with night/passenger restrictions
- **Evidence:** Consistent with best practices
- **Status:** Continue current policy

24.6 Step 4: Summary Dashboard

Total Expected Welfare Gain by Recommendation Type

Type	Recommendation	Income Effect	Health Effect	Grade
EN- ACT	Primary seat belt	+0.02 pp/yr	+0.15 years	A
EN- ACT	Universal helmet	+0.01 pp/yr	+0.08 years	A
RE- PLACE	Tobacco tax: \$1.41→\$2.50	-0.02 pp/yr	+0.25 years	A
RE- PLACE	Speed limit: 85→70 mph	+0.01 pp/yr	+0.06 years	B
MAIN- TAIN	DUI threshold, GDL	N/A	N/A	A
Total from changes		+0.02 pp/yr	+0.54 years	

Note: The tobacco tax shows a negative income effect but positive health effect, demonstrating the

two-metric framework's ability to surface genuine tradeoffs. MAINTAIN items confirm evidence alignment and require no action. REPEAL section empty for Texas; no harmful policies identified with strong evidence.

24.7 Interpretation

This example demonstrates how OPG transforms generic evidence into actionable, jurisdiction-specific recommendations using the two-metric framework:

If Texas adopted all recommendations, expected effects are:

- **+0.02 pp/year** net change to median income growth (modest positive after tradeoffs)
- **+0.54 years** added to median healthy life expectancy

Critical observation: The total income effect (+0.02 pp/yr) is much smaller than the total health effect (+0.54 years in relative terms) because the tobacco tax creates a genuine tradeoff. This is a feature, not a bug: it surfaces real policy tensions rather than hiding them behind combined scores.

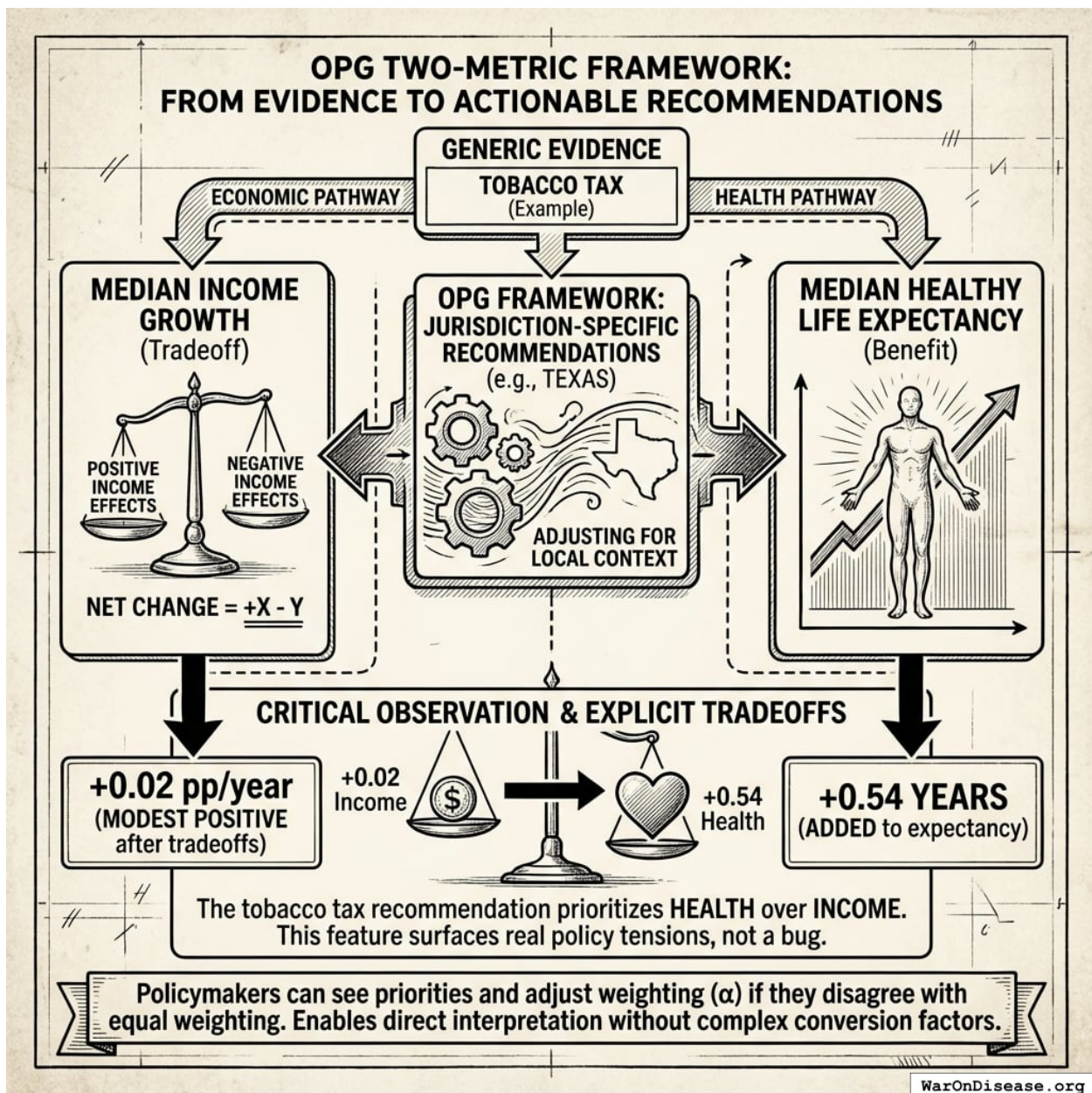


Figure 43: A comparison chart showing the disparate impacts of policy recommendations on income growth versus healthy life expectancy, highlighting the tradeoff between economic and health outcomes.

The two-metric format makes tradeoffs explicit and enables direct interpretation without complex conversion factors. Policymakers can see that the tobacco tax recommendation prioritizes health over income, and adjust their weighting () accordingly if they disagree with equal weighting.

25 Appendix B: OPG Analysis Workflow

25.1 Complete OPG Pipeline

OPTIMAL POLICY GENERATOR WORKFLOW

Phase 1: DATA COLLECTION

1. Policy database ingestion
 - Parse legislative text (LLM extraction)
 - Record implementation dates by jurisdiction
 - Classify policy type and category
 - Compute policy embeddings for similarity
2. Jurisdiction policy inventory
 - Pull current policy status for each jurisdiction
 - Record policy strength (for continuous policies)
 - Flag data quality and gaps
 - Identify last verification date
3. Outcome data collection
 - Pull from primary sources (World Bank, WHO, etc.)
 - Harmonize units and definitions
 - Identify missing data patterns
 - Flag measurement quality issues
4. Confounder data collection
 - Economic indicators (GDP, unemployment)
 - Demographic variables (age structure, education)
 - Political variables (regime type, election cycles)
 - Geographic variables (neighbors' policies)

Phase 2: EVIDENCE ANALYSIS (Quasi-Experimental)

5. Policy-outcome pair identification
 - Match policies to plausible outcome categories
 - Filter by minimum data requirements
 - Identify applicable quasi-experimental methods
6. Method selection
 - Synthetic control: single treated, good donors
 - Difference-in-differences: multiple treated, parallel trends
 - Regression discontinuity: sharp threshold exists
 - Event study: need dynamic effects
 - Interrupted time series: fallback

7. Effect estimation

- Run primary analysis
- Calculate standard errors (clustered)
- Compute confidence intervals
- Store jurisdiction-level results

8. Robustness checks

- In-time placebo tests
- In-space placebo tests
- Leave-one-out sensitivity
- Covariate adjustment sensitivity

Phase 3: AGGREGATION & PIS CALCULATION

9. Meta-analysis

- Pool jurisdiction estimates (random effects)
- Calculate I^2 , τ^2 , Q statistics
- Test for publication bias (funnel plot)
- Apply trim-and-fill if needed

10. Bradford Hill scoring

- Score each criterion (0-1)
- Apply criterion weights
- Compute CCS (causal confidence score)
- Document evidence for each criterion

11. PIS calculation

- Standardize effect estimate
- Calculate quality adjustment
- Compute final PIS
- Assign evidence grade (A-F)

Phase 4: RECOMMENDATION GENERATION

12. Policy gap analysis (per jurisdiction)

- Compare current inventory to evidence-supported
- Calculate gap magnitude
- Identify gap type (missing, harmful, suboptimal)
- Flag blocking factors

13. Context adjustment

- Adjust effect estimates for jurisdiction characteristics
- Widen confidence intervals for context uncertainty
- Identify similar jurisdictions for comparison
- Note implementation considerations

14. Priority scoring

- Rank by $|\text{Gap}| \times \text{Evidence Grade} \times \text{Impact}$

Assign to priority tiers (Quick Win, Major Reform, etc.)
 Generate enact/replace/repeal/maintain lists
 Calculate total expected welfare gain

Phase 5: OUTPUT GENERATION

15. Recommendation dashboard

Enact list (new policies to adopt)
 Replace list (existing policies to modify: current → optimal)
 Repeal list (harmful policies to remove)
 Maintain list (policies aligned with evidence)
 Jurisdictional level and tracking guidance for each

16. Two-metric reporting

Income effect (pp/year)
 Health effect (years)
 Combined welfare score (for ranking)

17. Documentation

Generate jurisdiction-specific reports
 Create methodology audit trail
 Version control all recommendations
 Publish to API/dashboard

25.2 Minimum Data Requirements Checklist

Before generating recommendations, verify:

- ☐ 4 pre-treatment periods in evidence base
- ☐ 2 post-treatment periods in evidence base
- ☐ 20 total outcome observations
- ☐ 5 control jurisdictions (for DiD-based evidence)
- ☐ Policy implementation dates documented
- ☐ Outcome variable has known valence
- ☐ Jurisdiction policy inventory verified within 2 years
- ☐ Data quality score 0.5 for target jurisdiction

26 Appendix C: Glossary

Brief definitions for quick reference. See referenced sections for full details.

Term	Definition	See Section
OPG	Optimal Policy Generator: produces jurisdiction-specific recommendations	Section 3
PIS	Policy Impact Score: effect × causal confidence × quality	Section 12

Term	Definition	See Section
CCS	Causal Confidence Score: weighted Bradford Hill average	Section 7.3
Evidence Grade	A-F rating based on PIS, heterogeneity, jurisdiction count	Section 13.4
Policy Gap	Difference between current and evidence-supported policy	Section 9
Blocking Factor	Constraint on adoption (constitutional, political, etc.)	Section 10.2
I²	Between-study heterogeneity (>75% = high)	Section 13
Synthetic Control	Weighted donor pool matching pre-treatment trajectory	Section 7.2
DiD	Difference-in-differences under parallel trends	Section 7.2
RDD	Regression discontinuity at eligibility threshold	Section 7.2

Recommendation types: ENACT (add new), REPLACE (modify level), REPEAL (remove harmful), MAINTAIN (keep current). See Section [10](#).

Bradford Hill criteria: Strength, Consistency, Temporality, Gradient, Experiment, Plausibility, Coherence, Analogy, Specificity. See Section [7.3](#).

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Data Availability: This specification describes a methodological framework. Policy databases referenced (V-Dem, Polity V, CPDS, World Bank WDI) are publicly available at URLs provided in Data Sources section. A complete replication package including data extraction scripts, analysis code, and recommendation generation algorithms will be deposited in a public repository upon system deployment.